

On the CCA security properties of a class of group-based linearly homomorphic encryption schemes^{*}

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Abstract. In a seminal paper at Eurocrypt’25, B. Libert solved the long-standing open question of the CCA1 security of the D  mgard-ElGamal scheme under standard assumptions. That paper also further showed that several other schemes, including a variant of Paillier-ElGamal with plaintext zero-padding, are CCA1 secure under falsifiable assumptions. However, all these schemes are only *somewhat* linearly homomorphic. In this paper, we first tackle the fundamental question of designing efficient “truly” linearly homomorphic encryption schemes achieving CCA1 security under standard and falsifiable assumptions. We generalize Libert’s approach to a large class of group-based PKE, that covers the aforementioned schemes, the Cramer-Shoup Lite scheme as well as new ones. In particular, we use our framework to show that a new variant of Paillier-ElGamal, which we call D  mgard-Paillier-ElGamal (DPEG) as its design follows a Knowledge-of-Exponent pattern, achieves CCA1 security solely under the standard DCR assumption. To the best of our knowledge, DPEG is the first “truly” linearly homomorphic encryption scheme that is proven CCA1 secure under a standard assumption. Furthermore, DPEG can be extended to support one level of multiplication while still preserving its CCA1 security under the same assumption. Second, we show that DPEG also achieves Manulis&Nguyen’s stronger notion of vCCA security under a non-falsifiable linear-only homomorphism assumption. We then connect vCCA security to a no-go Theorem of Gentry&Wichs and show that, under mild assumptions, vCCA security cannot be established from any falsifiable assumption. This explains why all currently known vCCA secure constructions, including ours, rely on non-falsifiable assumptions.

1 Introduction

“Beyond CPA” security for homomorphic schemes has been the subject of intense research over the last five years or so, with both theoretical advances

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(new security notions, impossibility results, or generic constructions) and more concrete constructions. Interestingly, there have been advances for both lattice-based and number-theoretic homomorphic schemes. However, going beyond CPA security often breaks algebraic (homomorphic) properties, typically by adding validity proofs in CCA1-secure public-key encryption (PKE) or even SNARKs in “beyond-CCA1”-secure fully homomorphic encryption (FHE). Even for linearly homomorphic encryption, which is already known to have significant practical applications, achieving security beyond CPA is difficult, with no known techniques achieving CCA1 security under falsifiable assumptions while *fully* preserving the homomorphic property.

Interestingly, several breakthroughs have recently occurred in the more “old-school” realm of number-theoretic linearly homomorphic encryption schemes, with several long-standing open questions finding either negative or positive answers. On the negative side, a recent impossibility result by S. Schäge [38] negatively answered the long-standing open question of the CCA1 security of ElGamal and Paillier under falsifiable assumptions [31] by showing that no black-box reduction can establish CCA1 security for a class of group-based linearly homomorphic schemes, termed SHPKE (“SH” stands for “semi-homomorphic” in Schäge’s terminology, to distinguish it from FHE), under such an assumption. This impossibility result has later been extended to cover the weaker notion of PCA1 security in [39]. On the positive side, Libert [28] recently achieved the tour de force of proving the CCA1 security of the Dămgard-ElGamal scheme [20] solely under the standard DDH assumption, while previous proofs required a non-falsifiable Knowledge-of-Exponent assumption. Interestingly, [28] also provides proofs of CCA1 security, again under (almost) standard assumptions, for variants of BGN and Paillier-ElGamal with sparsified ciphertext domains. However, these schemes achieve this sparsification by forcing λ plaintext bits to be zeroes and, as a result, are “only” *somewhat* linearly homomorphic: when a carry propagates into the zero padding during an homomorphic addition the resulting ciphertext decrypts to $\perp$ rather than the sum of the plaintexts of the two input ciphertexts. Beyond CCA1, [16] also investigates the connection between the non-falsifiable Linear-Only Homomorphism (LOH) assumption of Bitansky et al. [4,5] and the notion of vCCA security introduced by Manulis & Nguyen [30], and establishes the general result that any correct CPA-secure linearly homomorphic scheme with the LOH property achieves vCCA security. This yielded a simple and practical construction, based on the Paillier scheme, that achieves vCCA security at only twice the cost of achieving CPA security, albeit under a non-falsifiable assumption.

On the more theoretical side, It may be interesting to highlight that the large majority of works investigating (F)HE constructions achieving some degree of CCA security [8,12,29,30,10] require either the random oracle model or some non falsifiable assumption due to their reliance on general zk -SNARKs [25] or even stronger primitives such as iO. This is to the exception of the recent (Crypto’25) paper of Yang et. al [41] which achieves the tour de force of giving a construction that is IV-CCA secure (a stronger-than-CCA1 notion they define and study) in

the standard model, based solely on the LWE assumption. On the downside, their construction is intrinsically *non-compact*. The compactness requirement which is naturally expected from FHE, is however achieved by some of the construction blueprints that are proposed in [30,10] (namely Encrypt-then-Sign in [30] and Encrypt-then-Prove in [10]). However, these generic constructions are not easily amenable to efficient implementations in their full generality, essentially because they intimately require powerful SNARK machinery.

1.1 Our Objective

In this paper, we follow the former line of work on designing simple, elegant linearly homomorphic CCA1-secure encryption schemes and complete this line of work by proposing a framework for achieving truly linearly homomorphic functionality, without any validity conditions that would make the schemes only somewhat linearly homomorphic.

Our starting point was the Semi-Homomorphic PKE (SHPKE) class of schemes introduced by S. Schäge to prove his first impossibility result [38]. In a nutshell, this class includes group-based, perfectly correct, linearly homomorphic encryption schemes for which the image of the encryption function defines a group $(H, \cdot) \subseteq \mathcal{C}$, with $\mathcal{C}$ the ciphertext domain of the scheme. Then, the key property is that H is required to have an efficient public membership test or, equivalently, that valid ciphertexts under an SHPKE can be publicly efficiently recognized. On the other hand, B. Libert’s paper [28] shows that several group-based schemes for which *valid ciphertexts cannot be efficiently recognized*, and which therefore fall outside the SHPKE class, can be proven CCA1-secure under falsifiable assumptions, using relatively similar proof patterns. This naturally raises the following question:

Is it possible to define an SHPKE-like class of schemes such that all schemes in this class can be proven CCA1-secure under falsifiable assumptions?

We positively answer this question by defining a class of schemes obtained by *sparsifying* the ciphertext domain of any group-based linearly homomorphic scheme for which the image of the encryption function defines a group $(H, \cdot)$ that is embedded sparsely in a larger group $(H^*, \cdot) \subseteq \mathcal{C}$ such that random elements in H are computationally indistinguishable from random elements in H^* , and thus membership in H cannot be publicly verifiable. We call this class **sGPKE** for *spar-sified Group-based (linearly homomorphic) PKE* and, under certain generic conditions, we show that all schemes in this class are CCA1-secure. Interestingly, the sGPKE class covers a number of previously known schemes (Dåmgard-ElGamal, Cramer-Shoup-Lite and Libert’s variant of Paillier-ElGamal with plaintext zero-padding). We then use this framework to prove that a new convenient linearly homomorphic variant of Paillier-ElGamal achieves CCA1 security under the standard DCR assumption.

Our new variant, which we refer to as the D  mgard-Paillier-ElGamal scheme (or DPEG), is particularly interesting as it also achieves the stronger notion of

vCCA security under an additional previously known non-falsifiable assumption. Since all known constructions achieving vCCA security [30,10,16] rely on non-falsifiable assumptions, this naturally raises a second question:

Is it possible to prove a scheme vCCA-secure under a falsifiable assumption?

We answer this question negatively by showing, under relatively mild assumptions, that the existence of a linearly homomorphic encryption scheme achieving vCCA security under a falsifiable assumption would contradict the no-go theorem of Gentry&Wichs [25] on the existence of SNARGs provably secure under falsifiable assumptions.

The next section summarizes our contributions in more details.

1.2 Summary of contributions

The contributions of this paper are as follows:

- We first define the class of Group-based linearly homomorphic PKE, GPKE. In essence, for this class of schemes, the image of the encryption function defines a group $(H, \cdot)$ embedded in a larger group $(H^*, \cdot)$ such that uniformly picked elements of H and H^* are computationally indistinguishable.
- We then define the class of sparsified GPKE, sGPKE. An sGPKE is built by sparsifying the ciphertext domain of an inner GPKE scheme such that the plaintext domain of the former is 2^λ times smaller than the latter. The decryption function then returns $\perp$ whenever the decryption function of the inner scheme returns an element that is not in the smaller plaintext domain.
- *Our first main contribution* is then to establish that *any* sGPKE achieves CCA1 security, hence under the assumption under which the computational indistinguishability between random elements of H and H^* holds.
- To illustrate the generality of the sGPKE framework, we show that several known schemes satisfy the definition. This includes, Paillier-ElGamal with plaintext zero-padding [28] (under DCR), Dămgård-ElGamal [20,28] (under DDH) and Cramer-Shoup Lite (under DDH). All these schemes were already known to achieve CCA1 security under the same assumptions via different proofs, hence our framework paves the way for more easily proving the CCA1 security of new schemes. Interestingly, Paillier-ElGamal with plaintext zero-padding was only known to achieve CCA1 based solely under DCR by sacrificing 50% of the plaintext size. Our framework establishes that its CCA1 security holds solely under DCR without this restriction, closing an open question from [28]. We also emphasize that for reasons specific to each of them, all the above schemes are only *somewhat* linearly homomorphic.
- *Our second main contribution* is to design a new variant of Paillier-ElGamal with two message slots, DPEG, that implement the Bitansky et al. two-ciphertext blueprint [4,5]: under DPEG, when the first message slot in a ciphertext contains a value m , the second message slot is expected to contain ξm , for a secret integer ξ , and this linear relation is checked upon decryption.

Interestingly DPEG is “truly” (i.e. “non-somewhat”) linearly homomorphic. We show that DPEG is a sGPKE under DCR and thus establish that it is also CCA1 secure under that sole assumption. To the best of our knowledge, *DPEG is the first truly linearly homomorphic scheme that is proven CCA1 secure under a standard assumption* (closing an open problem from [22] which also describes interesting use-cases for such kind of schemes). Furthermore, DPEG achieves CCA1 security at only 1.5 times the cost of the CPA secure baseline Paillier-ElGamal scheme on which it is based.

- As a bonus, we also show that DPEG achieves the stronger notion of vCCA security recently introduced by Manulis&Nugyen under the *non-falsifiable* assumption that it has the Linear-Only Homomorphism (LOH) Property of Bitansky et al. [4,5]. As such, DPEG is the *first linearly homomorphic scheme that achieves vCCA security albeit under a non-falsifiable assumption while being CCA1 secure under a standard assumption*, while previous constructions [16,22] required the non-falsifiable assumption to close the gap from CPA to vCCA security rather than only from CCA1 to vCCA.
- Additionally, we also leverage on the Catalano-Fiore blueprint [14] to get one level of homomorphic of multiplication from DPEG. In doing so and to the best of our knowledge, we obtain *the first concrete scheme supporting both homomorphic additions and multiplications (even limited to one level) that is proven CCA1 secure under a standard assumption*. Furthermore, this scheme inherits its vCCA security from that of DPEG.
- *Our third main contribution* is to formally establish a connection between vCCA security and Gentry&Wichs’ no-go theorem on the existence of SNARGs that can be secure under falsifiable assumptions [25]. Indeed, we establish that, under relatively mild assumptions, no linearly homomorphic scheme can be proven vCCA secure under a falsifiable assumption (by means of a black-box reduction). Eventually, the whole class of sGPKE falls under this result, hence no sGPKE can be proven vCCA secure under a falsifiable assumption. In the sense that it achieves CCA1 security under a falsifiable assumption and then vCCA security under an additional non-falsifiable one, DPEG thus attains some form of “optimality” regarding the falsifiable nature of the assumptions under which its security properties hold.

1.3 Technical overview of our main result

We provide below an informal definition of the class of sGPKE schemes related to our main result that establishes that any scheme in this class achieves CCA1 security under a falsifiable computational indistinguishability assumption. Intuitively, any scheme in this class is obtained by *sparsifying* the ciphertext domain of another scheme that belong to the class of GPKE. The formal definitions are given in Sect. 3.1.

GPKE (Def. 9). The class of GPKE contains group-based linearly homomorphic schemes for which the plaintext domain defines a group $(G, +)$ and the image of the encryption function defines a group $(H, \cdot)$ that is embedded in a larger group

$(H^*, \cdot)$ such that random elements in H are computationally indistinguishable from random elements in H^* (property 3). From this class, we ask that the secret information that allows to distinguish between $\{z \xleftarrow{\$} H\}$ and $\{z \xleftarrow{\$} H^*\}$ is *not* required to decrypt (property 1) and further gives an efficient exact membership test for elements in H (property 8). From this, we will be able to use a random element from either H or H^* to generate a challenge ciphertext (using the homomorphic properties) while giving the decryption key to a simulator playing the CCA1 game against some adversary, without the adversary noticing. Thanks to the exact membership test, we will be able to define an alternative decryption function that returns $\perp$ for any ciphertext not in H . As additional natural properties, we require that GPKE are linearly homomorphic (property 5) and that evaluated ciphertexts are distributed like freshly encrypted ones (property 6). This latter property will allow us to build an alternative encryption function that uses the scheme homomorphic properties and that maps elements of G onto elements of H^* (rather than H), but cannot be distinguished from the original encryption function by an adversary. Lastly, given the public/private key pair of a GPKE, we ask that there exists many alternative secret keys that behave like the original one over H (property 7) while yielding independent outputs outside of it. This last technical property will then allow us to conveniently switch among secret keys without an adversary noticing.

sGPKE (Def. 10). The class of sGPKE schemes is built from that of GPKE ones. Given a GPKE over groups G , H and H^* , we obtain a sGPKE by considering a reduced plaintext domain $\mathcal{P}$ of size $2^{-\lambda}|G|$ such that there is an efficient membership test for elements in $\mathcal{P}$. The decryption function of the sGPKE scheme then returns $\perp$ whenever the decryption function of the GPKE one returns an element that is not in the smaller plaintext domain $\mathcal{P}$. Thanks to property 8 (Def. 9) of GPKE, we are also able to define an alternative decryption function for the sGPKE that returns $\perp$ for any ciphertext not in H . Then, the main additional property (property 1 in Def. 10) that is required from a sGPKE is that, given the public key and a decryption oracle running this alternative decryption function, a PPT adversary cannot produce a ciphertext that does not belong to H and such that the original decryption function does not return $\perp$. We will leverage on this property to prove that the decryption function conveniently does not leak any information to an adversary for ciphertexts not in H .

Proposition 1 (informal). *Any sGPKE is CCA1 secure (under a falsifiable computational indistinguishability assumption).*

Proof outline. To prove this result we first establish a convenient technical Lemma (Lemma 11) which establishes that a PPT adversary against an sGPKE cannot, given the public key and a decryption oracle, generate a ciphertext that is not well-formed (i.e. that does not belong to H) and that decrypts to other than $\perp$.

With this Lemma in our toolbox, the proof of the above result is not too hard. The idea of the proof relies on defining a new encryption function that

needs knowledge of the secret key and uses the scheme homomorphic properties. This new encryption function uniformly picks an element $c_r \in H$ as additional randomness. The proof then proceeds with the following steps:

- First, we show that the challenger can equivalently use the new encryption function to generate the challenge ciphertext.
- Second, we show that the challenger can be modified to pick the additional randomness c_r uniformly in H^* rather than in H . The adversary cannot notice this modification by reduction to the computational indistinguishability assumption in property 3 of Def. 9.
- The next step considers a further modified challenger that creates a new secret key $\widehat{\text{sk}}$ using property 7 in Def. 9 that behaves like sk over H but decrypts some $\mathfrak{h}' \xleftarrow{\$} H'$ to some $\mu \xleftarrow{\$} G$. From property 7, the challenger can then use $\widehat{\text{sk}}$ instead of sk for the whole game without the adversary noticing.
- Finally, we change the challenger such that it replies all the decryption requests from the adversary using sk and only the challenge ciphertext is generated with $\widehat{\text{sk}}$. Our technical Lemma 11 tells us that this change cannot be noticed by the adversary since it essentially tells that the adversary cannot get a non- $\perp$ answer to any of its decryption requests over H' , while sk and $\widehat{\text{sk}}$ yields the same answers over H . In this last step, the challenger picks $\mu \xleftarrow{\$} G$ and $\mathfrak{h}' \xleftarrow{\$} H'$ *after* the last decryption request from the adversary, such that the adversary's view is independent of both values.

With the last challenger in the sequence, the challenge ciphertext is a random element in H^* as a product of $\text{sPKE}.\text{Enc}(\text{pk}, m_\gamma - \mu)$ by $\mathfrak{h}'$. Hence, from the adversary's view the challenge ciphertext is independent of the challenge bit, yielding a null advantage. The formal proof is given in Sect. 3.2 (Prop. 12).

Interestingly, our framework covers several somewhat linearly homomorphic schemes that were previously known to achieve CCA1 security under DDH or DCR. In this framework, the core step is to define the plaintext domain of the sparsified scheme. For all these known schemes, the corresponding plaintext domains lose the group structure of the original scheme. Hence, they achieve only somewhat linear homomorphism. In contrast, we manage to define a new scheme, DPEG (Sect. 4), for which the group structure is preserved for the plaintext domain, thus achieving *true* linear homomorphism.

1.4 Paper organization

This paper is organized as follows. Section 2 covers the preliminaries. Then, Section 3 gives the definition of the class of sGPKE schemes, proves our main result on their CCA1 security and investigates several known schemes that fall into the class. Sect. 4 introduces our new variant of Paillier-ElGamal, DPEG, and establishes its CCA1 security under DCR, investigates its vCCA security under the LOH assumption and further study its Catalano-Fiore-style multiplicative variant and its security. Lastly, in Sect. 5, we investigate the connection between vCCA security and Gentry&Wichs' theorem.

2 Preliminaries

Given $l, u \in \mathbb{Z}$, we use $\llbracket l, u \rrbracket$ to denote the set $\{l, l+1, \dots, u-1, u\}$. Reduction modulo q is denoted as $[\cdot]_q$. We use this notation explicitly only when it avoid possible ambiguities.

2.1 Basic definitions

We define an encryption scheme $\mathcal{E} = (\text{KeyGen}, \text{Enc}, \text{Dec})$ over key space $\mathcal{K}$, plaintext domain $\mathcal{P}$ and ciphertext domain $\mathcal{C}$ as a triplet of PPT algorithms:

- **KeyGen**: on input 1^λ , outputs an encryption key ek and a decryption key sk . We then call (ek, sk) a *valid* key pair.
- **Enc**: on inputs $m \in \mathcal{P}$ and ek , outputs an encryption $c \in \mathcal{C}$ of m .
- **Dec**: on inputs $c \in \mathcal{C}$ and sk , outputs a¹ decryption $m \in \mathcal{P} \cup \{\perp\}$ of c .

Let COIN denote the randomness space of $\mathcal{E}$. We sometimes externalize the randomness used in the encryption function by means of the notation $\text{Enc}(\text{ek}, m; r)$, with $m \in \mathcal{P}$ and $r \in \text{COIN}$. In this latter case, the function $\text{Enc} : \mathcal{P} \times \text{COIN} \rightarrow \mathcal{C}$ is deterministic². When ek is public, we say that $\mathcal{E}$ is a *public-key* encryption scheme and use pk to denote ek .

When, for all $(\text{ek}, \text{sk}) \in \mathcal{K}$, all $m \in \mathcal{P}$ and all $r \in \text{COIN}$, we have that,

$$\mathcal{E}.\text{Dec}(\text{sk}, \mathcal{E}.\text{Enc}(\text{ek}, m; r)) = m, \quad (1)$$

we say that $\mathcal{E}$ is *perfectly correct* or simply *correct*³.

Given a function class $\mathcal{F}$, we define an homomorphic encryption (HE) scheme, $\mathcal{E}_H$, over $\mathcal{F}$ as an encryption scheme augmented by a PPT algorithm Eval which, on input $f \in \mathcal{F}$ and $c_0, \dots, c_{L-1} \in \mathcal{C}^L$, where L denotes the arity of function f , outputs a new *evaluated* ciphertext. When $\mathcal{E}_H$ satisfies condition (1) and when Eval is such that for all $(\text{ek}, \text{sk}) \in \mathcal{K}$, all $f \in \mathcal{F}$, all $m_0, \dots, m_{L-1} \in \mathcal{P}^L$ and all $r_0, \dots, r_{L-1} \in \text{COIN}^L$,

$$\mathcal{E}_H.\text{Dec}(\text{sk}, c_e) = f(m_0, \dots, m_{L-1}), \quad (2)$$

with,

$$c_e = \mathcal{E}_H.\text{Eval}(\text{pk}, \mathcal{E}_H.\text{Enc}(\text{ek}, m_0; r_0), \dots, \mathcal{E}_H.\text{Enc}(\text{ek}, m_{L-1}; r_{L-1})),$$

we also say that $\mathcal{E}_H$ is a *perfectly correct* or *correct* HE scheme or that $\mathcal{E}$ is homomorphic over $\mathcal{F}$. When this property is satisfied only under additional

¹ Decryption may not be deterministic.

² In this case we assume that r is uniformly picked in COIN to provide a seed that the encryption function then uses to sample from the proper distributions.

³ In this paper we do not consider more relaxed notions such as statistical correctness where equality (1) holds with overwhelming probability over the encryption randomness or others [21].

conditions, we say that $\mathcal{E}$ is *somewhat* homomorphic over $\mathcal{F}$. We will sometimes also use the prefix *truly* homomorphic when we want to put an emphasis on the fact that a given scheme is *not* somewhat homomorphic.

Lastly, when $|c_e| = O(\text{poly}(\lambda))$ and, in particular, is independent of $|f|$ and L , we say that $\mathcal{E}_H$ is *compact*.

2.2 Security notions

Unless otherwise stated, we consider only (single-challenge) indistinguishability-based security notions and thus omit the IND prefix. In all the security games we consider, the adversary has to guess with non-negligible advantage a random bit γ , privately chosen by its challenger, given some oracles including at least a challenge oracle. Given two distinct messages m_0 and m_1 , the challenge oracle returns an encryption c^* of m_γ .

CPA, CCA1 and CCA2 security.

Definition 2 (Generic indistinguishability experiment). *Given an encryption scheme $\mathcal{E} = (\text{KeyGen}, \text{Enc}, \text{Dec}, [\dots])$ and a PPT adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ with state S , we define the following generic indistinguishability experiment, $\text{Exp}_{\mathcal{E}}^{\mathcal{O}_1, \mathcal{O}_2}(\mathcal{A})$,*

$$\begin{aligned} (pk, sk) &\leftarrow \mathcal{E}.\text{KeyGen}(1^\lambda), \\ (m_0, m_1, S) &\leftarrow \mathcal{A}_1^{[\mathcal{E}.\text{Enc}(ek, \cdot)], \mathcal{O}_1}(pk), \\ \gamma &\xleftarrow{\$} \{0, 1\}, \\ c^* &\leftarrow \mathcal{E}.\text{Enc}(m_\gamma), \\ b' &\leftarrow \mathcal{A}_2^{[\mathcal{E}.\text{Enc}(ek, \cdot)], \mathcal{O}_2}(pk, S, c^*), \\ b &= b', \end{aligned}$$

where $\mathcal{O}_1$ and $\mathcal{O}_2$ respectively denote first and second step decryption oracles and with the oracle $\mathcal{E}.\text{Enc}(ek, \cdot)$ being provided only in the private key case.

Then, for a PPT adversary, $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, the CPA and CCA1 experiments, respectively $\text{Exp}_{\mathcal{E}}^{\text{CPA}}(\mathcal{A})$ and $\text{Exp}_{\mathcal{E}}^{\text{CCA1}}(\mathcal{A})$, respectively correspond to $\text{Exp}_{\mathcal{E}}^{\cdot}(\mathcal{A})$ and $\text{Exp}_{\mathcal{E}}^{\mathcal{E}.\text{Dec}(sk, \cdot), \cdot}(\mathcal{A})$. The CCA2 experiment is also defined as,

$$\text{Exp}_{\mathcal{E}}^{\text{CCA2}}(\mathcal{A}) \equiv \text{Exp}_{\mathcal{E}}^{\mathcal{E}.\text{Dec}(sk, \cdot), \mathcal{O}_{\text{CCA2}}(sk, c^*, \cdot)}(\mathcal{A}),$$

with,

$$\mathcal{O}_{\text{CCA2}}(c^*, c) = \begin{cases} \mathcal{E}.\text{Dec}(c) & \text{if } c^* \neq c, \\ \perp & \text{otherwise.} \end{cases}$$

A scheme $\mathcal{E}$ is then said to be CPA secure if for any PPT adversary $\mathcal{A}$,

$$\text{Adv}_{\mathcal{E}}^{\text{CPA}}(\mathcal{A}) = |P(\text{Exp}_{\mathcal{E}}^{\text{CPA}}(\mathcal{A}) = \text{True}) - 1/2| \leq \text{neg}(\lambda),$$

where λ is the security parameter, and likewise for CCA1 and CCA2.

Note that some cryptographers consider that CCA1 security is a historical security notion from the time before the first CCA2 public encryption schemes were proposed. While this is a legitimate point of view, CCA1 may still be considered a meaningful CCA security target for homomorphic schemes in order to circumvent the CCA2 impossibility to which malleable schemes are subject [29,13].

vCCA, vCCA^D (and IV-CCA) security. As introduced in [30], vCCA security is a single challenge security notion. As such, the vCCA game has two decryption oracles. The second step (post-challenge generation) oracle assumes the existence of a PPT witness extractor $\text{extract} : \mathcal{C} \times \mathcal{X} \rightarrow \mathcal{F} \times \mathcal{C}^*$, where $\mathcal{X}$ denotes a set of auxiliary data⁴ such that:

- For any ciphertext $c \in \mathcal{C}$ which is obtained by invoking $\text{Eval}(f, c_0, \dots, c_{L-1})$,

$$\text{extract}(c, \text{aux}) = (f, c_0, \dots, c_{L-1}).$$

- Otherwise, $\text{extract}(c, \text{aux}) = (\text{id}, c)$.

Definition 3 ($\mathcal{F}'$ -extraction algorithm [30]). Given an FHE scheme $\mathcal{E}_H = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{Eval})$ with plaintext domain $\mathcal{P}$, ciphertext domain $\mathcal{C}$ and function domain $\mathcal{F}$, let $\mathcal{F}' \subseteq \mathcal{F}$, an $\mathcal{F}'$ -extraction algorithm is a PPT algorithm $\text{extract} : \mathcal{C} \times \Omega \rightarrow \mathcal{F}' \times \mathcal{C}^L \cup \{\perp\}$, where Ω is a set of auxiliary data and where, for simplicity sake, all functions in $\mathcal{F}'$ are assumed to have the same arity L .

Following this, the vCCA experiment is defined as follows.

Definition 4 (vCCA experiment [30]).

$$\text{Exp}_{\mathcal{E}}^{\text{vCCA}}(\mathcal{A}) \equiv \text{Exp}_{\mathcal{E}}^{\mathcal{E}.Dec(sk, \cdot), \mathcal{O}_{\text{vCCA}}(sk, c^*, \cdot)}(\mathcal{A}),$$

with, letting $(f, c_0, \dots, c_{L-1}) \leftarrow \text{extract}(c, \text{aux})$,

$$\mathcal{O}_{\text{vCCA}}(sk, c^*, c) = \begin{cases} \mathcal{E}_H.Dec(sk, c) & \text{if } c^* \notin \{c_0, \dots, c_{L-1}\}, \\ \perp & \text{otherwise.} \end{cases}$$

Definition 5 ((IND-)vCCA security [30]). A scheme $\mathcal{E}_H$ is (IND-)vCCA secure with respect to $\mathcal{F}' \subseteq \mathcal{F}$ if there exists an $\mathcal{F}'$ -extraction algorithm extract such that for any PPT adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, the advantage of $\mathcal{A}$ winning the vCCA experiments is negligible in the security parameter λ .

⁴ The spirit of the vCCA security notion is (at least) to model construction blueprints which embed proof material in their ciphertexts and rely on a SNARK to enforce correct homomorphic evaluations over some input ciphertexts. In this context, the above extract thus corresponds to the extractor of that underlying SNARK which allows to retrieve a witness from the proven statement as well as auxiliary data forming the trace of the execution of the adversary, see [30,10] for more details.

In essence, the vCCA game is exactly the single challenge CCA2 game, with the second step decryption oracle being augmented in order to filter out *all* byproducts of the challenge ciphertext (rather than just the challenge ciphertext). In [30], vCCA security is defined and studied under a statistical correctness assumption on the underlying FHE scheme, and then further studied in [10] when that assumption is not satisfied. Then, [10] also defines the notion of vCCA^D security, which, in a nutshell, is a “CPA^D-style” [27] (also App. D.2) multiple challenge variant of vCCA in which the decryption oracle also accepts byproducts of the challenge ciphertexts as long as some associated left and right cleartext evaluations coincide (i.e. that the expected cleartext output does not leak the challenge bit).

Taken together, these two notions are the strongest CCA security notions so far achievable, respectively, by correct and approximate homomorphic schemes. However, the only known construction strategies intimately require advanced SNARK machinery, undermining their practicality. Following [10], vCCA and vCCA^D security are equivalent when the correctness assumption holds (under a mild assumption to which we will come back in Sect. 5). Despite this equivalence, we also provide a simplified definition for the (single challenge) vCCA^D indistinguishability experiment that is valid *only for correct schemes* and that will later conveniently help us.

Definition 6 (vCCA^D experiment [10]).

$$\text{Exp}_{\mathcal{E}}^{\text{vCCA}^D}(\mathcal{A}) \equiv \text{Exp}_{\mathcal{E}}^{\mathcal{E}.\text{Dec}(sk, \cdot), \mathcal{O}_{\text{vCCA}^D}(sk, c^*, m_0, m_1, \cdot)}(\mathcal{A}).$$

Given sk, c^* as well as m_0, m_1 , let $(f, c_0, \dots, c_{L-1}) \leftarrow \text{extract}(c, \text{aux})$. Then define,

$$\text{Dec}'(sk, c^*, m, c) = \begin{cases} m & \text{if } c = c^*, \\ \mathcal{E}_H.\text{Dec}(sk, c) & \text{otherwise,} \end{cases}$$

and,

$$L = f(\text{Dec}'(sk, c^*, m_0, c_0), \dots, \text{Dec}'(sk, c^*, m_0, c_{L-1})),$$

as well as,

$$R = f(\text{Dec}'(sk, c^*, m_1, c_0), \dots, \text{Dec}'(sk, c^*, m_1, c_{L-1})).$$

The second step vCCA^D decryption oracle, $\mathcal{O}_{\text{vCCA}^D}$, is then⁵,

$$\mathcal{O}_{\text{vCCA}^D}(sk, c^*, m_0, m_1, c) = \begin{cases} \mathcal{E}_H.\text{Dec}(sk, c) & \text{if } L = R, \\ \perp & \text{otherwise.} \end{cases}$$

To conclude this section, it is important to also mention the notion of IV-CCA (CCA1<IV-CCA<vCCA) security introduced in the recent (Crypto'25) paper of

⁵ As in [10, Sect. 3.3], we assume the convention that $f(m_1, \dots, m_K) = \perp$ when $\exists i : m_i = \perp$ so that if the left and right evaluations both give $\perp$, $L = R$ is satisfied and $\mathcal{E}_H.\text{Dec}(sk, c)$ is returned to $\mathcal{A}$.

Yang et. al [41]. However, the spirit of this notion is to model *non-compact* FHE. Indeed, in IV-CCA, the second step decryption oracle specification relies on a verification algorithm, which is part of the FHE scheme and which takes as input a ciphertext to be decrypted and *a set of input ciphertexts* that explains the former (via some legit homomorphic evaluation). At this price, they achieve the tour de force of giving a construction that is IV-CCA secure in the standard model, based solely on the LWE assumption (and the additional assumption that *perfectly* correct FHE can be built from LWE⁶).

2.3 The Linear-Only Homomorphism (LOH) assumption

Informally, for an encryption scheme $\mathcal{E}_H = (\text{KeyGen}, \text{Enc}, \text{ImVer}, \text{Dec}, \text{Eval})$, the Linear-Only Homomorphism (LOH) property [4,5] states that given polynomially-many ciphertexts $(c_0, \dots, c_{m-1})$ under $\mathcal{E}_H$ it is infeasible for an adversary to create a new ciphertext c' , which is in the image of the encryption function (as verified by ImVer) and cannot be expressed by (homomorphically) evaluating an affine combination of the ciphertexts in the previous list. In the above, ImVer is a function that verifies if a ciphertext is in the image of the encryption function with knowledge of sk (it is essentially the “verification” part of $\mathcal{E}_H.\text{Dec}$ and $\mathcal{E}_H.\text{ImVer}(c) = \text{True} \Leftrightarrow \mathcal{E}_H.\text{Dec}(c) \neq \perp$). The LOH property has been introduced in [4] to serve as the basis for several SNARK constructions in that paper and other subsequent works [7,24,32].

Formally, following [4,5], we have the following definition.

Definition 7 (LOH property, reproduced from [4]). *An encryption scheme $\mathcal{E}_H = (\text{KeyGen}, \text{Enc}, \text{ImVer}, \text{Dec}, \text{Eval})$ (with $\mathcal{P} = \mathbb{Z}_t$ ⁷) satisfies the Linear-only Homomorphism property if for every PPT adversary $\mathcal{A}$, there is a PPT extractor extract such that for any auxilliary input $\text{aux} \in \{0,1\}^{\text{poly}(\lambda)}$ and any plaintext generator $\mathcal{M}$,*

$$P \left(\begin{array}{l} \exists i \in [0, k-1] \text{ s. t.} \\ \text{ImVer}(c'_i) = \text{True} \\ \text{and} \\ \mathcal{E}_H.\text{Dec}(c'_i) \neq a'_i \end{array} \left| \begin{array}{l} (ek, sk) := \mathcal{E}_H.\text{KeyGen}(1^\lambda) \\ (a_0, \dots, a_{m-1}) := \mathcal{M}([ek]) \\ (c_0, \dots, c_{m-1}) := (\mathcal{E}_H.\text{Enc}(a_0), \dots, \mathcal{E}_H.\text{Enc}(a_{m-1})) \\ (c'_0, \dots, c'_{k-1}) := \mathcal{A}(c_0, \dots, c_{m-1}, [ek]; \text{aux}) \\ (\Pi, b) := \text{extract}(c_0, \dots, c_{m-1}, [ek]; \text{aux}) \\ (a'_0, \dots, a'_{k-1})^T := \Pi \cdot (a_0, \dots, a_{m-1})^T + b \end{array} \right. \right) \leq \text{neg}(\lambda). \quad (3)$$

where $\Pi \in \mathbb{Z}_t^{k \times m}$ and $b \in \mathbb{Z}_t^k$, and with the convention that row i of Π is left empty (i.e. $(\Pi_i, b_i) = \emptyset$) and $a'_i = \perp$ when c'_i was not generated by homomorphically evaluating an affine combination over the c_j 's. The notation $[ek]$ indicates that

⁶ Indeed, [41] proposes an LWE-based construction instantiating the Naor-Yung paradigm, which explicitly requires *perfect* correctness from the underlying (FHE) scheme [21]. Although no explicit construction is provided in [41], their intent is to achieve this strong property by using truncated discrete gaussians for the noise distribution in order to get exact L_∞ norm bounds [40].

⁷ The definition extends to the case where the plaintext space is a polynomial ring [4].

the encryption key ek is optionally provided, depending on whether the setting is private or public-key.

Remark that the c'_i do not have to be explicitly included as inputs to the extractor since they are built by the adversary and therefore implicitly included in aux . In summary, whenever $\mathcal{A}$ builds c'_i by doing “something equivalent to”,

$$c'_i = \mathcal{E}_H.\text{Add}(\mathcal{E}_H.\text{Eval}(\text{lincomb}_\pi, c_0, \dots, c_{m-1}), \underbrace{\mathcal{E}_H.\text{Eval}(\text{lincomb}_{\pi'}, c''_0, \dots, c''_{l-1})}_{\text{pub. key case only}}), \quad (4)$$

where $c''_0 = \mathcal{E}_H.\text{Enc}(\mu_0; \text{pk}), \dots, c''_{l-1} = \mathcal{E}_H.\text{Enc}(\mu_{l-1}; \text{pk})$, then $\Pi_i = \pi$ and $b_i = \sum_{j=0}^{l-1} \pi'_j \mu_j$. Conversely, $(\Pi_i, b_i) = \emptyset$ when this is not the case.

In our security proofs, we will use the following more convenient one-ciphertext notation for the above extractor,

$$(\pi, \beta) = \text{extract}(c, \text{aux}), \quad (5)$$

as a shortcut for $\Pi_{i:c'_i=c}$ and $b_{i:c'_i=c}$ (i.e. π , respectively β , is the row of Π , respectively the component of b , associated to ciphertext c). While a more rigorous notation would be to use $(\pi, \beta) = \text{extract}(c, \{c_0, \dots, c_{m-1}\}, \text{aux})$, we use Eq. (5) in the sequel as the set of ciphertexts given to the LOH adversary will always be clear from the nearby context. Remark that affine rather than linear combinations are considered in the above definition to account for the fact that, in the public-key setting, the adversary can create (from scratch) additional fresh well-formed ciphertexts on its own (a case that does not happen in the private-key setting) and homomorphically add them to homomorphic evaluations of linear combinations over the c_i 's.

On top of the above definition, [4] further proposes several heuristic approaches to build schemes satisfying the LOH property, starting from a *correct* CPA secure linear homomorphic scheme. An example, which has been considered in several works (e.g. [4, 23]), is the “two-ciphertexts” blueprint which consists in sparsifying the ciphertext domain of a correct linearly homomorphic scheme (e.g. Paillier) by following a Knowledge-of-Exponent (KEA) template⁸. To illustrate this approach with the Paillier scheme, one may consider that the encryption of a message m consists of a pair of ciphertexts $(\text{Enc}(m), \text{Enc}(\alpha \cdot m))$ under the same key material, for a secret random value $\alpha \in \mathbb{Z}_n$ (with n the RSA modulus of the scheme), and with the ImVer algorithm checking this linear relation by decrypting both ciphertexts. It is then assumed that this scheme satisfies Definition 7 as the underlying Paillier scheme only exhibits linear homomorphic properties (to the best of the research community’s knowledge), although the baseline Paillier scheme does not have the LOH property as also argued in [4] ([3], p. 33). Interestingly, this construction has recently been proven to achieve vCCA security in [16, 22].

⁸ This is a slight abuse of terminology. Yet, we will use it in this paper as the “two-ciphertexts” blueprint is a distant sibling of D  mgard original heuristic [20] and similar in spirit.

Although a number of precautions need to be taken, the “two-ciphertext” heuristic has also been applied following the same spirit to LWE-based schemes in several works since [4], e.g. [7,24,32,16].

2.4 Paillier-ElGamal

In this section, we specify the Paillier-ElGamal scheme that is used as a starting point in [28]. This scheme originates in the Camenisch-Shoup scheme [11] (which is CCA2 under DCR and, as such, non-malleable) as it is essentially equivalent to a variant of the latter in which the last out of three slots is removed. We also stress that there are several CPA secure schemes referred to as Paillier-ElGamal in the literature, e.g. [34,37,9,6], and they are not all equivalent. So, following [28], the baseline Paillier-ElGamal scheme (which may thus also be referred to as the Camenisch-Shoup Lite scheme) that we use may be defined as follows:

- PEG.KeyGen: choose a RSA modulus $N = (2p' + 1)(2q' + 1)$, with p' and q' two distinct Sophie Germain primes of suitable size. Then, uniformly pick $g \in \mathbb{Z}_{N^2}^*$, $x \in \llbracket 0, N(N-1)/4 \rrbracket$ and let $h = [g^{4Nx}]_{N^2}$. Then, let $\text{pk} = (N, g, h)$ and $\text{sk} = x$. Remark that (p', q') (that will later be denoted sk_0) is not required to decrypt.
- PEG.Enc: given $m \in \mathbb{Z}_N$ and pk , pick $r \in \llbracket 0, (N-1)/4 \rrbracket$ and return $(c_0, c_1) \in \mathbb{Z}_{N^2}^2$ such that,

$$c_0 = [g^{2Nr}]_{N^2}, \text{ and } c_1 = [(1+N)^m h^r]_{N^2}.$$

- PEG.Dec: given $(c_0, c_1) \in \mathbb{Z}_{N^2}^2$ and sk , let $\mu = [c_1 \cdot c_0^{-2x}]_{N^2}$. If μ is of the form $[(1+N)^m]_{N^2}$ for some $m \in \mathbb{Z}_N$, then return m and, otherwise, return $\perp$.

Remark that computing m in PEG.Dec may be done easily since $[(1+N)^m]_{N^2} = 1 + mN$. Hence, PEG.Dec can be equivalently written as returning $\perp$ when $[\mu - 1]_N \neq 0$ and $\frac{\mu-1}{N}$ otherwise.

This scheme is perfectly correct and linearly homomorphic offering operators:

- PEG.Add: given (c_0, c_1) and (c'_0, c'_1) , both in $\mathbb{Z}_{N^2}^2$, return $([c_0 c'_0]_{N^2}, [c_1 c'_1]_{N^2})$.
- PEG.MulByConst $_\alpha$: given $(c_0, c_1) \in \mathbb{Z}_{N^2}^2$ and $\alpha \in \mathbb{Z}_n$, return $([c_0^\alpha]_{N^2}, [c_1^\alpha]_{N^2})$.

It is well-know that it is CPA secure under the DCR assumption.

2.5 Paillier-ElGamal with plaintext zero padding

In [28], the following variant of PEG is considered. Let $2^\lambda \leq B = 2^t < \lfloor 2^{-\lambda} N \rfloor$,

- PEG $_z$.KeyGen: run PEG.KeyGen(1^λ) to get (pk, sk) .
- PEG $_z$.Enc: given $m \in \llbracket 0, B \rrbracket$ and pk , return PEG.Enc(pk, m).
- PEG $_z$.Dec: given $\text{ct} \in \mathbb{Z}_{N^2}^2$ and sk , if PEG.Dec(sk, ct) = $\perp$, then return $\perp$. Otherwise, let $m = \text{PEG.Dec}(\text{sk}, \text{ct})$, if $m > B$ then return $\perp$ and otherwise return m .

This scheme is only *somewhat* linearly homomorphic as it is no more correct:

$$\text{PEG}_z.\text{Dec}(\text{sk}, \text{PEG}_z.\text{Add}(\text{PEG}_z.\text{Enc}(\text{pk}, B), \text{PEG}_z.\text{Enc}(\text{pk}, 1))) = \perp \neq B + 1.$$

Following [28, Theorem 3], it is now known that this scheme is CCA1 secure under the DCR assumption (see Def. 39 in Appendix D.1) and an additional Composite Non-Invertibility assumption (see Def. 40 in Appendix D.1) which, although less standard than the former, is falsifiable⁹.

We also reproduce the following lemma from [28] that we use several times.

Lemma 8 (Lemma 3 in [28]). *Let integers $0 < \ell < L$. The statistical distance between the distributions $D_1 = \{[x]_\ell : x \xrightarrow{\$} \mathbb{Z}_L\}$ and $D_2 = \{x \xrightarrow{\$} \mathbb{Z}_\ell\}$ is bounded by*

$$\frac{\min(\ell, 2[L]_\ell)}{L}.$$

3 On a new class of IND-CCA1 secure schemes

3.1 Sparsified Group-based PKE (sGPKE)

We now define the class of sGPKE schemes related to our main result that establishes that any scheme in this class achieves CCA1 security under a falsifiable computational indistinguishability assumption. Intuitively, any scheme in this class is obtained by *sparsifying* the ciphertext domain of another scheme that belong to the class of GPKE. Please refer to Sect. 1.3 for an intuitive overview.

Definition 9 (GPKE). *A public-key encryption scheme $\text{PKE} = (\text{KeyGen}, \text{Enc}, \text{Dec})$ is a Group-based linearly homomorphic Public-Key Encryption (sGPKE) scheme if the following properties are fulfilled.*

1. *There exists functions KeyGen_0 and KeyGen_1 such that $\text{KeyGen}_0(1^\lambda)$ outputs (pk_0, sk_0) and $\text{KeyGen}_1(pk_0)$ outputs (pk, sk) . Then, PKE.KeyGen is defined as $\text{PKE.KeyGen}(1^\lambda) = \text{KeyGen}_1(\text{KeyGen}_0(1^\lambda).pk_0)$.*
2. *For any security parameter λ and all secret key/public key pairs $(pk, sk) \leftarrow \text{PKE.KeyGen}(1^\lambda)$, it is possible to define finite groups $(G, +)$ and $(H, \cdot)$ such that G is the plaintext space, and $H = \text{PKE.Enc}(pk, G; D)$ is embedded in the ciphertext space $\mathcal{C}$.*
3. *Given pk_0 , $(sk, pk) = \text{PKE.KeyGen}_1(pk_0)$, there exists groups $(H^*, \cdot)$ and $(H', \cdot)$ such that $H^* = H \cdot H'$ with $H \cap H' = 1$, $\mathcal{C} \supseteq H^* \supset H$ and such that the distributions $\{z \xrightarrow{\$} H\}$ and $\{z \xrightarrow{\$} H^*\}$ are computationally indistinguishable without the knowledge of sk_0 . Additionally, for all $c \notin H^*$, $\text{PKE.Dec}(sk, c) = \perp$.*
4. *The set G grows exponentially in the security parameter λ .*

⁹ Still, reliance on this latter assumption may be avoided under some conditions [28] on which we will come back later.

5. *PKE is perfectly correct and linearly homomorphic: Let $m, m' \in G$ and take any scalar $t \in \mathbb{N}$. Then it holds that*

$$\begin{aligned} \text{PKE.Dec}(sk, \text{PKE.Enc}(pk, m') \cdot \prod_{i=1}^t \text{PKE.Enc}(pk, m)) &= m' + \sum_{i=1}^t m \\ &= m' + tm. \end{aligned}$$

6. *For all $m, m' \in G$ and $r \in \text{COIN}$, the distributions of $\text{PKE.Enc}(pk, m; \text{COIN}) \cdot \text{PKE.Enc}(pk, m'; r)$ and $\text{PKE.Enc}(pk, m + m'; \text{COIN})$ are identical.*
7. *For all secret key/public key pairs $(pk, sk) \leftarrow \text{PKE.KeyGen}(1^\lambda)$, for $h' \in H'$ and $\mu \xleftarrow{\$} G$, there is a PPT algorithm taking as input (pk, sk, h', μ) that outputs, with overwhelming probability over H' , a valid pair $(pk, \widehat{sk})$ such that $\text{PKE.Dec}(\widehat{sk}, h') = \mu$ and the distribution of $\widehat{sk}$ is indistinguishable from that of the output of KeyGen . Note that the correctness of the scheme implies that, as pk is the same in both key pairs, $\text{PKE.Dec}(sk, c) = \text{PKE.Dec}(\widehat{sk}, c)$ for all $c \in H$.*
8. *Given sk and sk_0 , there exists an efficient exact membership test for elements in H . Equivalently, there exists an alternative decryption function $\text{PKE}.\widehat{\text{Dec}}$ such that, for any $c \in \mathcal{C}$, $\text{PKE}.\widehat{\text{Dec}}(sk, sk_0, c) = \perp$, if $c \notin H$, and $\text{PKE}.\widehat{\text{Dec}}(sk, sk_0, c) = \text{PKE.Dec}(sk, c)$, otherwise ($c \in H$).*

Definition 10 (sGPKE). *Given a GPKE scheme $\text{PKE} = (\text{KeyGen}, \text{Enc}, \text{Dec})$, define $\text{sPKE} = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{ImVer})$, with plaintext domain $\text{sPKE}.\mathcal{P} \subset \text{PKE}.G$ with $|\text{sPKE}.\mathcal{P}| \leq 2^{-\lambda} |\text{PKE}.G|$ and ciphertext domain $\text{sPKE}.\mathcal{C} = \text{PKE}.\mathcal{C} = \mathcal{C}$, as follows,*

- *sPKE.KeyGen : run PKE.KeyGen to get $(\text{PKE}.pk, \text{PKE}.sk)$. Let pk' and sk' denote any additional key material, return $((\text{PKE}.pk, pk'), (\text{PKE}.sk, sk'))$ as $(\text{sPKE}.pk, \text{sPKE}.sk)$.*
- *sPKE.Enc : given $m \in \text{sPKE}.\mathcal{P}$ and $\text{sPKE}.pk$, return $\text{PKE.Enc}(\text{PKE}.pk, m)$.*
- *sPKE.ImVer : given $c \in \mathcal{C}$ and $\text{sPKE}.sk$, let $\mu = \text{PKE.Dec}(\text{PKE}.sk, c)$, if $\mu \in \text{sPKE}.\mathcal{P}$, then return *True* and otherwise return *False*.*
- *sPKE.Dec : given $c \in \mathcal{C}$ and $\text{sPKE}.sk$, if $\text{sPKE.ImVer}(sk, c) = \text{True}$ then return $\text{PKE.Dec}(sk, c)$, otherwise return $\perp$.*

Additionally, from Property 8 in Def. 9, we also define,

- *$\text{sPKE}.\widehat{\text{Dec}}$: given $c \in \mathcal{C}$, $\text{sPKE}.sk$ and $\text{PKE}.sk_0$, if $\text{sPKE.ImVer}(sk, c) = \text{True}$ then return $\text{PKE}.\widehat{\text{Dec}}(sk, \text{PKE}.sk_0, c)$, otherwise return $\perp$. Hence, for any $c \in \mathcal{C}$, $\text{sPKE}.\widehat{\text{Dec}}(sk, \text{PKE}.sk_0, c) = \perp$, if $c \notin H$, and $\text{sPKE}.\widehat{\text{Dec}}(sk, \text{PKE}.sk_0, c) = \text{sPKE.Dec}(sk, c)$ otherwise ($c \in H$).*

Then sPKE is a sparsified GPKE or sGPKE scheme when ImVer satisfies,

1. *For any PPT adversary $\mathcal{A}$,*

$$P \left(\text{sPKE.ImVer}(sk, c) = \text{True} \mid \begin{array}{l} (pk, sk) := \text{sPKE.KeyGen}(1^\lambda) \\ c \leftarrow \mathcal{A}^{\text{sPKE}.\widehat{\text{Dec}}(sk, \cdot)}(pk) \end{array} \right) \leq \text{neg}(\lambda). \quad (6)$$

Remark that although GPKE's are correct and linearly homomorphic by definition, sGPKE's may not be correct and may only be somewhat linearly homomorphic. We will emphasize this when considering concrete examples of schemes.

3.2 A general result on sGPKE's CCA1 security

We first prove a convenient technical Lemma which establishes that a PPT adversary against an sGPKE cannot, given the public key and a decryption oracle, generate a ciphertext that is not well-formed (i.e. that does not belong to H) and that decrypts to other than $\perp$. This property will be useful later in the proof of our main result and in several other places in this paper.

Lemma 11. *Given any sGPKE scheme $sPKE = (KeyGen, Enc, Dec)$, for any PPT adversary $\mathcal{A}$,*

$$P \left(\begin{array}{c} sPKE.Dec(sk, c) \neq \perp \\ \wedge c \notin H \end{array} \middle| \begin{array}{c} (pk, sk) := sPKE.KeyGen(1^\lambda) \\ c := \mathcal{A}^{sPKE.Dec(sk, \cdot)}(pk) \end{array} \right) \leq neg(\lambda), \quad (7)$$

Proof. Consistently with Eq. (7), we define G_0 as the game in which a challenger first runs $sPKE.KeyGen(1^\lambda)$ to get (pk, sk) and gives pk to some PPT adversary $\mathcal{A}$. The challenger then answers polynomially many decryption requests from $\mathcal{A}$. The adversary then wins the game if, for its last decryption request, it submits a ciphertext $c \notin H$ and get $sPKE.Dec(sk, c) \neq \perp$ in return.

First recall that, by Property 3 in Def. 9, for any $c \notin H^*$, $PKE.Dec(PKE.sk, c) = \perp$ and, hence, $sPKE.Dec(sPKE.sk, c) = \perp$. So $\mathcal{A}$ cannot win G_0 by submitting a ciphertext $c \notin H^*$ for its last decryption request.

For notational convenience, H_0 is the same game as G_0 .

Hybrid argument. We now consider the sequence of games H_i ($i \in \llbracket 1, Q \rrbracket$) where Q is the number of decryption requests performed by the adversary. In game H_i , we proceed as follows,

- Up to the i -th decryption query, the challenger uses secret keys sk and sk_0 with the modified decryption algorithm $sPKE.\widehat{Dec}$ (from Def. 10) to reply to the adversary's decryption requests, i.e. let c_j , $j \leq i$, be the j -th ciphertext submitted by the adversary, then $sPKE.\widehat{Dec}(sk, sk_0, c_j)$ is returned.
- After the i -th decryption query, the challenger answers the remaining $Q - i$ decryption requests by means of the normal decryption function $sPKE.Dec$, i.e. let c_j , $j > i$, be the j -th ciphertext submitted by the adversary, then $sPKE.Dec(sk, c_j)$ is returned.

Indistinguishability between games H_i and H_{i-1} . Clearly, from the adversary's view, games H_i (where the i -th decryption request is replied to by means of $sPKE.\widehat{Dec}(sk, sk_0, \cdot)$) and H_{i-1} (where the i -th decryption request is replied to by means of $sPKE.Dec(sk, \cdot)$) are indistinguishable unless the i -th decryption query involves a ciphertext that is rejected in the former but not the latter. Let, again, c_i denote the ciphertext submitted by $\mathcal{A}$ at the i -th decryption query, we then consider the following cases:

- $c_i \in H$: in this case $sPKE.\widehat{Dec}(sk, sk_0, c_i) = sPKE.Dec(sk, c_i)$, by Property 8 in Def. 9, and both games proceed identically.

- $c_i \notin H$: in this case, $\text{sPKE}.\widehat{\text{Dec}}(\text{sk}, \text{sk}_0, c_i) = \perp (H_i)$. Also, $\text{sPKE}.\text{ImVer}(\text{sk}, c_i) = \text{False}$ with overwhelming probability, by Property 1 in Def. 10. So, both games proceed identically since $\text{sPKE}.\text{Dec}(\text{sk}, c) = \perp$ with overwhelming probability.

It thus follows from the standard hybrid argument that $H_0 \equiv G_0$ is indistinguishable from game H_{Q-1} , in which all but the last of the adversary's decryption requests are replied to by means of $\text{sPKE}.\widehat{\text{Dec}}(\text{sk}, \text{sk}_0, \cdot)$ and the last one is replied to by means of $\text{sPKE}.\text{Dec}(\text{sk}, \cdot)$.

Game G_1 . Let $G_1 \equiv H_{Q-1}$. In this last game, when $\mathcal{A}$ submits its last decryption request over ciphertext $c \in H^* \setminus H$, Property 1 in Def. 10 implies that $\text{sPKE}.\text{ImVer}(\text{sk}, c) = \text{True}$ with negligible probability and, by definition of $\text{sPKE}.\text{Dec}$, that $\text{sPKE}.\text{Dec}(\text{sk}, c) \neq \perp$ with the same probability. $\square$

We then turn to our main result which establishes that any sGPKE is CCA1 secure. Please refer to Sect. 1.3 for an outline of the proof.

Proposition 12. *Any sGPKE, $\text{sPKE} = (\text{KeyGen}, \text{Enc}, \text{ImVer}, \text{Dec}, \widehat{\text{Dec}})$, is CCA1 secure.*

Proof. We denote W_i the event that an adversary $\mathcal{A}$ wins the game G_i , then

$$\text{Adv}_{\text{sPKE}}^{\text{CCA1}}(\mathcal{A}) = |\Pr[W_0] - 1/2|.$$

We now define a new encryption function that needs knowledge of the secret key as it uses the decryption function of the GPKE, PKE, on which sPKE is built,

$$\text{sPKE}.\text{Enc}'(\text{sk}, \text{pk}, m; (r, c_r)) = \text{sPKE}.\text{Enc}(\text{pk}, m - \text{PKE}.\text{Dec}(\text{sk}, c_r); r) \cdot c_r, \quad (8)$$

with $c_r \in H^*$. For all (pk, sk) , all $m \in G$ and $c_r \in H$, the distributions $\text{sPKE}.\text{Enc}(\text{pk}, m)$ and $\text{sPKE}.\text{Enc}'(\text{sk}, \text{pk}, m; c_r)$ are identical. Indeed,

$$\begin{aligned} \text{sPKE}.\text{Enc}'(\text{sk}, \text{pk}, m; (r, c_r)) &= \text{sPKE}.\text{Enc}(\text{pk}, m - \text{PKE}.\text{Dec}(\text{sk}, c_r); r) \\ &\quad \cdot \text{sPKE}.\text{Enc}(\text{pk}, \text{PKE}.\text{Dec}(\text{sk}, c_r); r'), \end{aligned} \quad (9)$$

$$\begin{aligned} &= \text{sPKE}.\text{Enc}(\text{pk}, m - \text{PKE}.\text{Dec}(\text{sk}, c_r) + \text{PKE}.\text{Dec}(\text{sk}, c_r); r''), \\ &= \text{sPKE}.\text{Enc}(\text{pk}, m; r''), \end{aligned} \quad (10)$$

with r'' is distributed uniformly in COIN by Property 6.

Game G_0 This is the standard CCA1 game. $\mathcal{A}$ can ask an oracle for decryption of any ciphertext $c \in \mathcal{C}$. He then chooses m_0, m_1 and receives $c^* = \text{sPKE}.\text{Enc}(\text{pk}, m_\gamma; r)$, $\gamma \xleftarrow{\$} \{0, 1\}$ in return from the challenger, output a guess γ' for γ and wins the game when $\gamma = \gamma'$.

Game G_1 . In this game, the challenger returns,

$$c^* = \text{sPKE}.\text{Enc}'(\text{sk}, \text{pk}, m_\gamma; (r, c_r)),$$

with c_r chosen uniformly in H . In G_1 , the challenge ciphertext is, in $\mathcal{A}$'s view, identically distributed to one from G_0 by Eq. (10). It implies that G_0 and G_1 are identical and $\Pr[W_0] = \Pr[W_1]$.

Game G_2 . This game is the same as G_1 except that c_r is chosen uniformly in H^* (in G_2) rather than H (in G_1). The indistinguishability between G_1 and G_2 is given by Property 3 in Def. 9.

Indeed, given an adversary $\mathcal{A}_0$ that can distinguish between G_1 and G_2 , there exists an adversary $\mathcal{B}_0$ that breaks the indistinguishability assumption in Property 3 using $\mathcal{A}_0$ as a subroutine, and which achieves the same advantage as $\mathcal{A}_0$. The reduction $\mathcal{B}_0$ then proceeds as follows. First, after it gets pk_0 from its challenger which runs $\text{PKE.KeyGen}_0(1^\lambda)$, $\mathcal{B}_0$ runs $\text{sPKE.KeyGen}_1(\text{pk}_0)$ to get (pk, sk) . It then sends pk to $\mathcal{A}_0$ as well as (pk, sk) to its challenger. The reduction then simply answers $\mathcal{A}_0$'s decryption requests by running $\text{sPKE.Dec}(\text{sk}, \cdot)$ on its own. Finally, when $\mathcal{A}_0$ issues its challenge request over $m_0, m_1 \in \text{sPKE.P}$, $\mathcal{B}_0$ ask its challenger for a single sample

$$c_r \stackrel{\$}{\leftarrow} \begin{cases} H & \text{if } \beta = 0, \\ H^* & \text{if } \beta = 1, \end{cases}$$

where β is the challenger's secret bit. Finally, it returns,

$$c^* = \text{sPKE.Enc}'(\text{sk}, \text{pk}, m_\gamma; (r, c_r)),$$

for some random bit γ , to $\mathcal{A}_0$ and eventually replies as $\mathcal{A}_0$ does. Clearly, $\mathcal{B}_0$ perfectly simulates G_1 when $\beta = 0$ and G_2 when $\beta = 1$.

So we have $\text{Adv}^{G_1/G_2}(\mathcal{A}_0) \leq \text{Adv}(\mathcal{B}_0)$, and so $|\Pr[W_1] - \Pr[W_2]| \leq \text{Adv}(\mathcal{B}_0)$.

Game G_3 . In this game, the challenger picks $\mu \stackrel{\$}{\leftarrow} G$ and $\mathfrak{h}' \stackrel{\$}{\leftarrow} H'$ and use Property 7 in Def. 9 to generate a new secret key $\widehat{\text{sk}}$ which is used in lieu of sk . Both games are indistinguishable since, also from that property, the distribution of $\widehat{\text{sk}}$ is indistinguishable from that of the output of KeyGen . Hence, $|\Pr[W_2] - \Pr[W_3]| \leq \text{neg}(\lambda)$.

Game G_4 . In G_4 all the adversary's decryption requests are replied to by means of $\text{sPKE.Dec}(\text{sk}, \cdot)$ and only the challenge ciphertext is generated by means of $\widehat{\text{sk}}$. Clearly, game G_3 and G_4 cannot be distinguished unless the adversary can issue a decryption request over a ciphertext $c \notin H$ that does not decrypt to $\perp$ under $\widehat{\text{sk}}$ (respectively $\widehat{\text{sk}}$) as c would decrypt to $\perp$ with overwhelming probability under $\widehat{\text{sk}}$ (respectively sk). By Lemma 11, this occurs with negligible probability in the security parameter. As a result, in G_4 , μ and $\mathfrak{h}'$ can be uniformly picked in G and H' respectively after all the adversary's requests.

Let $c_r = (\mathfrak{h}, \mathfrak{h}') \in H \times H'$. In this last game, the challenge ciphertext is thus,

$$\begin{aligned}
c^* &= \text{sPKE.Enc}'(\widehat{\text{sk}}, \text{pk}, m_\gamma; (r, c_r)), \\
&= \text{sPKE.Enc}(\text{pk}, m_\gamma - \text{PKE.Dec}(\widehat{\text{sk}}, c_r); r) \cdot c_r, \\
&= \text{sPKE.Enc}(\text{pk}, m_\gamma - \text{PKE.Dec}(\widehat{\text{sk}}, \mathfrak{h}) - \text{PKE.Dec}(\widehat{\text{sk}}, \mathfrak{h}'); r) \cdot c_r, \\
&= \text{sPKE.Enc}(\text{pk}, m_\gamma - \text{PKE.Dec}(\widehat{\text{sk}}, \mathfrak{h}) - \mu; r) \cdot c_r, \\
&= \text{sPKE.Enc}(\text{pk}, m_\gamma - \text{PKE.Dec}(\widehat{\text{sk}}, \mathfrak{h}) + \text{PKE.Dec}(\text{sk}, \mathfrak{h}) - \mu; r) \cdot \mathfrak{h}', \\
&= \text{sPKE.Enc}(\text{pk}, m_\gamma - \mu; r) \cdot \mathfrak{h}'.
\end{aligned}$$

Since μ and $\mathfrak{h}'$ have, in this game, been chosen *independently* after all of $\mathcal{A}$'s queries, it implies that $m_\gamma - \mu$ is uniformly distributed in G from $\mathcal{A}$'s view, in the statistical sense. Hence, c^* is a random element in H^* independent of m_γ and $\Pr[W_4] = 1/2$.

It thus follows from the above sequence of game hops that $\text{Adv}_{\text{sPKE}}^{\text{CCA1}}(\mathcal{A})$ is negligible. $\square$

3.3 Examples of known sGPKE

In this section, we illustrate that our sGPKE framework allows to establish the CCA1 security (under a falsifiable indistinguishability assumption) of several known schemes. As all the schemes in this section are already known to achieve the same, we defer the proofs to Appendix A.

Paillier-ElGamal with plaintext zero-padding (PEG_z, Sect. 2.5). In Appendix A.1, Lemma 24 first establishes that PEG is a GPKE under DCR and Proposition 25 that PEG_z belongs to the sGPKE class. Thus Proposition 12 implies that PEG_z is CCA1 secure solely under DCR. Interestingly, this result establishes this without reliance on the Composite Non-Invertibility assumption (Def. 40 in Appendix D.1) contrary to [28, Theorem 3] which requires this extra assumption in some parameter regime (see Remark 33, p. 49). This closes an open question from that paper: “An interesting open question is to prove the CCA1 security of the scheme in Section 5.1 under the sole DCR assumption and without sacrificing more than 50% of the message space” [28, Sect. 5.4].

Cramer-Shoup Lite. We recall the definition of the Cramer-Shoup Lite scheme [19] (denoted CSL) in Appendix A.2. To comply with Def. 10 we split the definition in two schemes: CSL₀ and CSL. In that section, Lemma 27 thus first establishes that CSL₀ is a GPKE under DDH and Proposition 28 that CSL belongs to the sGPKE class. Thus Proposition 12 implies that CSL is CCA1 secure under DDH a fact which was already established in [19] by a (seminal) dedicated proof.

We emphasize that CSL is truly multiplicatively homomorphic but that its additive variant, which consists in encrypting a message m as $c_2 = g^m A^r$ rather

than mA^r , then requires solving a discrete-log in the decryption function. So, its effective plaintext size is limited by the constraint that the decryption algorithm should be PPT. As a consequence, a bound B has to be enforced over the messages. Hence, *this additive variant of the scheme is “only” somewhat linearly homomorphic* and do not satisfy correctness (recall Eq. 2). Interestingly, the D  mgard-ElGamal scheme is equivalent to the three first slots of CSL. Since we now know that this latter scheme is CCA1 secure under DDH, due to the recent proof by Libert [28, Theorem 1], it means that the term c_3 can be dropped in the additive-variant of CSL.

D  mgard’s additively homomorphic ElGamal. We recall the definition of the D  mgard-ElGamal scheme [28,20] (denoted DEG) in Appendix A.3. To comply with Def. 10 we split the definition in two schemes: DEG_0 and DEG. In that section, Lemma 29 thus first establishes that DEG_0 is a GPKE under DDH and Proposition 30 that DEG belongs to the sGPKE class. Thus Proposition 12 implies that DEG is CCA1 secure under DDH a fact which also follows from Libert’s recent proof [28, Theorem 1] (previous proofs of CCA1 security for that scheme required a non-falsifiable Knowledge-of-Exponent assumption).

Concluding remarks. Overall, in this section, we have illustrated that our sGPKE framework has allowed to unify the proofs of CCA1 security (under falsifiable assumptions) for three schemes that were previously known to achieve the same by means of separate dedicated proofs. Furthermore, our proof technique removed an assumption for one of them since the proof of CCA1 security of PEG_z in [28, Theorem 3] requires an additional Composite Non-Invertibility assumption that our proof technique allows to bypass.

In the next section we will now use our sGPKE framework to design a new CCA1-secure scheme with desirable properties. Amongst these latter, and by opposition to all the schemes in the present section, our new scheme will be *truly* (i.e. *non-somewhat*) linearly homomorphic.

4 A new CCA1 (and vCCA) secure variant of Paillier-ElGamal

The intuition behind our new scheme is very similar to Bitansky et al. two-ciphertext Paillier heuristic that has already been considered in several works [4,3,5,23,16,22]: encrypt a message m as a pair of Paillier ciphertexts ($\text{Enc}(m)$, $\text{Enc}(\xi \cdot m)$) under the same key material, for a secret random value $\xi \in \mathbb{Z}_N$, and check this linear relation at decryption.

Following this idea, we now define a two message slots variant of PEG, PEG_2 , in which the two message slots are under independent secret keys. Because of this we can reuse the randomness across these two slots. We then sparsify the plaintext domain of the resulting scheme following D  mgard’s Knowledge-of-Exponent template. This yields a more compact scheme which we call D  mgard-Paillier-ElGamal as it is very similar in spirit to the D  mgard-ElGamal scheme.

To follow our sGPKE blueprint we first define the associated GPKE (PEG₂) in the next section.

4.1 Paillier-ElGamal with two message slots (PEG₂)

Following the definition of PEG (Sect. 2.4), PEG₂ is simply defined as follows:

- PEG₂.KeyGen: choose a RSA modulus $N = (2p' + 1)(2q' + 1)$, with p' and q' two distinct Sophie Germain primes. Then, uniformly pick $g \in \mathbb{Z}_{N^2}^*$, $x, x' \in \llbracket 0, N(N-1)/4 \rrbracket$ and let $h = [g^{4Nx}]_{N^2}$ and $h' = [g^{4Nx'}]_{N^2}$. Then, $\text{pk} = (N, g, h, h')$, $\text{sk} = (x, x')$ and $\text{sk}_0 = (p', q')$ (following Def. 9).
- PEG₂.Enc: given $(m, m') \in \mathbb{Z}_N^2$ and pk , pick $r \in \llbracket 0, (N-1)/4 \rrbracket$ and return $(c_0, c_1, c_2) \in \mathbb{Z}_{N^2}^3$ such that,

$$(c_0, c_1, c_2) = ([g^{2Nr}]_{N^2}, [(1+N)^m h^r]_{N^2}, [(1+N)^{m'} h'^r]_{N^2}).$$

- PEG₂.Dec: given $(c_0, c_1, c_2) \in \mathbb{Z}_{N^2}^3$ and sk , let $\mu = [c_1 \cdot c_0^{-2x}]_{N^2}$ and $\mu' = [c_2 \cdot c_0^{-2x'}]_{N^2}$. If μ is of the form $[(1+N)^m]_{N^2}$ for some $m \in \mathbb{Z}_N$ and μ' is of the form $[(1+N)^{m'}]_{N^2}$ for some $m' \in \mathbb{Z}_N$, then return (m, m') . Otherwise, return $\perp$.

As PEG, this scheme is perfectly correct and trivially linearly homomorphic offering operators,

- PEG₂.Add: given (c_0, c_1, c_2) and (c'_0, c'_1, c'_2) , both in $\mathbb{Z}_{N^2}^3$, return $([c_0 c'_0]_{N^2}, [c_1 c'_1]_{N^2}, [c_2 c'_2]_{N^2})$.
- PEG₂.MulByConst _{α} : given $\alpha \in \mathbb{Z}_N$ and $(c_0, c_1, c_2) \in \mathbb{Z}_{N^2}^3$, an encryption of (m, m') , return $([c_0^\alpha]_{N^2}, [c_1^\alpha]_{N^2}, [c_2^\alpha]_{N^2})$, an encryption of $(\alpha m, \alpha m')$.

The scheme is also CPA secure under the DCR assumption as its CPA security is easy to establish from that of PEG. As a sanity check, we give a proof of this in appendix Sect. B.1 (Prop. 31).

4.2 D  mgard-Paillier-ElGamal (DPEG)

We then apply the Knowledge-of-Exponent pattern within the two message slots of a PEG₂ ciphertext. Like in D  mgard-ElGamal, ciphertexts then consists in only 3 group elements. From the definition of PEG₂ in the previous section, our new scheme DPEG is defined as follows:

- DPEG.KeyGen: run PEG₂.KeyGen(1^λ) to get PEG₂.pk = (N, g, h, h') and (x, x') . Further pick ξ uniformly at random in $\mathbb{Z}_N$ and let,

$$\text{ct}^\Delta = \text{PEG}_2.\text{Enc}(\text{PEG}_2.\text{pk}, (1, \xi)).$$

The public key is $\text{pk} = (N, g, h, h', \text{ct}^\Delta)$ while the secret key is (x, x', ξ) .

- DPEG.Enc: given $m \in \mathbb{Z}_N$ and pk , return,

$$\text{PEG}_2.\text{Add}(\text{PEG}_2.\text{MulByConst}_m(\text{ct}^\Delta), \text{PEG}_2.\text{Enc}(\text{PEG}_2.\text{pk}, 0)). \quad (11)$$

- DPEG.Dec: given $\text{ct} \in \mathbb{Z}_{N^2}^3$ and sk , if $\text{PEG}_2.\text{Dec}(\text{sk}, \text{ct}) = \perp$, then return $\perp$. Otherwise, let $(m, m') = \text{PEG}_2.\text{Dec}(\text{sk}, \text{ct})$, if $[\xi m]_N \neq m'$ then return $\perp$ and otherwise return m .

Remark that, since ξ is private, the only well-formed ciphertexts that can be created without using ct^Δ are encryptions of 0.

We now prove a first simple Lemma that establishes that for any $m \in \mathbb{Z}_N$ the distribution $\{\text{DPEG.Enc}(\text{DPEG.pk}, m; r) : r \xleftarrow{\$} \llbracket 0, (N-1)/4 \rrbracket\}$ cannot be distinguished from $\{\text{PEG}_2.\text{Enc}(\text{PEG}_2.\text{pk}, (m, \xi m); r) : r \xleftarrow{\$} \llbracket 0, (N-1)/4 \rrbracket\}$.

Lemma 13. *Let $\text{ct} = \text{DPEG.Enc}(\text{pk}, m)$, then,*

$$\text{ct} = ([g^{2Nr}]_{N^2}, [(1+N)^m h^r]_{N^2}, [(1+N)^{\xi m} h'^r]_{N^2})$$

with $r \xleftarrow{\$} \llbracket 0, (N-1)/4 \rrbracket$.

Proof. From Lemma 8, the distribution $\{z \xleftarrow{\$} \llbracket 0, (N-1)/4 \rrbracket\}$ is within $2^{-\lambda}$ statistical distance of the distribution $\{z \xleftarrow{\$} \mathbb{Z}_{p'q'}\}$, so that these two distributions can be used interchangeably. From Eq. (11), we have,

$$\begin{aligned} \text{ct} &= \text{PEG}_2.\text{Add}(\text{PEG}_2.\text{MulByConst}_m(\text{ct}^\Delta), \text{PEG}_2.\text{Enc}(\text{pk}, (0, 0); r_0)) \\ &= ([g^{2Nr_\Delta m}]_{N^2}, [(1+N)^m h^{r_\Delta m}]_{N^2}, [(1+N)^{\xi m} h'^{r_\Delta m}]_{N^2}) \cdot ([g^{2Nr_0}]_{N^2}, [h_0^r]_{N^2}, [h'^{r_0}]_{N^2}), \\ &= ([g^{2N[r_\Delta m + r_0]_{p'q'}}]_{N^2}, [(1+N)^m h^{[r_\Delta m + r_0]_{p'q'}}]_{N^2}, [(1+N)^{\xi m} h'^{[r_\Delta m + r_0]_{p'q'}}]_{N^2}), \\ &= ([g^{2Nr}]_{N^2}, [(1+N)^m h^r]_{N^2}, [(1+N)^{\xi m} h'^r]_{N^2}), \end{aligned}$$

with, following DPEG.KeyGen , $\text{ct}^\Delta = \text{PEG}_2.\text{Enc}(\text{pk}, (1, \xi); r_\Delta)$. Since r_0 is uniformly picked in $\mathbb{Z}_{p'q'}$ at encryption, $r = [r_\Delta m + r_0]_{p'q'}$ is also uniform in $\mathbb{Z}_{p'q'}$. $\square$

This Lemma also follows from Property 6 in Def. 9 since, as we shall now see, PEG_2 is a GPKE (Lemma 14).

4.3 CCA1 security of DPEG under DCR

To establish the CCA1 security of DPEG, we proceed in two steps. First we establish that PEG_2 is a GPKE (under the DCR assumption) and then that DPEG is a sGPKE (under the same assumption). The CCA1 security of DPEG, then follows from Proposition 12.

Lemma 14. *PEG_2 is a GPKE under the DCR assumption.*

Proof. This proof is quite mechanical so we defer it to appendix Sect. B.2. In that proof, we provide the definition of the alternative decryption function, $\text{PEG}_2.\widehat{\text{Dec}}$ (Def. 9, property 7), that uses the factorization of N given by $\text{sk}_0 = (p', q')$,

- $\text{PEG}_2.\widehat{\text{Dec}}$: given $\text{ct} = (c_0, c_1, c_2) \in \mathbb{Z}_{N^2}^3$, sk and $\text{sk}_0 = (p', q')$, return $\perp$ when $[c_0^{2p'q'}]_{N^2} \neq 1$ and, otherwise, return $\text{PEG}_2.\text{Dec}(\text{sk}, \text{ct})$.

□

Proposition 15. *DPEG is a sGPKE under the DCR assumption.*

Proof. From its definition (Sect. 4.2) DPEG uses PEG_2 as its inner scheme and, from Lemma 14 above, that scheme is a GPKE under the DCR assumption.

$\text{DPEG}.\mathcal{P} = \{(m, \xi m) : m \in \mathbb{Z}_N\} \cong \mathbb{Z}_N$, hence $2^\lambda |\text{DPEG}.\mathcal{P}| \leq |\text{PEG}_2.G| = N^2$ since $N \gg 2^\lambda$.

From Sect. 4.2 (p. 22), $\text{DPEG}.\text{Dec}$ can equivalently be written as,

- $\text{DPEG}.\text{ImVer}$: given $\text{ct} \in \mathbb{Z}_{N^2}^3$ and $\text{DPEG}.\text{sk}$, if $\text{PEG}_2.\text{Dec}(\text{sk}, \text{ct}) = \perp$, then return **False**. Otherwise, let $(\mu, \mu') = \text{PEG}_2.\text{Dec}(\text{sk}, \text{ct})$, if $[\xi\mu]_N \neq \mu'$ then return **False** and otherwise return **True**.
- $\text{PEG}_2.\text{Dec}$: given $\text{ct} \in \mathbb{Z}_{N^2}^3$ and $\text{DPEG}.\text{sk}$, if $\text{DPEG}.\text{ImVer}(\text{sk}, \text{ct}) = \text{False}$, then return $\perp$. Otherwise, let $(\mu, \mu') = \text{PEG}_2.\text{Dec}(\text{sk}, \text{ct})$, return μ .

To establish the claim, there remains to prove that DPEG satisfies Eq. (6).

First recall that in $\text{sk} = (x, x')$, x and x' are picked uniformly in $\llbracket 0, N(N-1)/4 \rrbracket$. Since, by Lemma 8, the statistical distance between $\{[z]_{Np'q'} : z \xleftarrow{\$} \llbracket 0, N(N-1)/4 \rrbracket\}$ and $\{z \xleftarrow{\$} \mathbb{Z}_{Np'q'}\}$ is negligible in λ , both x and x' are indistinguishable from uniform over $\mathbb{Z}_{Np'q'}$. Furthermore, $[x]_N$ is independent from $[x]_{p'q'}$ and so is $[x']_N$ from $[x']_{p'q'}$. Now,

- In $\text{pk} = (N, g, h, h')$, $h = [g^{4Nx}]$ is the only term that depends on x and it only depends on $[x]_{p'q'}$. Similarly, $h' = [g^{4Nx'}]$ is the only term that depends on x' and it only depends on $[x']_{p'q'}$.
- Additionally, for any ciphertext $\text{ct} \in \mathbb{Z}_{N^2}^3$, $\text{PEG}_2.\widehat{\text{Dec}}(\text{ct})$ also depends only on $[x]_{p'q'}$ and $[x']_{p'q'}$.

Hence, for any (pk, sk) and any $\mathcal{A}^{\text{DPEG}.\widehat{\text{Dec}}(\text{sk}, \cdot)}(\text{pk})$, $\mathcal{A}$'s view is independent from both $[x]_N$ and $[x']_N$. Assume the adversary against (6) submits a ciphertext in H^* ¹⁰, i.e. of the form,

$$\text{ct} = ((1+N)^{a_0} g^{2Nr}, (1+N)^{a_1} h^r, (1+N)^{a_2} h'^r)$$

for some a_0, a_1, a_2, r . If $\text{ct} \notin H$, then $a_0 \neq 0$ and $\text{PEG}_2.\text{Dec}(\text{sk}, \text{ct})$ returns $\mu = [a_1 - 2[x]_N a_0]_N$ and $\mu' = [a_2 - 2[x']_N a_0]_N$. We know that $[x]_N$ and $[x']_N$ are independent from the adversary's view. This implies that both $[a_1 - 2xa_0]_N$ and $[a_2 - 2x'a_0]_N$ are uniformly distributed in $\mathbb{Z}_N$ from $\mathcal{A}$'s view. Hence, $[\xi\mu]_N = \mu'$ only with probability $\text{neg}(\lambda)$. Note that if a_0 is not invertible in $\mathbb{Z}_N$, let say a multiple of p , then $[a_1 - 2xa_0]_N$ and $[a_2 - 2x'a_0]_N$ also respectively appear uniformly distributed in $[a_1 + p\mathbb{Z}_q]_N$ and $[a_2 + p\mathbb{Z}_q]_N$. Since $q \gg 2^\lambda$ (and likewise for p), it is also the case that $[\xi\mu]_N = \mu'$ with probability $\text{neg}(\lambda)$. □

Proposition 16. *DPEG is CCA1 secure under the DCR assumption.*

¹⁰ $\text{PEG}_2.\text{Dec}$ returns $\perp$ for ciphertexts not in H^* .

Proof. Directly follows from Prop. 15 and Prop. 12. $\square$

For the sake of it, we provide in Appendix B.3 another direct proof of the above proposition that closely follows that of [28, Theorem 3].

To the best of our knowledge, *DPEG is thus the first truly linearly homomorphic scheme that is proven CCA1 secure under a standard assumption.*

4.4 vCCA security of DPEG

Since DPEG instantiates the Bitansky et al. two-ciphertext heuristic [5] it is natural to assume that it has the LOH property that we recalled in Sect. 2.3 (p. 12). Under this assumption, there are several avenues for proving its vCCA security. First, [16, Proposition 2] provides the general result that any correct, CPA secure public-key linearly homomorphic scheme that has the LOH property is vCCA secure, so the vCCA security of DPEG trivially follows from that result. For two-ciphertext Paillier, [22, Proposition 15] also gives a direct proof of vCCA security (by reduction to the CPA security of Paillier) that also works for proving the vCCA security of DPEG only at the cost of a few minor adaptations.

For self-containedness, we thus provide a proof that follows this latter strategy in appendix B.4. Remark that the fact that we have established the CCA1 security of DPEG (Prop. 16) does not help for proving vCCA security. Indeed, a reduction towards a CCA1 challenger against DPEG would still have to simulate a second step decryption oracle on its own since the challenger provides none.

Proposition 17. *DPEG is vCCA secure under the assumption that it has the LOH property and that it is CPA secure.*

Proof. Deferred to appendix Sect. B.4 (p. 49). $\square$

Although [16, Sect. 5.2] and [22, Sect. 6.3] already establish that Bitansky et al.’s two-ciphertext Paillier is vCCA secure under the same assumptions, it is a desirable feature that DPEG achieves CCA1 security under a falsifiable standard assumption and the stronger notion of vCCA security by adding a non-falsifiable one. Indeed, for the former scheme, the non-falsifiable LOH assumption bridges the much wider gap between CPA and vCCA security, a bolder leap of faith. It is also interesting to note that DPEG (as Bitansky et al.’s two-ciphertext Paillier) supports multi-hop (linear) homomorphic calculations while the known generic construction blueprints achieving vCCA security [30,10] are all single-hop.

4.5 A depth 1 multiplicative extension

In this section, we follow [22, Sect. 4.2] and apply the Catalano-Fiore blueprint [14] to get one level of homomorphic multiplication out of DPEG.

Indeed, in [15], Catalano and Fiore have proposed a clever and simple trick to get one level of multiplication from the Paillier scheme, or any other *truly* linearly homomorphic scheme. In a nutshell, considering Paillier, the trick consists in encrypting a message m as a pair $(c_0, c_1) = ([m-b]_N, \text{Pai.Enc}(b))$ for a uniformly

chosen mask $b \in \mathbb{Z}^N$. Then, given two such ciphertexts (c_0, c_1) and (c'_0, c'_1) , one may obtain an encryption of mm' by building a triplet of the form

$$(\text{Pai.Enc}(c_0 c'_0) c_1^{c'_0 c'_1 c_0}, c_1, c'_1),$$

which first term is thus an encryption of $(m-b)(m'-b') + b(m'-b') + b'(m-b) = mm' - bb'$ under Pai. Although still linearly homomorphic to some extent, the post-multiplication ciphertext structure is different from the pre-multiplication one and no procedure is known to switch from the latter form to the former, hence the limitation to one level of multiplication.

We can then apply this blueprint using DPEG as the underlying scheme, yielding scheme **mDPEG** which definition is deferred to Appendix C as the application of the above blueprint is direct and does not present any difficulty.

The CCA1 security of **mDPEG** follows by a simple reduction towards that of DPEG (Prop. 36, App. C, p. 53) and therefore also holds under DCR. To the best of our knowledge, *mDPEG is thus the first concrete scheme being both truly linearly homomorphic and supporting multiplications (even limited to one-level) that is proven CCA1 secure under a standard assumption* (we emphasize that the BGN variant that is proven CCA1 secure under DDH in [28] uses the same plaintext zero padding trick as PEG_z is therefore only somewhat homomorphic).

Likewise, the vCCA security of **mDPEG** also easily reduces to that of DPEG (Prop. 37, App. C, p. 54) and therefore holds under the same LOH assumption. Although [22, Sect. 6.3] already presents a closely related scheme that supports both homomorphic additions and (one level of) multiplications that is proven vCCA secure under the same assumptions (that scheme is essentially based on the Catalano-Fiore blueprint from the Bitansky et al. two-ciphertext Paillier scheme rather than DPEG), it is again a desirable feature that **mDPEG** achieves CCA1 security under a falsifiable standard assumption and then the stronger notion of vCCA security by adding a non-falsifiable one. Indeed, for the scheme in [22, Sect. 6.3], the non-falsifiable LOH assumption bridges the much wider gap between CPA and vCCA security.

5 vCCA security under falsifiable assumptions?

As already emphasized, the spirit of the vCCA security notion is (at least) to model construction blueprints which embed proof material in their ciphertexts and rely on a SNARK to enforce correct homomorphic evaluations over some input ciphertexts. In this context, **extract** in the decryption oracle thus corresponds to the extractor of the underlying SNARK which allows to retrieve a witness from the proven statement as well as auxiliary data forming the trace of the execution of the adversary.

Because of this intimate connection with Proof-of-Knowledge approaches, several works [16, 41, 22] have speculated that vCCA security may not be achievable under non-falsifiable assumptions due to the seminal no-go result of Gentry&Wichs [25] which essentially establishes that no black-box reduction exists

to prove a SNARG or SNARK construct secure based on a falsifiable assumption. In the next section, we confirm this intuition by showing that, under relatively mild assumptions, any vCCA secure linearly homomorphic scheme can be used to build a SNARK for any language in $\mathcal{NP}$, thus formally establishing a connection between vCCA security and this latter impossibility result.

In a nutshell, our proof approach is as follows. We first establish that the existence of a vCCA secure FHE scheme implies the existence of a certain SNARK. We then use this to show that the existence of a vCCA secure linearly homomorphic HE scheme allows to build a scheme with the LOH property. Our next step is then to leverage on a result from Bitansky et al. [4,5] giving a SNARK construct for any language in $\mathcal{NP}$ from any scheme with the LOH Property. Finally, we thus get that the existence of a linearly homomorphic scheme that is vCCA secure under a falsifiable assumption contradicts Gentry&Wichs no-go Theorem. As a last corollary, we show that all sGPKE satisfy the assumptions under which this latter result holds. As a consequence, the whole sGPKE class is covered by our impossibility result.

5.1 A no-go result for vCCA security under falsifiable assumptions

Given an FHE scheme $\mathcal{E}_H = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{Eval})$ with plaintext domain $\mathcal{P}$, function domain $\mathcal{F}$ and ciphertext domain $\mathcal{C}$ we denote by $\mathcal{C}_F$ the set of fresh well-formed ciphertexts, i.e.,

$$\mathcal{C}_F = \{c : \exists m \in \mathcal{P}, r \in \text{COIN}, c = \mathcal{E}_H.\text{Enc}(m, r; \text{pk})\}$$

and by $\mathcal{C}_E \supseteq \mathcal{C}_F$ the set of well-formed evaluated ciphertexts, i.e.,

$$\mathcal{C}_E = \{c_e : \exists f \in \mathcal{F}, \exists c_0, \dots, c_{L-1} \in \mathcal{C}_F, c_e = \mathcal{E}_H.\text{Eval}(f, c_0, \dots, c_{L-1})\},$$

with $L = O(\text{poly}(\lambda))$. The reason why $\mathcal{C}_E \supseteq \mathcal{C}_F$ is to account for the convention that $\text{extract}(c) = \{\text{id}, c\}$ for $c \in \mathcal{C}_F$. For self-containedness, we give the basic formal definitions for SNARG and SNARK systems in appendix Sect. D.5.

Proposition 18. *If there exists an FHE scheme $\mathcal{E}_H = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{Eval})$ with function domain $\mathcal{F}$ and ciphertext domain $\mathcal{C} \supset \mathcal{C}_E \supseteq \mathcal{C}_F$ that is vCCA secure, correct, compact and such that,*

$$P\left(\begin{array}{l} \mathcal{E}_H.\text{Dec}(c_e) \neq \perp \\ \wedge c_e \notin \mathcal{C}_E \end{array} \middle| \begin{array}{l} ([pk], sk) := \mathcal{E}_H.\text{KeyGen}(1^\lambda) \\ c_e := \mathcal{A}^{[\text{Enc}, \text{Dec}]}([pk]) \end{array}\right) \leq \text{neg}(\lambda), \quad (12)$$

for any PPT adversary $\mathcal{A}$, then there exists a designated-verifier SNARK $\Pi = (\text{Setup}, \text{Prove}, \text{Verify})$ for language,

$$L = \{c_e | \exists f \in \mathcal{F}, \exists c_0, \dots, c_{L-1} \in \mathcal{C}_F, c_e = \mathcal{E}_H.\text{Eval}(f, c_0, \dots, c_{L-1})\}. \quad (13)$$

Proof. Consider the SNARK Π such that proofs under Π are ciphertexts under $\mathcal{E}_H$ and such that $\Pi.\text{Setup}$ is $\mathcal{E}_H.\text{KeyGen}$, $\Pi.\text{Prove}$ is $\mathcal{E}_H.\text{Eval}$ and

$$\Pi.\text{Verify}(c; sk) = \begin{cases} \text{True} & \text{if } \mathcal{E}_H.\text{Dec}(c; sk) \neq \perp, \\ \text{False} & \text{otherwise.} \end{cases} \quad (14)$$

First, Π 's *succinctness*, follows directly from $\mathcal{E}_H$'s *compactness*.

Second, Π 's *completeness* follows from $\mathcal{E}_H$'s *correctness*. Indeed, the correctness of $\mathcal{E}_H$ implies that,

$$P(\mathcal{E}_H.\text{Dec}(c; \text{sk}) = \perp | c \in \mathcal{C}_E) \leq \text{neg}(\lambda),$$

or, equivalently (following Eq. 14 above), that

$$P(\Pi.\text{Verify}(c; \text{sk}) = \text{False} | c \in \mathcal{C}_E) \leq \text{neg}(\lambda).$$

Lastly, Π 's *knowledge soundness* follows from $\mathcal{E}_H$'s *vCCA security*. Indeed, the vCCA extractor associated to $\mathcal{E}_H$ has to satisfy the following two properties:

- *Input soundness*: Assume that the adversary is able to find some (non-constant) function $f \in \mathcal{F}$ such that, for any ciphertexts $c_0, \dots, c_{L-1}$,

$$c_e = \mathcal{E}_H.\text{Eval}(f; c_0, \dots, c_{L-1}),$$

yields, with non negligible probability, $(g, c'_0, \dots, c'_{L-1}) = \text{extract}(c_e)$ with $\{c_0, \dots, c_{L-1}\} \neq \{c'_0, \dots, c'_{L-1}\}$. Then, this contradicts the vCCA security of $\mathcal{E}_H$ as the adversary may then perform an evaluation of f involving the challenge ciphertext and eventually get a decryption of the resulting ciphertext (with non negligible probability) which thus betrays the challenge bit.

- *Function soundness*: since the vCCA decryption oracle does not use the function that is output by the extractor, it appears that achieving vCCA security does not require any soundness on this output. However, this is not the case for vCCA^D security (recall Def. 6) which, *under the correctness assumption* as well as assumption (12), is equivalent to vCCA security [10, Prop. 7]. Now assume that the adversary is able to find some distinct (non-constant) functions $f, g \in \mathcal{F}$ such that, for any ciphertexts $c_0, \dots, c_{L-1}$,

$$c_e = \mathcal{E}_H.\text{Eval}(f; c_0, \dots, c_{L-1}),$$

yields, with non negligible probability, $(g, c_0, \dots, c_{L-1}) = \text{extract}(c_e)$. Assume that it can further find $m_0^* \neq m_1^*$ as well as $m_1, \dots, m_{L-1}$ such that,

$$f(m_0^*, m_1, \dots, m_{L-1}) \neq f(m_1^*, m_1, \dots, m_{L-1}), \quad (15)$$

but,

$$g(m_0^*, m_1, \dots, m_{L-1}) = g(m_1^*, m_1, \dots, m_{L-1}). \quad (16)$$

Then, from (15), if the adversary chooses m_0^* and m_1^* as test messages, it may retrieve the challenge bit from a decryption of,

$$c_e^* = \mathcal{E}_H.\text{Eval}(f; c^*, \mathcal{E}_H.\text{Enc}(m_1), \dots, \mathcal{E}_H.\text{Enc}(m_{L-1})),$$

which is accepted by the vCCA^D decryption oracle by cond. (16). This thus also contradicts the initial assumption that $\mathcal{E}_H$ is vCCA secure.

Input and function soundness above then yield the expected Knowledge Soundness property for Π ,

$$P \left(\begin{array}{l} \Pi.\text{Verif}(c_e) = \text{True} \wedge \\ c_e \neq \mathcal{E}_H.\text{Eval}(f, c_0, \dots, c_{L-1}; \text{pk}) \\ \wedge \exists i : c_i \notin \mathcal{C}_F \end{array} \middle| \begin{array}{l} (\text{pk}, \text{sk}) := \mathcal{E}_H.\text{KeyGen}(1^\lambda) \\ c_e := \mathcal{A}(\text{pk}) \\ (f, c_0, \dots, c_{L-1}) := \text{extract}(c_e, \text{aux}) \end{array} \right) \leq \text{neg}(\lambda),$$

for any PPT adversary $\mathcal{A}$. $\square$

Proposition 19. *Let $\mathcal{E}_H = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{Eval})$ denote a vCCA secure homomorphic scheme with plaintext domain $\mathbb{Z}_t$ for some modulus t , function domain $\mathcal{F}$ restricted to linear functions and ciphertext domain $\mathcal{C} \supset \mathcal{C}_E \supseteq \mathcal{C}_F$, with $|\mathcal{C}_E| \leq 2^{-\lambda}|\mathcal{C}|$, that is correct, compact and satisfies property (12). Under the additional assumptions that ciphertexts under $\mathcal{E}_H$ are indistinguishable from uniform over $\mathcal{C}$ and that $(\mathcal{C}, \oplus)$ is an Abelian group, where $\oplus$ denotes $\mathcal{E}_H$'s homomorphic addition operator, $\mathcal{E}_H$, yields a scheme $\mathcal{E}'_H$ that has the LOH property.*

Proof. We build $\mathcal{E}'_H$ from $\mathcal{E}_H$ by modifying the following:

- $\mathcal{E}'_H.\text{KeyGen}$: run $\mathcal{E}_H.\text{KeyGen}(1^\lambda)$ to get $(\mathcal{E}_H.\text{pk}, \mathcal{E}_H.\text{sk})$. Let $K = 4\lambda + \log_2 |\mathcal{C}|$, then $\mathcal{E}'_H.\text{sk} = ((\mathcal{E}_H.\text{pk}, \mathcal{E}_H.\text{sk}))^{11}$ and $\mathcal{E}'_H.\text{pk}$ is $(\text{ct}^\Delta, \text{ct}_0^{(0)}, \dots, \text{ct}_{K-1}^{(0)})$, where,

$$\text{ct}^\Delta = \mathcal{E}_H.\text{Enc}(\mathcal{E}_H.\text{pk}, 1) \text{ and } \text{ct}_i^{(0)} = \mathcal{E}_H.\text{Enc}(\mathcal{E}_H.\text{pk}, 0), \text{ for } i \in [0, K-1]. \quad (17)$$

- $\mathcal{E}'_H.\text{Enc}$: given $m \in \mathcal{P}$ and $\mathcal{E}'_H.\text{pk}$, pick a random subset E of $[0, K-1]$ and return,

$$\mathcal{E}_H.\text{MulByConst}_m(\text{ct}^\Delta) \oplus \bigoplus_{i \in E} \text{ct}_i^{(0)}.$$

Note that $\mathcal{E}'_H$'s CPA security follows from our choice for K and [36, Claim 5.3]¹² which implies that ciphertexts under $\mathcal{E}'_H$ are indistinguishable from uniform over $\mathcal{C}$. Also remark that $\mathcal{E}'_H$ vCCA security follows by a straightforward reduction to that of $\mathcal{E}_H$ (see Lemma 34 in Sect. B.5).

We now show that $\mathcal{E}'_H$ has the LOH property (Def. 7). We prove this by using the extractor from the SNARK that is induced by $\mathcal{E}$ (from Prop. 18), which we denote $\text{extract}_{\text{vCCA}}$, to build $\mathcal{E}'_H$'s LOH extractor, $\text{extract}_{\text{LOH}}$, as required by Eq. (3). Let $\mathcal{A}$ denote the adversary against the LOH Property of $\mathcal{E}'_H$ in Eq. (3),

¹¹ This is a slight abuse of notations as we have to make the distinction between a scheme public parameters and its public key which we have not yet needed in the paper. For example, for PEG, the set of public parameters is the modulus N and the public key is (g, h) . The key point is that the public parameters alone do not allow to build valid ciphertexts, although they may be needed for the homomorphic operators.

In the above construction, $\mathcal{E}'_H$ is assumed to have the same public parameters as $\mathcal{E}_H$.
¹² Claim 5.3 in [36] establishes that, given a finite Abelian group G and any N elements $g_0, \dots, g_{N-1}$ in G , the probability that the statistical distance between the uniform distribution on G and the distribution given by the sum of a random subset of $g_0, \dots, g_{N-1}$ exceeds $\sqrt[4]{|G|/2^N}$ is bounded by $\sqrt[4]{|G|/2^N}$.

- Since, in $\mathcal{E}'_H$, $\mathcal{E}_H.\text{pk}$ is not available to $\mathcal{A}$, the only ciphertexts from $\mathcal{E}_H.\mathcal{C}_F$ it can use as inputs to homomorphic evaluations are the elements of $\mathcal{E}'_H.\text{pk}$. Therefore, any well-formed evaluated ciphertext produced by the adversary must be a linear combination of these elements (Lemma 35, Sect. B.5).
- By definition, the vCCA extractor works *only* on ciphertexts generated by the adversary as the extraction is performed by parsing its auxiliary data input, aux , which essentially contains the trace of execution of $\mathcal{A}$.

Recall Eq. (3)’s notations, let $\Omega = (c_0, \dots, c_{m-1}) = (\mathcal{E}'_H.\text{Enc}(a_0), \dots, \mathcal{E}'_H.\text{Enc}(a_{m-1}))$ denote the set of ciphertexts given to $\mathcal{A}$ and $\Omega' = (c'_0, \dots, c'_{k-1})$ denote the set of ciphertexts output by the adversary. Let $c' \in \Omega'$. We distinguish two cases:

- $c' \in \mathcal{E}_H.\mathcal{C}_E$: it then follows from Prop. 18 that $\text{extract}_{\text{vCCA}}(c, \text{aux})$ outputs a vector $\rho \in \mathbb{N}^{m+K+1}$ such that,

$$c' = \bigoplus_{i=0}^{m-1} \bigoplus_{j=1}^{\rho_i} c_i \oplus \bigoplus_{j=1}^{\rho_m} \text{ct}^\Delta \oplus \bigoplus_{i=0}^{K-1} \bigoplus_{j=1}^{\rho_{i+m+1}} \text{ct}_i^{(0)},$$

and, from the correctness assumption,

$$\mathcal{E}'_H.\text{Dec}(\mathcal{E}'_H.\text{sk}, c') = \mathcal{E}_H.\text{Dec}(\mathcal{E}_H.\text{sk}, c') = \sum_{i=0}^{m-1} \rho_i a_i + \rho_m.$$

Hence, setting $\text{extract}_{\text{LOH}}(c', \text{aux}) = (\pi, b)$ with $\pi \in \mathcal{P}^m : \pi_i = [\rho_i]_t, i \in \llbracket 0, m-1 \rrbracket$ and $b = [\rho_m]_t$ gives the expected LOH extractor output.

- $c' \notin \mathcal{E}_H.\mathcal{C}_E$: from Prop. 18, $\text{extract}_{\text{vCCA}}(c', \text{aux}) = (\text{id}, c')$ in this case. Then, if $c' \neq c_i, i \in \llbracket 0, m-1 \rrbracket$, $c' \neq \text{ct}^\Delta$ and $c' \neq \text{ct}_i^{(0)}, i \in \llbracket 0, K-1 \rrbracket$ (as all these ciphertexts are in $\mathcal{E}_H.\mathcal{C}_E$), $\emptyset$ is returned as the output for the LOH extractor. Consistently, it follows from property (12) that $\mathcal{E}'_H.\text{Dec}(\mathcal{E}'_H.\text{sk}, c') = \mathcal{E}_H.\text{Dec}(\mathcal{E}_H.\text{sk}, c') = \perp$, so this also is the expected LOH extractor output. $\square$

To prove our next claim. We will need the following two Theorems.

Theorem 20 ([4,5], Construction 6.1). *If there exists a homomorphic scheme with the LOH property, then there exists a designated-verifier (preprocessing) SNARK for any language in $\mathcal{NP}$.*

We then reproduce Gentry&Wichs no-go Theorem [25] with respect to building SNARKs (or SNARGs) which security holds under falsifiable assumptions (stricto sensu, this results holds under the mild assumption that there exists some sub-exponentially hard subset-membership problem in $\mathcal{NP}$).

Theorem 21 ([25], Corollary 5.2). *Any candidate publicly verifiable or designated-verifier SNARK construction for any $\mathcal{NP}$ -complete language cannot have a black-box proof of security based on a falsifiable assumption, unless the assumption is false.*

Proposition 22. *Any candidate homomorphic scheme with plaintext domain $\mathbb{Z}_t$ for some modulus t , function domain $\mathcal{F}$ restricted to linear functions and ciphertext domain $\mathcal{C} \supset \mathcal{C}_E \supseteq \mathcal{C}_F$, with $|\mathcal{C}_E| \leq 2^{-\lambda}|\mathcal{C}|$ and with $(\mathcal{C}, \oplus)$ an Abelian group, that is correct, compact, achieves ciphertext pseudorandomness and satisfies property (12), cannot have a black-box proof of vCCA security based on a falsifiable assumption.*

Proof. Otherwise, such a scheme would have the LOH property grounded in a falsifiable assumptions (Prop. 19), which would then allow to build a SNARK for any language in $\mathcal{NP}$ (Theorem 20) which security would also be based on the same assumption. Thus a contradiction to Theorem 21 would result. $\square$

Corollary 23 (Prop. 22). *Any correct sGPKE that achieves ciphertext pseudorandomness and which ciphertext domain $(\mathcal{C}, \cdot)$ is an Abelian group, cannot have a black-box proof of vCCA security based on a falsifiable assumption.*

Proof. By definition, for a correct sGPKE, sPKE, $\mathcal{C}_E = \mathcal{C}_F \subset H \subset \mathcal{C}$, with $|\mathcal{C}_E| \leq |H| \leq 2^{-\lambda}|H^*| \leq |\mathcal{C}|$. By definition, also, all sGPKE are compact. Any correct sGPKE is linearly homomorphic. Lastly, all sGPKE satisfy (12) by Lemma 11. $\square$

6 Conclusion and perspectives.

We believe that achieving “truly” linearly homomorphic encryption with CCA1 security is an important question, beautiful in theory and meaningful in practice. We have developed a framework toward this goal and leave as an open challenge the design of schemes with improved or optimal efficiency.

While our results primarily concern linearly homomorphic encryption, they naturally raise questions about the feasibility of achieving CCA security for stronger forms of homomorphism, in particular fully homomorphic encryption (FHE), under standard falsifiable assumptions.

On the positive side, our construction shows that it is possible to obtain linearly homomorphic encryption supporting one level of multiplication together with CCA1 security from falsifiable assumptions. Although still far from FHE, this can be viewed as a promising step toward constructing CCA1-secure FHE, or at least somewhat homomorphic schemes, whose security relies on falsifiable assumptions.

On the negative side, our impossibility result for vCCA security directly extends to FHE schemes and suggests that FHE achieving stronger CCA1-type security is unlikely to be provable under purely falsifiable assumptions. An important open question therefore remains to determine the optimal security guarantees achievable from falsifiable assumptions for FHE.

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A Examples of known sGPKE

In this section, we illustrate that our sGPKE framework allows to establish the CCA1 security (under a falsifiable indistinguishability assumption) of several known schemes. All the schemes in this section are also already known to achieve the same, however with separate proofs.

A.1 Paillier-ElGamal with plaintext zero-padding (PEG_z , Sect. 2.5)

First, we consider the Paillier-ElGamal variant of Libert [28], that we defined in Sect. 2.5.

Lemma 24. *PEG is a GPKE under the DCR assumption.*

Proof. *Property 1.* Following the definition of PEG (Sect. 2.4, p. 14),

- KeyGen_0 : on input 1^λ , choose a RSA modulus $N = (2p' + 1)(2q' + 1)$, with p' and q' two distinct Sophie Germain primes of suitable size. Then $\text{pk}_0 = N$ and $\text{sk}_0 = (p', q')$.
- KeyGen_1 : on input $\text{pk}_0 = N$, uniformly pick $g \in \mathbb{Z}_{N^2}^*$, $x \in \llbracket 0, N(N-1)/4 \rrbracket$ and let $h = [g^{4Nx}]_{N^2}$. Then, $\text{pk} = (N, g, h)$ and $\text{sk} = x$.

Property 2. $G = \mathbb{Z}_N$ and

$$H = \{(c_0, c_1) \in \mathbb{Z}_{N^2}^2 : c_0 \in \langle g^{2N} \rangle, c_1 \in \langle (1+N)c_0^{2x} \rangle\}.$$

For every key pair (pk, sk) , let ct be a element of H , then

$$\text{ct} = (g^{2Nr}, (1+N)^m \cdot g^{4Nrx}) = \text{PEG.Enc}(\text{pk}, m; r),$$

for some $r \in \mathbb{Z}_{p'q'} \subset \llbracket 0, (N-1)/4 \rrbracket$ and $m \in G$. Hence, $\text{PEG.Enc}(\text{pk}, G; \text{COIN}) = H$.

Property 3. Given $\text{pk}_0 = N$, $\text{PEG.KeyGen}_1(\text{pk}_0) = (\text{sk}, \text{pk}) = (x, (N, g, h))$, let $H' = \langle ((1+N), (1+N)^{2x}) \rangle$. The indistinguishability between $H^* = H \cdot H'$ and H , follows by the DCR assumption, as it is equivalent to distinguish between,

$$\{[z^{2N}]_N : z \xleftarrow{\$} \mathbb{Z}_N\} \text{ and } \{[z^2]_{N^2} : z \xleftarrow{\$} \mathbb{Z}_{N^2}\},$$

without knowledge of the factorization of N , i.e. $\text{sk}_0 = (p', q')$. Additionally, for any $c = (c_0, c_1) \notin H^*$, we have that $c_1 c_0^{-2x} \notin \langle 1+N \rangle$ and by definition, $\text{PEG.Dec}(c) = \perp$.

Property 4. $G = \mathbb{Z}_N$ with $N \gg 2^\lambda$.

Property 5. Let $m, m' \in G$, $t \in [|G|]$, further let,

$$\text{ct} = \text{PEG.Enc}(\text{pk}, m') \cdot \prod_{i=1}^t \text{PEG.Enc}(\text{pk}, m),$$

then,

$$\begin{aligned} \text{PEG.Dec}(\text{sk}, \text{ct}) &= \text{PEG.Dec}(\text{sk}, (g^{2Nr'}, (1+N)^{m'} \cdot h^{r'})) \\ &\quad \cdot \prod_{i=1}^t (g^{2Nr}, (1+N)^m \cdot h^r), \\ &= \text{PEG.Dec}(\text{sk}, (g^{2Nr'}, (1+N)^{m'} \cdot h^{r'})) \\ &\quad \cdot (g^{2Ntr}, (1+N)^{tm} \cdot h^{tr}), \\ &= \text{PEG.Dec}(\text{sk}, (g^{2N(r'+tr)}, (1+N)^{m'+tm} \cdot h^{r'+tr})), \\ &= m' + tm. \end{aligned}$$

Property 6. Let $\text{ct} = \text{PEG.Enc}(\text{pk}, m; r)$ and $\text{ct}' = \text{PEG.Enc}(\text{pk}, m'; r')$,

$$\begin{aligned} \text{ct} \cdot \text{ct}' &= ([g^{2Nr}]_{N^2}, [(1+n)^m h^r]_{N^2}) \cdot ([g^{2Nr'}]_{N^2}, [(1+N)^{m'} h^{r'}]_{N^2}), \\ &= ([g^{2N[r+r']}_{p'q'}]_{N^2}, [(1+N)^{(m+m')} h^{[r+r']}_{p'q'}]_{N^2}), \\ &= ([g^{2Nr''}]_{N^2}, [(1+N)^{(m+m')} h^{r''}]_{N^2}), \end{aligned}$$

with either or both r and r' uniformly random in $\mathbb{Z}_{p'q'}$, $r'' = [r + r']_{p'q'}$ is also uniform in $\mathbb{Z}_{p'q'}$. Now, recall that the encryption randomness is picked uniformly in $\llbracket 0, (N-1)/4 \rrbracket$. Then Lemma 8 implies that the distributions $\{[g^{2Nr}]_{N^2} : r \xleftarrow{\$} \mathbb{Z}_{p'q'}\}$ and $\{[g^{2Nr}]_{N^2} : r \xleftarrow{\$} \llbracket 0, (N-1)/4 \rrbracket\}$ are within $2^{-\lambda}$ statistical distance since the subgroup of $2N$ -th residues in $\mathbb{Z}_{N^2}^*$ is a cyclic group of order $p'q'$ and $[g^{2N}]_{N^2}$ is with overwhelming probability a generator for the latter [28].

Property 7. Given a key pair $(\text{pk}, \text{sk}) = ((N, g, h), x)$, $\text{PEG.Dec} : H^* \rightarrow G$ takes (c_0, c_1) and returns $m \in G$ such that $1 + mN = [c_1 c_0^{-2x}]_{N^2}$. Now consider $\mathfrak{h}' = ((1+N)^a, (1+N)^{2xa}) \in H'$ for $a \in \mathbb{Z}_N$ (note that a is invertible in $\mathbb{Z}_N$ with overwhelming probability) and $\mu \xleftarrow{\$} G$. Since h and $c_1 c_0^{-2x}$ only depend on $[x]_{p'q'}$, we can compute another valid key pair $(\text{pk}, \widehat{\text{sk}}) = ((N, g, h), \widehat{x})$, such that $[\widehat{x}]_{p'q'} = [x]_{p'q'}$. Then, we have $\text{PEG.Dec}(\widehat{\text{sk}}, \mathfrak{h}') = [2(x - \widehat{x})a]_N$, and we can choose $\widehat{\text{sk}}$ such that $[\widehat{x}]_N = [x - \mu(2a)^{-1}]_N$ to get $\text{PEG.Dec}(\widehat{\text{sk}}, \mathfrak{h}') = \mu$. By Lemma 8, the statistical distance between $\{z \xleftarrow{\$} \mathbb{Z}_{Np'q'}\}$ and $\{z \xleftarrow{\$} \llbracket 0, N(N-1)/4 \rrbracket\}$ (used for key generation) is negligible in λ . Since, $\mu \xleftarrow{\$} G$, $[\widehat{x}]_N$ appears uniform in $\mathbb{Z}_N$ and $[\widehat{x}]_{p'q'} = [x]_{p'q'}$ is also uniform in $\mathbb{Z}_{p'q'}$ (from key generation). It follows that $\widehat{x}$ is indistinguishable from uniform in $\llbracket 0, N(N-1)/4 \rrbracket$.

Property 8. For $c = (c_0, c_1) \in H$, c_0 is by definition of the form $[g^{2Nr}]_{N^2}$. For g a generator of $(\mathbb{Z}_{N^2}, \cdot)$, $(\langle g^{2N} \rangle, \cdot)$ is the only subgroup of $(\mathbb{Z}_{N^2}, \cdot)$ of order $2p'q'$. It follows that $c_0 \in (\langle g^{2N} \rangle, \cdot)$ if and only if $[c_0^{2p'q'}]_{N^2} = 1$. We can thus define an alternative decryption function that uses the factorization of N given by $\text{sk}_0 = (p', q')$,

- $\text{PEG}.\widehat{\text{Dec}}$: given $c = (c_0, c_1) \in \mathbb{Z}_{N^2}^2$, sk and $\text{sk}_0 = (p', q')$, return $\perp$ when $[c_0^{2p'q'}]_{N^2} \neq 1$ and, otherwise, return $\text{PEG.Dec}(\text{sk}, c)$.

□

Proposition 25. PEG_z is a sGPKE under the DCR assumption.

Proof. From its definition (Sect. 2.5) PEG_z uses PEG as its GPKE and, from Lemma 24 above, that scheme is indeed a GPKE under the DCR assumption.

$\text{PEG}_z.\mathcal{P} = \llbracket 0, B \rrbracket$, with $2^\lambda \leq B = 2^t < \lfloor 2^{-\lambda} N \rfloor$, hence $2^\lambda |\text{PEG}_z.\mathcal{P}| \leq |\text{PEG}.G| = N$.

From Sect. 2.5 (p. 14), $\text{PEG}_z.\text{Dec}$ can equivalently be written as,

- $\text{PEG}_z.\text{ImVer}$: given $\text{ct} \in \mathbb{Z}_{N^2}^2$ and $\text{PEG}_z.\text{sk}$, if $\text{PEG.Dec}(\text{sk}, \text{ct}) = \perp$, then return False. Otherwise, let $\mu = \text{PEG.Dec}(\text{sk}, \text{ct})$, if $\mu > B$ then return False and otherwise return True.
- $\text{PEG}_z.\text{Dec}$: given $\text{ct} \in \mathbb{Z}_{N^2}^2$ and $\text{PEG}_z.\text{sk}$, if $\text{PEG}_z.\text{ImVer}(\text{sk}, \text{ct}) = \text{False}$, then return $\perp$. Otherwise, return $\text{PEG.Dec}(\text{sk}, \text{ct})$.

To establish the claim, there remain to prove that PEG_z satisfies Eq. (6).

First recall that $\text{sk} = x$ is picked uniformly in $\llbracket 0, N(N-1)/4 \rrbracket$. Since, by Lemma 8, the statistical distance between $\{[x]_{Np'q'} : x \xleftarrow{\$} \llbracket 0, N(N-1)/4 \rrbracket\}$ and

$\{x \xleftarrow{\$} \mathbb{Z}_{Np'q'}\}$ is negligible in λ , x is indistinguishable from uniform over $\mathbb{Z}_{Np'q'}$. Furthermore, $[x]_N$ is independent from $[x]_{p'q'}$. Now,

- In $\text{pk} = (N, g, h)$, $h = [g^{4Nx}]$ is the only term that depends on x and it only depends on $[x]_{p'q'}$.
- Additionally, for any ciphertext $\text{ct} \in \mathbb{Z}_{N^2}^2$, $\text{PEG}.\widehat{\text{Dec}}(\text{ct})$ also depends only on $[x]_{p'q'}$.

Hence, for any (pk, sk) and any $\mathcal{A}^{\text{sPKE}.\widehat{\text{Dec}}(\text{sk}, \cdot)}(\text{pk})$, $\mathcal{A}$'s view is independent from $[x]_N$. Assume the adversary against (6) submits a ciphertext in H^* , i.e. of the form,

$$\text{ct} = ((1 + N)^{a_0} g^{2Nr}, (1 + N)^{a_1} h^r)$$

for some a_0, a_1, r . If $\text{ct} \notin H$, then $a_0 \neq 0$ and $\text{PEG}.\text{Dec}(\text{ct})$ returns $m = [a_1 - 2xa_0]_N$. We know that $[x]_N$ is independent from the adversary's view. This implies that $[a_1 - 2xa_0]_N$ is uniformly distributed in G from $\mathcal{A}$'s view, and is in $\llbracket 0, B \rrbracket$ with probability $\text{neg}(\lambda)$, as $B \leq N \cdot 2^{-\lambda}$. Note that if a_0 is not invertible in $\mathbb{Z}_N$, let say a multiple of p , then $[a_1 - 2xa_0]_N$ also appears uniformly distributed in $[a_1 + p\mathbb{Z}_q]_N$, and is still in $\llbracket 0, B \rrbracket$ with probability $\frac{B/p}{N/p} = \frac{B}{N}$. $\square$

The following result then directly follows from Prop. 12 and Prop. 25 above.

Proposition 26. *PEG_z is CCA1 secure under the DCR assumption.*

A.2 Cramer-Shoup Lite

We now turn our attention to The Cramer-Shoup Lite scheme, as defined in [19]. Following our sGPKE construction blueprint (Def. 10), we first define the following GPKE scheme:

- $\text{CSL}_0.\text{KeyGen}$: on input 1^λ , choose a group $(\mathcal{G}, \cdot)$ of prime order $q \gg 2^\lambda$ and g a generator of $\mathcal{G}$, then pick $x_1, x_2, y_1, y_2, \alpha \xleftarrow{\$} \llbracket 0, q-1 \rrbracket$ and compute $h = g^\alpha, A = g^{x_1} h^{x_2}$ and $B = g^{y_1} h^{y_2}$. Then $\text{pk} = (q, g, h, A, B)$ and $\text{sk} = (x_1, x_2, y_1, y_2)$.
- $\text{CSL}_0.\text{Enc}$: given $m, m' \in \mathcal{G}$ and pk , choose $r \xleftarrow{\$} \llbracket 0, q-1 \rrbracket$ and return $c = (g^r, h^r, mA^r, m'B^r)$.
- $\text{CSL}_0.\text{Dec}$: given $c = (c_0, c_1, c_2, c_3)$, and sk , return $m = c_2 / (c_0^{x_1} c_1^{x_2})$ and $m' = c_3 / (c_0^{y_1} c_1^{y_2})$.

Then, we define the associated sGPKE, CSL, as follows,

- $\text{CSL}.\text{KeyGen}$: on input 1^λ , run $\text{CSL}_0.\text{KeyGen}(1^\lambda)$.
- $\text{CSL}.\text{Enc}$: given $m \in \mathcal{G}$ and pk , run $\text{CSL}_0.\text{Enc}(\text{pk}, m, m' = 1)$.
- $\text{CSL}.\text{Dec}$: given $c = (c_0, c_1, c_2, c_3)$, and sk , run $\text{CSL}_0.\text{Dec}(\text{sk}, c)$ to get (m, m') and if $m' = 1$ return m , otherwise return $\perp$.

Lemma 27. *CSL₀ is a GPKE under the DDH assumption.*

Proof. Property 1. Following the definition of CSL₀ above,

- KeyGen_0 : on input 1^λ , choose a group $(\mathcal{G}, \cdot)$ of prime order $q \gg 2^\lambda$ and g a generator of $\mathcal{G}$, then pick $\alpha \xleftarrow{\$} \llbracket 0, q-1 \rrbracket$ and compute $h = g^\alpha$. Then $\text{pk}_0 = (q, g, h)$ and $\text{sk}_0 = \alpha$.
- KeyGen_1 : on input $\text{pk}_0 = (q, g, h)$, pick $x_1, x_2, y_1, y_2 \xleftarrow{\$} \llbracket 0, q-1 \rrbracket$ and compute $A = g^{x_1} h^{x_2}$ and $B = g^{y_1} h^{y_2}$. Then $\text{pk} = (q, g, h, A, B)$ and $\text{sk} = (x_1, x_2, y_1, y_2)$.

Property 2. $G = \mathcal{G}^2$ and,

$$H = \{(c_0, c_1, c_2, c_3) \in \mathcal{G}^4 : c_1 = c_0^\alpha\}.$$

For every key pair (pk, sk) , let ct be a element of H , then,

$$\text{ct} = (g^r, g^{\alpha r}, m A^r, m' B^r) = \text{CSL}_0.\text{Enc}(\text{pk}, m, m'; r)$$

for $r \in \llbracket 0, q-1 \rrbracket$ and $m, m' \in \mathcal{G}$. Hence, $\text{CSL}_0.\text{Enc}(\text{pk}, G; \text{COIN}) = H$.

Property 3. Given $\text{pk}_0 = (q, g, h)$, $\text{CSL}.\text{KeyGen}_1(\text{pk}_0)$ outputs,

$$(\text{pk}, \text{sk}) = ((q, g, h, A, B), (x_1, x_2, y_1, y_2)).$$

Let $H' = \langle (1, g, g^{x_2}, g^{y_2}) \rangle$. The indistinguishability between $H^* = H \cdot H'$ and H , follows by the DDH assumption, as it is equivalent to distinguish between,

$$\{(g^\alpha, g^r, g^{\alpha r}) : r \xleftarrow{\$} \llbracket 0, q-1 \rrbracket\} \text{ and } \{(g^\alpha, g^r, g^{r'}) : r, r' \xleftarrow{\$} \llbracket 0, q-1 \rrbracket\},$$

without knowledge of $\text{sk}_0 = \alpha$. Remark that $\mathcal{C} = H^*$.

Property 4. $G = \mathcal{G}^2$ with $\mathcal{G}$ of order $q \gg 2^\lambda$.

Property 5. Let $m = (m_0, m_1), m' = (m'_0, m'_1) \in G$, $t \in \llbracket |G| \rrbracket$, then

$$\begin{aligned} & \text{CSL}_0.\text{Dec}(\text{sk}, \text{CSL}_0.\text{Enc}(\text{pk}, m') \cdot \prod_{i=1}^t \text{CSL}_0.\text{Enc}(\text{pk}, m)), \\ &= \text{CSL}_0.\text{Dec}(\text{sk}, (g^{r'}, h^{r'}, m'_0 A^{r'}, m'_1 B^{r'}) \cdot \prod_{i=1}^t (g^r, h^r, m_0 A^r, m_1 B^r)), \\ &= \text{CSL}_0.\text{Dec}(\text{sk}, (g^{r'}, h^{r'}, m'_0 A^{r'}, m'_1 B^{r'}) \cdot (g^{tr}, h^{tr}, m_0^t A^{tr}, m_1^t B^{tr})), \\ &= \text{CSL}_0.\text{Dec}(\text{sk}, (g^{r'+tr}, h^{r'+tr}, m'_0 m_0^t A^{r'+tr}, m'_1 m_1^t B^{r'+tr})), \\ &= m' m^t, \end{aligned}$$

Property 6. Let $\text{ct} = \text{CSL}_0.\text{Enc}(\text{pk}, m_0, m_1; r)$ and $\text{ct}' = \text{CSL}_0.\text{Enc}(\text{pk}, m'_0, m'_1; r')$,

$$\begin{aligned} \text{ct} \cdot \text{ct}' &= (g^r, h^r, A^r \cdot m_0, B^r \cdot m_1) \cdot (g^{r'}, h^{r'}, A^{r'} \cdot m'_0, B^{r'} \cdot m'_1), \\ &= (g^{r+r'}, h^{r+r'}, A^{r+r'} \cdot m_0 m'_0, B^{r+r'} \cdot m_1 m'_1), \\ &= (g^{r''}, h^{r''}, A^{r''} \cdot m_0 m'_0, B^{r''} \cdot m_1 m'_1), \end{aligned}$$

with either or both r and r' uniformly random in $\llbracket 0, q-1 \rrbracket$, $r'' = r + r'$ is also uniform in $\llbracket 0, q-1 \rrbracket$.

Property 7. Given a key pair $(\mathbf{pk}, \mathbf{sk}) = ((q, g, h, A, B), (x_1, x_2, y_1, y_2))$, $\text{CSL}_0.\text{Dec} : H^* \rightarrow G$ takes (c_0, c_1, c_2, c_3) and returns $(m, m') \in G$ such that $m = c_2 / (c_0^{x_1} c_1^{x_2})$ and $m' = c_3 / (c_0^{y_1} c_1^{y_2})$. Now consider $\mathbf{h}' = (1, g^a, g^{ax_2}, g^{ay_2}) \in H'$ for $a \in \llbracket 0, q-1 \rrbracket$ and $\mu = (g^b, g^c) \xleftarrow{\$} G$. Since A and B only depend on $x_1 + \alpha x_2$ and $y_1 + \alpha y_2$ respectively, we can compute another valid key pair $(\mathbf{pk}, \widehat{\mathbf{sk}}) = ((q, g, h, A, B), (\hat{x}_1, \hat{x}_2, \hat{y}_1, \hat{y}_2))$, such that $\hat{x}_1 + \alpha \hat{x}_2 = x_1 + \alpha x_2$ and $\hat{y}_1 + \alpha \hat{y}_2 = y_1 + \alpha y_2$. Then, we have $\text{CSL}_0.\text{Dec}(\widehat{\mathbf{sk}}, \mathbf{h}') = (g^{(x_2 - \hat{x}_2)a}, g^{(y_2 - \hat{y}_2)a})$, we can choose $\widehat{\mathbf{sk}}$ such that $\hat{x}_2 = x_2 - ba^{-1}$ and $\hat{y}_2 = y_2 - ca^{-1}$ to get $\text{CSL}_0.\text{Dec}(\widehat{\mathbf{sk}}, \mathbf{h}') = \mu$.

Property 8. For $c = (c_0, c_1, c_2, c_3) \in H$, by definition $c_1 = c_0^\alpha$, We can thus define an alternative decryption function that uses $\mathbf{sk}_0 = \alpha$,

- $\text{CSL}_0.\widehat{\text{Dec}}$: given $c = (c_0, c_1, c_2, c_3) \in \mathcal{G}^4$, $\mathbf{sk}$ and $\mathbf{sk}_0 = \alpha$, return $\perp$ when $c_0^\alpha \neq c_1$ and return $m = c_2 / (c_0^{x_1 + \alpha x_2})$ and $m' = c_3 / (c_0^{y_1 + \alpha y_2})$ otherwise.

It is obvious that $\text{CSL}_0.\widehat{\text{Dec}}(\mathbf{sk}, \mathbf{sk}_0, \text{ct}) = \text{CSL}_0.\text{Dec}(\mathbf{sk}, \text{ct})$ for any $\text{ct} \in H$. $\square$

Proposition 28. *CSL is a sGPKE under the DDH assumption.*

Proof. From its definition CSL uses CSL_0 as its GPKE and, from the Lemma above, that scheme is indeed a GPKE under the DDH assumption.

$\text{CSL}.\mathcal{P} = \mathcal{G}$, hence $2^\lambda |\text{CSL}.\mathcal{P}| \leq |\text{CSL}_0.G| = \mathcal{G}^2$.

From its definition, $\text{CSL}.\text{Dec}$ can equivalently be written as,

- $\text{CSL}.\text{ImVer}$: given $\text{ct} \in \mathcal{G}^4$ and $\text{CSL}.\mathbf{sk}$, if $\text{CSL}_0.\text{Dec}(\mathbf{sk}, \text{ct}) = (m, m')$ with $m' \neq 1$, then return **False**. Otherwise, return **True**.
- $\text{CSL}.\text{Dec}$: given $\text{ct} \in \mathcal{G}^4$ and $\text{CSL}.\mathbf{sk}$, if $\text{CSL}.\text{ImVer}(\mathbf{sk}, \text{ct}) = \text{False}$, then return $\perp$. Otherwise, return m from $(m, m') = \text{CSL}_0.\text{Dec}(\mathbf{sk}, \text{ct})$.

To establish the claim, there remain to prove that CSL satisfies Eq. (6).

First recall that $\mathbf{sk} = (x_1, x_2, y_1, y_2)$ is picked uniformly in $\llbracket 0, q-1 \rrbracket^4$. Now,

- In $\mathbf{pk} = (q, g, h, A, B)$, $A = g^{x_1} h^{x_2}$ and $B = g^{y_1} h^{y_2}$ are the only term that depends on $\mathbf{sk}$ and it only depends on $x_1 + \alpha x_2$ and $y_1 + \alpha y_2$.
- Additionally, for any ciphertext $\text{ct} \in \mathcal{G}^4$, $\text{CSL}_0.\widehat{\text{Dec}}(\text{ct})$ also depends only on $x_1 + \alpha x_2$ and $y_1 + \alpha y_2$.

Hence, for any $(\mathbf{pk}, \mathbf{sk})$ and any $\mathcal{A}^{\text{sPKE}.\widehat{\text{Dec}}(\mathbf{sk}, \cdot)}(\mathbf{pk})$, $\mathcal{A}$'s view is independent from x_2 and y_2 . Assume the adversary against (6) submits a ciphertext in H^* , i.e. of the form,

$$\text{ct} = (g^r, h^{r'}, a_0 \cdot A^r, a_1 \cdot B^r)$$

for some a_0, a_1, r and r' . If $\text{ct} \notin H$, then $r \neq r'$ and $\text{CSL}_0.\text{Dec}(\text{ct})$ returns (m, m') with especially $m' = a_1 \cdot h^{y_2(r-r')}$. We know that y_2 is independent from the adversary's view. This implies that $a_1 \cdot h^{y_2(r-r')}$ is uniformly distributed in $\mathcal{G}$ from $\mathcal{A}$'s view, and is equal to 1 with probability $\text{neg}(\lambda)$, as $2^\lambda \ll q$. $\square$

The CCA1 security of CSL under DDH, which of course was already established in [19] by a (seminal) dedicated proof, thus also follows from Prop. 28 and 12.

A.3 D  mgard's additively homomorphic ElGamal

We now turn to the D  mgard-ElGamal scheme, as described in [28,20]. Following our sGPKE blueprint (Def. 10), we first define the following GPKE scheme, DEG_0 :

- $\text{DEG}_0.\text{KeyGen}$: on input 1^λ , choose a group $(\mathcal{G}, \cdot)$ of prime order $q \gg 2^\lambda$ and g generator of $\mathcal{G}$, then pick $x_1, x_2, \alpha \xleftarrow{\$} [0, q-1]$ and compute $h = g^\alpha$ and $A = g^{x_1} h^{x_2}$. Then $\text{pk} = (q, g, h, A)$ and $\text{sk} = (x_1, x_2)$.
- $\text{DEG}_0.\text{Enc}$: given $m \in \mathcal{G}$ and pk , choose $r \xleftarrow{\$} [0, q-1]$ and return $c = (c_0, c_1, c_2) = (g^r, h^r, g^m A^r)$.
- $\text{DEG}_0.\text{Dec}$: given $c = (c_0, c_1, c_2)$, and sk , compute $M = c_2 / (c_0^{x_1} c_1^{x_2})$. If there exists an integer $m \in [0, q-1]$ such that $M = g^m$, return m . Otherwise return $\perp$.¹³

Then, we define the associated sGPKE, DEG , as follows,

- $\text{DEG}.\text{KeyGen}$: on input 1^λ and $t = O(\log(\lambda))$, run $\text{DEG}_0.\text{KeyGen}(1^\lambda)$ and return $\text{pk} = (\text{DEG}_0.\text{pk}, B = 2^t)$ and $\text{sk} = \text{DEG}_0.\text{sk}$. The plaintext domain is thus $[0, B]$.
- $\text{DEG}.\text{Enc}$: given $m \in [0, B]$ and pk , run $\text{DEG}_0.\text{Enc}(\text{pk}, m)$.
- $\text{DEG}.\text{Dec}$: given $c = (c_0, c_1, c_2)$, and sk , compute $M = c_2 / (c_0^{x_1} c_1^{x_2})$. If there exists an integer $m \in [0, B]$ such that $M = g^m$, return m . Otherwise return $\perp$.

Lemma 29. *DEG_0 is a GPKE under the DDH assumption.*

Proof. *Property 1.* Following the definition of DEG_0 above,

- KeyGen_0 : Given a parameter λ , choose a group $(\mathcal{G}, \cdot)$ of prime order $q \gg 2^\lambda$ and g generator of $\mathcal{G}$, then pick $\alpha \xleftarrow{\$} [0, q-1]$ and compute $h = g^\alpha$. Then $\text{pk}_0 = (q, g, h)$ and $\text{sk}_0 = (\alpha)$.
- KeyGen_1 : on input $\text{pk}_0 = (q, g, h)$, pick $x_1, x_2 \xleftarrow{\$} [0, q-1]$ and compute $A = g^{x_1} h^{x_2}$. Then $\text{pk} = (q, g, h, A)$ and $\text{sk} = (x_1, x_2)$.

Property 2. $G = [0, q-1]$ and

$$H = \{(c_0, c_1, c_2) \in \mathcal{G}^3 : c_1 = c_0^\alpha\}.$$

For every key pair (pk, sk) , let ct be a element of H , then

$$\text{ct} = (g^r, g^{r\alpha}, g^m A^r) = \text{DEG}_0.\text{Enc}(\text{pk}, m; r)$$

for $r, m \in [0, q-1]$. Hence, $\text{DEG}_0.\text{Enc}(\text{pk}, G; \text{COIN}) = H$.

Property 3. Given $\text{pk}_0 = (q, g, h)$, $\text{DEG}.\text{KeyGen}_1(\text{pk}_0)$ outputs,

$$(\text{pk}, \text{sk}) = ((q, g, h, A), (x_1, x_2)).$$

¹³ Remark that $\text{DEG}_0.\text{Dec}$ is not a PPT algorithm, but it poses does not create any issue with the proof of security of DEG .

Let $H' = \langle (1, g, g^{x_2}) \rangle$. The indistinguishability between $H^* = H \cdot H'$ and H , follows by the DDH assumption, as it is equivalent to distinguish between,

$$\{(g^\alpha, g^r, g^{\alpha r}) : r \xleftarrow{\$} [0, q-1]\} \text{ and } \{(g^\alpha, g^r, g^{r'}) : r, r' \xleftarrow{\$} [0, q-1]\},$$

without knowledge of $sk_0 = \alpha$. Remark that $\mathcal{C} = H^*$.

Property 4. $G = [0, q-1]$ with $q \gg 2^\lambda$.

Property 5. Let $m, m' \in G$, $t \in [G]$, then

$$\begin{aligned} & \text{DEG}_0.\text{Dec}(\text{sk}, \text{DEG}_0.\text{Enc}(\text{pk}, m') \cdot \prod_{i=1}^t \text{DEG}_0.\text{Enc}(\text{pk}, m)), \\ &= \text{DEG}_0.\text{Dec}(\text{sk}, (g^{r'}, h^{r'}, g^{m'} A^{r'}) \cdot \prod_{i=1}^t (g^r, h^r, g^m A^r)), \\ &= \text{DEG}_0.\text{Dec}(\text{sk}, (g^{r'}, h^{r'}, g^{m'} A^{r'}) \cdot (g^{tr}, h^{tr}, g^{tm} A^{tr})), \\ &= \text{DEG}_0.\text{Dec}(\text{sk}, (g^{r'+tr}, h^{r'+tr}, g^{m'+tm} A^{r'+tr})), \\ &= m' + tm, \end{aligned}$$

Property 6. Let $\text{ct} = \text{DEG}_0.\text{Enc}(\text{pk}, m; r)$ and $\text{ct}' = \text{DEG}_0.\text{Enc}(\text{pk}, m'; r')$,

$$\begin{aligned} \text{ct} \cdot \text{ct}' &= (g^r, h^r, g^m A^r) \cdot (g^{r'}, h^{r'}, g^{m'} A^{r'}), \\ &= (g^{r'+r}, h^{r'+r}, g^{m'+m} A^{r'+r}), \\ &= (g^{r''}, h^{r''}, g^{m+m'} A^{r''}), \end{aligned}$$

with either or both r and r' uniformly random in $[0, q-1]$, $r'' = r + r'$ is also uniform in $[0, q-1]$.

Property 7. Given a key pair $(\text{pk}, \text{sk}) = ((q, g, h, A), (x_1, x_2))$, $\text{DEG}_0.\text{Dec} : H^* \rightarrow G$ takes (c_0, c_1, c_2) and returns $m \in G$ such that $g^m = c_2 / (c_0^{x_1} c_1^{x_2})$.

Now consider $\mathfrak{h}' = (1, g^a, g^{ax_2}) \in H'$ for $a \in [0, q-1]$ and $\mu \xleftarrow{\$} G$. Since A only depends on $x_1 + \alpha x_2$, we can compute another valid key pair $(\text{pk}, \widehat{\text{sk}}) = ((q, g, h, A), (\hat{x}_1, \hat{x}_2))$, such that $\hat{x}_1 + \alpha \hat{x}_2 = x_1 + \alpha x_2$. Then, we have $\text{DEG}_0.\text{Dec}(\widehat{\text{sk}}, \mathfrak{h}') = (x_2 - \hat{x}_2)a$, we can choose $\widehat{\text{sk}}$ such that $\hat{x}_2 = x_2 - \mu a^{-1}$ to get $\text{DEG}_0.\text{Dec}(\widehat{\text{sk}}, \mathfrak{h}') = \mu$.

Property 8. For $c = (c_0, c_1, c_2) \in H$, by definition $c_1 = c_0^\alpha$. We can thus define an alternative decryption function that uses $sk_0 = \alpha$,

- $\text{DEG}_0.\widehat{\text{Dec}}$: given $c = (c_0, c_1, c_2) \in \mathcal{G}^3$, sk and $sk_0 = \alpha$, return $\perp$ when $c_0^\alpha \neq c_1$ and return $m = c_2 / c_0^{x_1 + \alpha x_2}$ otherwise.

It is obvious that $\text{DEG}_0.\widehat{\text{Dec}}(\text{sk}, sk_0, \text{ct}) = \text{DEG}_0.\text{Dec}(\text{sk}, \text{ct})$ for any $\text{ct} \in H$. $\square$

Proposition 30. *DEG is a sGPKE under the DDH assumption.*

Proof. From its definition DEG uses DEG_0 as its GPKE and, from the Lemma above, that scheme is indeed a GPKE under the DDH assumption.

$\text{DEG}.\mathcal{P} = [0, B]$, hence $2^\lambda |\text{DEG}.\mathcal{P}| \leq |\text{DEG}_0.G| = \mathcal{G}$ as $q \gg 2^\lambda$ and $t = O(\log(\lambda))$.

From its definition, $\text{DEG}.\text{Dec}$ can equivalently be written as,

- **DEG.ImVer**: given $\text{ct} \in \mathcal{G}^3$ and DEG.sk , if $\text{DEG}_0.\text{Dec}(\text{sk}, \text{ct}) = m$ with $m \notin \llbracket 0, B \rrbracket$, then return **False**. Otherwise, return **True**.
- **DEG.Dec**: given $\text{ct} \in \mathcal{G}^3$ and DEG.sk , if $\text{DEG.ImVer}(\text{sk}, \text{ct}) = \text{False}$, then return $\perp$. Otherwise, return $\text{DEG}_0.\text{Dec}(\text{sk}, \text{ct})$.

To establish the claim, there remain to prove that **DEG** satisfies Eq. (6).

First recall that $\text{sk} = (x_1, x_2)$ is picked uniformly in $\llbracket 0, q-1 \rrbracket^2$. Now,

- In $\text{pk} = (q, g, h, A)$, $A = g^{x_1} h^{x_2}$ is the only term that depends on sk and it only depends on $x_1 + \alpha x_2$.
- Additionally, for any ciphertext $\text{ct} \in \mathcal{G}^3$, $\text{DEG}_0.\widehat{\text{Dec}}(\text{ct})$ also depends only on $x_1 + \alpha x_2$.

Hence, for any (pk, sk) and any $\mathcal{A}^{\text{sPKE}.\widehat{\text{Dec}}(\text{sk}, \cdot)}(\text{pk})$, $\mathcal{A}$'s view is independent from x_2 . Assume the adversary against (6) submits a ciphertext in H^* , i.e. of the form,

$$\text{ct} = (g^r, h^{r'}, g^{a_0} A^r)$$

for some a_0, r and r' . If $\text{ct} \notin H$, then $r \neq r'$ and $\text{DEG}_0.\text{Dec}(\text{ct})$ returns m with $m = a_0 + x_2(r - r')$. We know that x_2 is independent from the adversary's view. This implies that $a_0 + x_2(r - r')$ is uniformly distributed in $\mathcal{G}$ from $\mathcal{A}$'s view, and is in $\llbracket 0, B \rrbracket$ with probability $\text{neg}(\lambda)$. $\square$

The CCA1 security of **DEG** under DDH, which was recently already established in [28] by a delicate proof, thus also follows from Prop. 30 and 12.

B Additional proofs

B.1 CPA security of PEG_2

In this section, we prove the CPA security of PEG_2 (Sect. 4.1). This result is easy to establish under the assumption that PEG is CPA secure. Since the latter holds under the DCR assumption, the CPA security of PEG_2 also holds under the same assumption.

Proposition 31. *PEG_2 is CPA secure under the assumption that PEG is.*

Proof. The proof proceeds by a quite conventional game hopping argument. Let $\text{CPA-}\gamma$ be CPA game against PEG_2 where the challenger picks the bit γ . Because, PEG_2 has two message slots, i.e. its plaintext domain is $\mathbb{Z}_N^2$, the adversary submits $(m_0, m'_0) \in \mathbb{Z}_N^2$ and $(m_1, m'_1) \in \mathbb{Z}_N^2$ as a challenge request and gets $\text{PEG}_2.\text{Enc}(\text{pk}, (m_\gamma, m'_\gamma))$ in return.

First game hop. Let G_0 be the CPA-0 game against PEG_2 and let G_1 be the same as G_0 except that the challenge ciphertext is,

$$([g^{2Nr}]_{N^2}, [(1+N)^{m_0} h^r]_{N^2}, [(1+N)^{m'_1} h'^r]_{N^2}),$$

rather than (in G_0),

$$([g^{2Nr}]_{N^2}, [(1+N)^{m_0} h^r]_{N^2}, [(1+N)^{m'_0} h'^r]_{N^2}).$$

We now prove that the two games cannot be distinguished under the assumption that PEG is CPA secure. Indeed, let $\mathcal{A}_0$ be an adversary that can distinguish between G_0 and G_1 , then we can build as follows an adversary $\mathcal{B}_0$ against the CPA security of PEG that achieves at least $\mathcal{A}_0$'s advantage. First $\mathcal{B}_0$ gets (N, g, h') from its CPA challenger against PEG (so the challenger operates slots 0 and 2 in the ciphertexts under PEG_2), then it picks x uniformly in $\llbracket 0, N(N-1)/4 \rrbracket$ (following $\text{PEG}_2.\text{KeyGen}$), computes $h = [g^{4Nx}]_{N^2}$ and forwards (N, g, h, h') to $\mathcal{A}_0$. Because we are in the public key setting, the reduction only has to handle challenge requests. It does this as follows:

- When $\mathcal{A}_0$ issues a challenge request over $(m_0, m'_0), (m_1, m'_1) \in \mathbb{Z}_N^2$, $\mathcal{B}_0$ first issues a challenge request over messages m'_0 and m'_1 towards its challenger to get PEG ciphertext,

$$(c_0^*, c_2^*) = ([g^{2Nr}]_{N^2}, [(1+N)^{m'_\gamma} h'^r]_{N^2}).$$

for unknown γ and r . Then, the reduction computes,

$$c_1^* = (1+N)^{m_0} c_0^{*2x} = (1+N)^{m_0} g^{4Nrx} = (1+N)^{m_0} h^r,$$

and returns (c_0^*, c_1^*, c_2^*) to $\mathcal{A}_0$.

Finally, when $\mathcal{A}_0$ outputs a guess $\hat{b}$, $\mathcal{B}_0$ does the same. Clearly, $\mathcal{B}_0$ perfectly simulates game G_0 when $b = 0$ and game G_1 when $b = 1$. Therefore,

$$\text{Adv}^{G_0/G_1}(\mathcal{A}_0) \leq \text{Adv}_{\text{PEG}}^{\text{CPA}}(\mathcal{B}_0). \quad (18)$$

Second game hop. We now consider game G_2 which is the same as G_1 except that the challenge ciphertext is now,

$$([g^{2Nr}]_{N^2}, [(1+N)^{m_1} h^r]_{N^2}, [(1+N)^{m'_1} h'^r]_{N^2}),$$

rather than (in G_1),

$$([g^{2Nr}]_{N^2}, [(1+N)^{m_0} h^r]_{N^2}, [(1+N)^{m'_1} h'^r]_{N^2}).$$

We can also prove that the two games cannot be distinguished under the assumption that PEG is CPA secure. Indeed, let $\mathcal{A}_1$ be an adversary that can distinguish between G_1 and G_2 , then we can build an adversary $\mathcal{B}_1$ against the CPA security of PEG that achieves at least $\mathcal{A}_1$'s advantage, with a reduction similar to that in the first game hop. First $\mathcal{B}_1$ gets (N, g, h) from its CPA challenger against PEG (so, this time, the challenger operates slots 0 and 1 in the ciphertexts under PEG_2), then it picks x' uniformly in $\llbracket 0, N(N-1)/4 \rrbracket$, computes $h' = [g^{4Nx'}]_{N^2}$ and forwards (N, g, h, h') to $\mathcal{A}_1$. Then $\mathcal{B}_1$ proceeds as follows:

- When $\mathcal{A}_1$ issues a challenge request over $(m_0, m'_0), (m_1, m'_1) \in \mathbb{Z}_N^2$, $\mathcal{B}_1$ first forward the request as is to its challenger to get ciphertext,

$$(c_0^*, c_1^*) = ([g^{2Nr}]_{N^2}, [(1+N)^{m_\gamma} h^r]_{N^2}).$$

for unknown γ and r . Then, $\mathcal{B}_1$ computes,

$$c_2^* = (1 + N)^{m'_1} c_0^{*2x'} = (1 + N)^{m'_1} g^{4Nr x'} = (1 + N)^{m'_1} h'^r,$$

and finally returns (c_0^*, c_1^*, c_2^*) to $\mathcal{A}_1$.

Again, when $\mathcal{A}_1$ outputs a guess $\hat{b}$, $\mathcal{B}_1$ does the same. Clearly, $\mathcal{B}_1$ perfectly simulates game G_1 when $b = 0$ and game G_2 when $b = 1$. Therefore,

$$\text{Adv}^{G_0/G_1}(\mathcal{A}_1) \leq \text{Adv}_{\text{PEG}}^{\text{CPA}}(\mathcal{B}_1). \quad (19)$$

Since G_0 is CPA-0 and G_2 is CPA-1, the claim follows by gathering Eqs. (18) and (19). $\square$

B.2 Proof of Lemma 14

The proof of Lemma 14 (PEG₂ is a GPKE) is very similar to that of Lemma 24 (for PEG).

Proof. *Property 1.* Following the definition of PEG₂,

- **KeyGen₀**: on input 1^λ , choose a RSA modulus $N = (2p' + 1)(2q' + 1)$, with p' and q' two distinct Sophie Germain primes of suitable size. Then $\text{pk}_0 = N$ and $\text{sk}_0 = (p', q')$.
- **KeyGen₁**: on input $\text{pk}_0 = N$, uniformly pick $g \in \mathbb{Z}_{N^2}^*$, $x, x' \in \llbracket 0, N(N-1)/4 \rrbracket$ and let $h = [g^{4Nx}]_{N^2}$ and $h' = [g^{4Nx'}]_{N^2}$. Then, $\text{pk} = (N, g, h, h')$ and $\text{sk} = (x, x')$.

Property 2. $G = \mathbb{Z}_N^2$ and

$$H = \{(c_0, c_1, c_2) \in \mathbb{Z}_{N^2}^3 : c_0 \in \langle g^{2N} \rangle, c_1 \in \langle 1 + N \rangle c_0^{2x}, c_2 \in \langle 1 + N \rangle c_0^{2x'}\}.$$

For every key pair (pk, sk) , let ct be a element of H , then,

$$\text{ct} = (g^{2Nr}, (1 + N)^m \cdot g^{4Nrx}, (1 + N)^{m'} \cdot g^{4Nrx'}) = \text{PEG}_2.\text{Enc}(\text{pk}, (m, m'); r),$$

for some $r \in \mathbb{Z}_{p'q'} \subset \llbracket 0, (N-1)/4 \rrbracket$ and $(m, m') \in G$. Hence, $\text{PEG}_2.\text{Enc}(\text{pk}, G; \text{COIN}) = H$.

Property 3. Given $\text{pk}_0 = N$, $\text{PEG}_2.\text{KeyGen}_1(\text{pk}_0) = (\text{sk}, \text{pk}) = (x, (N, g, h, h'))$, let $H' = \{((1 + N), (1 + N)^{2x}, (1 + N)^{2x'})\}$. The indistinguishability between $H^* = H \cdot H'$ and H , follows by the DCR assumption, as it is equivalent to distinguish between,

$$\{[z^{2N}]_N : z \xleftarrow{\$} \mathbb{Z}_N\} \text{ and } \{[z^2]_{N^2} : z \xleftarrow{\$} \mathbb{Z}_{N^2}\},$$

without knowledge of the factorization of N , i.e. $\text{sk}_0 = (p', q')$. Additionally, for any $c = (c_0, c_1, c_2) \notin H^*$, we have that either $c_1 c_0^{-2x} \notin \langle 1 + N \rangle$ or $c_2 c_0^{-2x'} \notin \langle 1 + N \rangle$ and by definition, $\text{PEG}_2.\text{Dec}(c) = \perp$.

Property 4. $G = \mathbb{Z}_N^2$ with $N \gg 2^\lambda$.

Property 5. Let $(m_0, m_1), (m'_0, m'_1) \in G$, $t \in [|G|]$, further let

$$\text{ct} = \text{sk}, \text{PEG}_2.\text{Enc}(\text{pk}, (m'_0, m'_1)) \cdot \prod_{i=1}^t \text{PEG}_2.\text{Enc}(\text{pk}, (m_0, m_1)),$$

then,

$$\begin{aligned} \text{PEG}_2.\text{Dec}(\text{ct}) &= \text{PEG}_2.\text{Dec}(\text{sk}, (g^{2Nr'}, (1+N)^{m'_0} \cdot h^{r'}, (1+N)^{m'_1} \cdot h^{r'})) \\ &\quad \cdot \prod_{i=1}^t (g^{2Nr}, (1+N)^{m_0} \cdot h^r, (1+N)^{m_1} \cdot h^{r'}), \\ &= \text{PEG}_2.\text{Dec}(\text{sk}, (g^{2Nr'}, (1+N)^{m'_0} \cdot h^{r'}, (1+N)^{m'_1} \cdot h^{r'})) \\ &\quad \cdot (g^{2Ntr}, (1+N)^{tm_0} \cdot h^{tr}, (1+N)^{tm_1} \cdot h^{tr'}), \\ &= \text{PEG}_2.\text{Dec}(\text{sk}, (g^{2N(r'+tr)}, (1+N)^{m'_0+tm_0} \cdot h^{r'+tr}, (1+N)^{m'_1+tm_1} \cdot h^{r'+tr'})), \\ &= (m'_0 + tm_0, m'_1 + tm_1). \end{aligned}$$

Property 6. Let $\text{ct} = \text{PEG}_2.\text{Enc}(\text{pk}, (m_0, m_1); r)$ and $\text{ct}' = \text{PEG}.\text{Enc}(\text{pk}, (m'_0, m'_1); r')$,

$$\begin{aligned} \text{ct} \cdot \text{ct}' &= ([g^{2Nr}]_{N^2}, [(1+n)^{m_0} h^r]_{N^2}, [(1+n)^{m_1} h^{r'}]_{N^2}) \\ &\quad \cdot ([g^{2Nr'}]_{N^2}, [(1+N)^{m'_0} h^{r'}]_{N^2}, [(1+N)^{m'_1} h^{r'}]_{N^2}), \\ &= ([g^{2N[r+r']}_{p'q'}]_{N^2}, [(1+N)^{(m_0+m'_0)} h^{[r+r']_{p'q'}}]_{N^2}, [(1+N)^{(m_1+m'_1)} h^{[r+r']_{p'q'}}]_{N^2}), \\ &= ([g^{2Nr''}]_{N^2}, [(1+N)^{(m_0+m'_0)} h^{r''}]_{N^2}, [(1+N)^{(m_1+m'_1)} h^{r''}]_{N^2}), \end{aligned}$$

with either or both r and r' uniformly random in $\mathbb{Z}_{p'q'}$, $r'' = [r + r']_{p'q'}$ is also uniform in $\mathbb{Z}_{p'q'}$. Now, recall that the encryption randomness is picked uniformly in $[0, (N-1)/4]$. Then Lemma 8 implies that the distributions,

$$\{[g^{2Nr}]_{N^2} : r \xleftarrow{\$} \mathbb{Z}_{p'q'}\} \text{ and } \{[g^{2Nr}]_{N^2} : r \xleftarrow{\$} [0, (N-1)/4]\},$$

are within $2^{-\lambda}$ statistical distance since the subgroup of $2N$ -th residues in $\mathbb{Z}_{N^2}^*$ is a cyclic group of order $p'q'$ and $[g^{2N}]_{N^2}$ is with overwhelming probability a generator for the latter [28].

Property 7. Given a key pair $(\text{pk}, \text{sk}) = ((N, g, h, h'), x, x')$, $\text{PEG}_2.\text{Dec} : H^* \rightarrow G$ takes (c_0, c_1, c_2) and returns $(m, m') \in G$ such that $1 + mN = [c_1 \cdot c_0^{-2x}]_{N^2}$ and $1 + m'N = [c_2 \cdot c_0^{-2x'}]_{N^2}$. Now consider $\mathfrak{h}' = ((1+N)^a, (1+N)^{2xa}, (1+N)^{2x'a}) \in H'$, for some $a \in \mathbb{Z}_N$ (note that a is invertible in $\mathbb{Z}_N$ with overwhelming probability), and $(\mu, \mu') \xleftarrow{\$} G$. Since h and $c_1 c_0^{-2x}$ only depend on $[x]_{p'q'}$ and since h' and $c_2 c_0^{-2x'}$ only depend on $[x']_{p'q'}$, we can compute another valid key pair $(\text{pk}, \widehat{\text{sk}}) = ((N, g, h, h'), \widehat{x}, \widehat{x}')$, such that $[\widehat{x}]_{p'q'} = [x]_{p'q'}$ and $[\widehat{x}']_{p'q'} = [x']_{p'q'}$. Then, we have $\text{PEG}_2.\text{Dec}(\widehat{\text{sk}}, \mathfrak{h}') = ([2(x - \widehat{x})a]_N, [2(x' - \widehat{x}')a]_N)$, and we can choose $\widehat{\text{sk}}$ such that $[\widehat{x}]_N = [x - \mu(2a)^{-1}]_N$ and $[\widehat{x}']_N = [x' - \mu'(2a)^{-1}]_N$ to get $\text{PEG}_2.\text{Dec}(\widehat{\text{sk}}, \mathfrak{h}') = (\mu, \mu')$. By Lemma 8, the statistical distance between $\{z \xleftarrow{\$} \mathbb{Z}_{Np'q'}\}$ and $\{z \xleftarrow{\$} [0, N(N-1)/4]\}$ (used for key generation) is negligible in λ . Since, $(\mu, \mu') \xleftarrow{\$} G$,

both $[\hat{x}]_N$ and $[\hat{x}']_N$ appear uniform in $\mathbb{Z}_N$. Also, $[\hat{x}]_{p'q'} = [x]_{p'q'}$ and $[\hat{x}']_{p'q'} = [x']_{p'q'}$ are uniform in $\mathbb{Z}_{p'q'}$ (from key generation). It follows that both $\hat{x}$ and $\hat{x}'$ are indistinguishable from uniform in $\llbracket 0, N(N-1)/4 \rrbracket$.

Property 8. For $c = (c_0, c_1, c_2) \in H$, c_0 is by definition of the form $[g^{2Nr}]_{N^2}$. For g a generator of $(\mathbb{Z}_{N^2}, \cdot)$, $(\langle g^{2N} \rangle, \cdot)$ is the only subgroup of $(\mathbb{Z}_{N^2}, \cdot)$ of order $2p'q'$. It follows that $c_0 \in (\langle g^{2N} \rangle, \cdot)$ if and only if $[c_0^{2p'q'}]_{N^2} = 1$. We can thus define an alternative decryption function that uses the factorization of N given by $\text{sk}_0 = (p', q')$,

- $\text{PEG}_2.\widehat{\text{Dec}}$: given $c = (c_0, c_1, c_2) \in \mathbb{Z}_{N^2}^3$, sk and $\text{sk}_0 = (p', q')$, return $\perp$ when $[c_0^{2p'q'}]_{N^2} \neq 1$ and, otherwise, return $\text{PEG}_2.\text{Dec}(\text{sk}, c)$.

□

B.3 Alternative proof of Prop. 16

In this section, we provide an alternative direct proof of Prop. 16 by porting the proof of [28, Theorem 3] to that scheme. We thus closely follows the proof [28, Theorem 3] such that our adaptations can easily be spotted when reading both proofs. It turns out that our adaptations are relatively minor. Interestingly, for DPEG we can rely only on DCR and avoid the additional reliance on the CNI assumption in Libert's original proof for PEG_z (we further discuss this point in Remark 33, p. 49).

Proof. Game G_0 . This is the CCA1 game against DPEG. Recall that DPEG is a private key scheme, to the adversary may perform encryption requests, a single challenge request and decryption requests over arbitrary elements $\mathbb{Z}_{N^2}^3$ until the output of its challenge request is given to it. For a challenge request, the adversary choose $m_0, m_1 \in \mathbb{Z}_n$ and, following Lemma 13, gets ciphertext

$$\text{ct}^* = (c_0^*, c_1^*, c_2^*) = ([g^{2Nr}]_{N^2}, [(1+N)^{m_\gamma} h^r]_{N^2}, [(1+N)^{\xi m_\gamma} h'^r]_{N^2}), \quad (20)$$

for unknown bit γ , randomness $r \in \llbracket 0, (N-1)/4 \rrbracket$ and ξ .

Game G_1 . The first game hop is identical to that of the proof of [28, Theorem 3]: G_1 is the same as G_0 except that the challenge ciphertext generation is modified to be,

$$\text{ct}^* = (c_0^*, c_1^*, c_2^*) = ([g^{2Nr}]_{N^2}, [(1+N)^{m_\gamma} c_0^{*2x}]_{N^2}, [(1+N)^{\xi m_\gamma} c_0^{*2x'}]_{N^2}), \quad (21)$$

rather than Eq. (20). This clearly does not change anything since $h^r = (g^{4Nx})^r = (g^{2Nr})^{2x} = c_0^{*2x}$, and similarly so for h'^r .

Game G_2 . The second game hop is also identical to that of the proof of [28, Theorem 3]: G_2 is the same game as G_1 except that the challenge generation is further modified so that c_0^* is now obtained by sampling a random N -th residue $z = [z_0^N]_{N^2}$, for some z_0 uniformly picked in $\mathbb{Z}_N^*$, and doing,

$$\text{ct}^* = (c_0^*, c_1^*, c_2^*) = ([z^2]_{N^2}, [(1+N)^{m_\gamma} c_0^{*2x}]_{N^2}, [(1+N)^{\xi m_\gamma} c_0^{*2x'}]_{N^2}), \quad (22)$$

rather than Eq. (21). The two games are indistinguishable as the statistical distance between the distribution of $[z^2]_{N^2}$ in Eq. (22) and that of $[g^{2Nr}]_{N^2}$ in Eq. (21) is bounded by $2^{-\lambda}$. This follows from a simple argument that we do not reproduce here [28, Lemma 3].

Game G_3 . The third game hop remains identical to that of the proof of [28, Theorem 3]: G_3 is identical to G_2 except that c_0 in Eq. (22) is no longer an N -th residue but generated by picking z uniformly in $\mathbb{Z}_{N^2}^*$. The DCR assumption then implies that G_2 and G_3 cannot be distinguished. From the definition of the Paillier scheme (App. E) recall that $\{[(1+N)^m g^{2Nr}]_{N^2} : m \in \mathbb{Z}_N, r \in \mathbb{Z}_{4p'q'}\}$ (with $y = [g^2]_{N^2}$ in the notation of App. E) is isomorphic to $\mathbb{Z}_{N^2}^*$. Therefore, $\{[(1+N)^m g^{2Nr}]_{N^2} : m \in \mathbb{Z}_N, r \in \mathbb{Z}_{p'q'}\}$ is isomorphic to $\{z^2 : z \in \mathbb{Z}_{N^2}^*\}$ and c_0^* may be interpreted as a Paillier encryption,

$$c_0^* = [(1+N)^\alpha g^{2N\beta}]_{N^2} \quad (23)$$

for some $\alpha \in \mathbb{Z}_N$ and $\beta \in \mathbb{Z}_{p'q'}$. Then,

$$c_1^* = [(1+N)^{[m_\gamma + 2\alpha x]_N} g^{[4N\beta x]_{p'q'}}]_{N^2}, \quad (24)$$

and,

$$c_2^* = [(1+N)^{[\xi m_\gamma + 2\alpha x']_N} g^{[4N\beta x']_{p'q'}}]_{N^2}. \quad (25)$$

Game G_4 . Here we depart from the proof of [28, Theorem 3] as, in our case, G_4 is identical to G_3 . We however keep this step such that our game numbering remains consistent with that in the proof of [28, Theorem 3]. See the post proof Remark 33 (p. 49).

Game G_5 . We are still progressing following the proof of [28, Theorem 3]. This time we modify the generation of x which is sampled uniformly in $\mathbb{Z}_{Np'q'}$ (in G_5) rather than in $\llbracket 0, N(N-1)/4 \rrbracket$ (in G_4). Then, the distinguishing advantage between G_4 and G_5 is bounded by the statistical distance between $\{[x]_{Np'q'} : x \xleftarrow{\$} \llbracket 0, N(N-1)/4 \rrbracket\}$ and $\{x \xleftarrow{\$} \mathbb{Z}_{Np'q'}\}$ which is negligible in λ [28, Lemma 3].

Game $G_{5'}$. We slightly deviate from the the proof of [28, Theorem 3] and apply the same modification to the generation of x' which, likewise, is now sampled uniformly in $\mathbb{Z}_{Np'q'}$ (in $G_{5'}$) rather than in $\llbracket 0, N(N-1)/4 \rrbracket$ (in G_5). The exact same argument yields a distinguishing advantage between G_5 and $G_{5'}$ negligible in the security parameter λ .

We now adapt the hybrid argument in the proof of [28, Theorem 3]. First, for notational convenience, $G_{5,0}$ is the same game as $G_{5'}$.

Game $G_{5,i}$ ($i \in \llbracket 1, Q \rrbracket$). Like in the original proof, we modify the key generation by letting $h = g^{4N\beta_x}$ and $h' = g^{4N\beta_{x'}}$ for both β_x and $\beta_{x'}$ uniformly drawn in $\mathbb{Z}_{p'q'}$. We then define an alternative secret key $\text{sk}' = (\beta_x, \beta_{x'}, \xi, p, q)$. To reply the i first decryption queries, the following modified decryption algorithm is used:

- **DPEG.Dec'**: on inputs (c_0, c_1, c_2) and sk' , return $\perp$ if $[c_0^{2p'q'}]_{N^2} \neq 1$. When this is not the case, compute $\mu = [c_1 \cdot c_0^{-2\beta_x}]_{N^2}$ and $\mu' = [c_2 \cdot c_0^{-2\beta_{x'}}]_{N^2}$. If there exists $m, m' \in \mathbb{Z}_N$ such that $\mu = [(1+N)^m]_{N^2}$, $\mu' = [(1+N)^{m'}]_{N^2}$ and $[\xi m]_N = m'$, return m . Otherwise return $\perp$.

After the i -th decryption query, the challenger picks α_x and $\alpha_{x'}$, both uniformly in $\mathbb{Z}_N$. It then chooses x in $\mathbb{Z}_{Np'q'}$ such that $\alpha_x = [x]_N$ and $\beta_x = [x]_{p'q'}$, and similarly so for $x' \in \mathbb{Z}_{Np'q'}$ i.e. $\alpha_{x'} = [x']_N$ and $\beta_{x'} = [x']_{p'q'}$. Then it uses $\text{sk} = (x, x', \xi)$ and the original decryption function, DPEG.Dec , to answer the remaining $Q - i$ decryption requests.

Lemma 32 implies that no adversary can distinguish between $G_{5,i}$ and $G_{5,i-1}$ so that, by the standard hybrid argument, $G_{5'} = G_{5,0}$ is indistinguishable from the last game in the sequence, $G_{5,Q}$, in which all the adversary's decryption requests are replied to by means of the above modified decryption algorithm.

Game G_6 . This is the last game hop. Game G_6 is identical to $G_{5,Q}$ except that the challenge ciphertext is computed as,

$$\begin{aligned} c_0^* &= [(1+N)^\alpha g^{2N\beta}]_{N^2}, \\ c_1^* &= [(1+N)^u g^{4N\beta[x]_{p'q'}}]_{N^2}, \\ c_2^* &= [(1+N)^{[\xi u]_N} g^{4N\beta[x']_{p'q'}}]_{N^2}, \end{aligned}$$

with α , β and u uniformly picked in $\mathbb{Z}_N$, $\mathbb{Z}_{p'q'}$ and $\mathbb{Z}_N$, respectively. Now, remark that in G_6 (as in $G_{5,Q}$), all the adversary's decryption requests are replied to by means of the modified decryption algorithm, $\text{DPEG.Dec}'$, and, most importantly, by means of $\text{sk}' = (\beta_x, \beta_{x'}, \xi, p, q) = ([x]_{p'q'}, [x']_{p'q'}, \xi, p, q)$. Hence, the challenger's replies are independent of $[x]_N$ and $[x']_N$ (which are further drawn after the last decryption request from the adversary). Now, recall that in $G_{5,Q}$, following Eqs. (23), (24) and (25), the challenge ciphertext is of the form,

$$\begin{aligned} c_0^* &= [(1+N)^\alpha g^{2N\beta}]_{N^2}, \\ c_1^* &= [(1+N)^{m_\gamma + 2\alpha[x]_N} g^{4N\beta x}]_{p'q'}]_{N^2}, \\ c_2^* &= [(1+N)^{\xi m_\gamma + 2\alpha[x']_N} g^{4N\beta x'}]_{p'q'}]_{N^2}, \end{aligned}$$

with α and β uniformly picked in $\mathbb{Z}_N$ and $\mathbb{Z}_{p'q'}$, respectively. Since $\gcd(\alpha, N) = 1$ (with overwhelming probability) and $\gcd(2, N) = 1$, the terms $[m_\gamma + 2\alpha[x]_N]_N$ and $[\xi m_\gamma + 2\alpha[x']_N]_N$ are, from $\mathcal{A}$'s view, both indistinguishable from uniform over $\mathbb{Z}_N$. The indistinguishability between $G_{5,Q}$ and G_6 thus follows.

The overall claim finally follows from the fact that, in G_6 , the challenge ciphertext is now independent from m_γ and, hence, from the challenge bit γ . $\square$

The following lemma is a natural adaptation of [28, Lemma 5] to the case of DPEG. Most of the proof is identical to that of the aforementioned lemma and reproduced for self-containedness. Only the end of the argument is slightly modified.

Lemma 32 (Adaptation of Lemma 5 in [28]). *For each $i \in \llbracket 1, Q \rrbracket$, $G_{5,i}$ is indistinguishable from $G_{5,i-1}$.*

Proof. From the adversary's view, games $G_{5,i}$ (where the i -th decryption request is replied to by means of the *modified* decryption algorithm) and $G_{5,i-1}$

(where the i -th decryption request is replied to by means of the *original* decryption algorithm) are indistinguishable unless the i -th decryption query involves a ciphertext that is rejected in the former but not in the latter.

For any ciphertext (c_0, c_1, c_2) such that $[c_0^{2p'q'}]_{N^2} = 1$, both algorithms yield the same output since,

$$[c_0^{-2\beta_x}]_{N^2} = [c_0^{-2[x]_{p'q'}}]_{N^2} \text{ and } [c_0^{-2\beta_{x'}}]_{N^2} = [c_0^{-2[x']_{p'q'}}]_{N^2}.$$

Then, on a ciphertext (c_0, c_1, c_2) such that $[c_0^{2p'q'}]_{N^2} \neq 1$, DPEG.Dec' returns $\perp$. So the bottom line is to assess the probability that the original decryption function, DPEG.Dec, does not return the same. For such a ciphertext, we have,

$$\begin{aligned} c_0 &= [\zeta_0(1+N)^{\alpha_0} g^{2N\beta_0}]_{N^2}, \\ c_1 &= [\zeta_1(1+N)^{\alpha_1} g^{2N\beta_1}]_{N^2}, \\ c_2 &= [\zeta_2(1+N)^{\alpha_2} g^{2N\beta_2}]_{N^2}, \end{aligned}$$

for some elements, ζ_0, ζ_1 and ζ_2 , of order at most 2 in $\mathbb{Z}_{N^2}^*$ and arbitrary values $\alpha_0 \in \mathbb{Z}_N \setminus \{0\}$, $\alpha_1 \in \mathbb{Z}_N$, $\alpha_2 \in \mathbb{Z}_N$ as well as β_0, β_1 and β_2 all in $\mathbb{Z}_{p'q'}$. Then, in $G_{5,i-1}$, DPEG.Dec computes,

$$\begin{aligned} \mu &= c_1 c_0^{-2x} \mod N^2, \\ &= c_1 (\zeta_0(1+N)^{\alpha_0} g^{2N\beta_0})^{-2x} \mod N^2, \\ &= c_1 (1+N)^{-2\alpha_0[x]_N} g^{-4N\beta_0[x]_{p'q'}} \mod N^2, \\ &= \zeta_1(1+N)^{\alpha_1-2\alpha_0[x]_N} g^{-2N(-\beta_1+2\beta_0[x]_{p'q'})} \mod N^2, \end{aligned}$$

and, similarly,

$$\begin{aligned} \mu' &= [c_2 c_0^{-2x'}]_{N^2}, \\ &= c_2 (1+N)^{-2\alpha_0[x']_N} g^{-4N\beta_0[x']_{p'q'}} \mod N^2, \\ &= \zeta_2(1+N)^{\alpha_2-2\alpha_0[x']_N} g^{-2N(-\beta_2+2\beta_0[x']_{p'q'})} \mod N^2. \end{aligned}$$

It may then be assumed that $\zeta_1 = \zeta_2 = 1$, that $\beta_1 = 2\beta_0[x]_{p'q'}$ and that $\beta_2 = 2\beta_0[x']_{p'q'}$ since both decryption functions return $\perp$ unless μ and μ' are such that there exists m and m' in $\mathbb{Z}_N$ with $\mu = [(1+N)^m]_{N^2}$ and $\mu' = [(1+N)^{m'}]_{N^2}$. Additionally, $\alpha_x = [x]_N$ and $\alpha_{x'} = [x']_N$ are picked uniformly in $\mathbb{Z}_N$ after $\mathcal{A}$ submitted its i -th decryption query over (c_0, c_1, c_2) , thus after $\mathcal{A}$'s fixing of α_0, α_1 and α_2 . Since $[x]_N$ and $[x']_N$ are respectively independent of $[x]_{p'q'}$ and $[x']_{p'q'}$, $-2\alpha_0[x]_N$ and $-2\alpha_0[x']_N$ are independent from $\mathcal{A}$'s view. To finalize the proof, we have to handle the following cases:

1. $\gcd(\alpha_0, N) = 1$: in this case, $\gcd(2\alpha_0, N) = 1$. It then follows that $[\alpha_1 - 2\alpha_0[x]_N]_N$ and $[\alpha_2 - 2\alpha_0[x']_N]_N$ are both uniform in $\mathbb{Z}_N$ (and independent of each other). For this case, the claim thus follows from the fact that $[\xi(\alpha_1 - 2\alpha_0[x]_N)]_N \neq [\alpha_2 - 2\alpha_0[x']_N]_N$ with overwhelming probability, so DPEG.Dec also returns $\perp$ at the i -th decryption request.

2. $\gcd(\alpha_0, N) > 1$ and $[\xi\alpha_1]_p = [\alpha_2]_p$: we assume wlog that $\alpha_0 = kp$. Then $[2\alpha_0[x]_N]_N$ and $[2\alpha_0[x']_N]_N$ are uniform multiples of p (and independent of each other). For this case, the claim follows from the fact that $[\xi 2\alpha_0[x]_N]_N \neq [2\alpha_0[x']_N]_N$ with overwhelming probability, so DPEG.Dec also returns $\perp$ at the i -th decryption request.
3. $\gcd(\alpha_0, N) > 1$ and $[\xi\alpha_1]_p \neq [\alpha_2]_p$: in this case, $[\xi(\alpha_1 - 2\alpha_0[x]_N)]_N$ and $[\alpha_2 - 2\alpha_0[x']_N]_N$ cannot be equal. So, again, DPEG.Dec also returns $\perp$ at the i -th decryption request.

□

Remark 33. In the original proof of [28, Theorem 3], G_4 introduces the following modification in the decryption oracle of the challenger. Indeed, G_4 is the same as G_3 except that decryption requests of the form (c_0, c_1, c_2) , where c_0 is a *Paillier* encryption of some $\alpha \in \mathbb{Z}_N$ such that $1 < \gcd(\alpha, N) < N$, are rejected. To do so, the challenger runs $\text{Pai.Dec}(c_0)$ (thus, with the knowledge of the factorization of $N = pq$), checks the latter condition and returns $\perp$ whenever it is true. Then, G_3 and G_4 are indistinguishable by reduction to the Composite Non-Invertibility (CNI) assumption [28, Lemma 4] (we recall the CNI assumption in Def. 40, Sect. D.1).

Although the distributions $\{\alpha \xleftarrow{\$} \mathbb{Z}_N\} \times \{\beta \xleftarrow{\$} \mathbb{Z}_{p'q'}\}$ and $\{\alpha \xleftarrow{\$} \mathbb{Z}_N^*\} \times \{\beta \xleftarrow{\$} \mathbb{Z}_{p'q'}\}$ are within a statistical distance negligible in the security parameter, reliance on the CNI assumption eventually occurs in the proof of [28, Lemma 5]. Indeed, due to the condition in $\text{PEG}_z.\text{Dec}$, the final argument in the proof of [28, Lemma 5] works under the restriction that $\gcd(\alpha_0, N) = 1$ over a c_0 that is generated by the adversary (which is not restricted to pick it at random).

In contrast, we explicitly deal with the case where $1 < \gcd(\alpha_0, N) < N$ in the proof of Lemma 32 above (cases 2 and 3 at the end of the proof): even when α_0 is a multiple of (wlog) p , $[2\alpha_0[x]_N]_N$ and $[2\alpha_0[x']_N]_N$ appear (from $\mathcal{A}$'s view) uniformly distributed in $[p\mathbb{Z}_q]_N$ so that $[\xi 2\alpha_0[x]_N]_N = [2\alpha_0[x']_N]_N$ only with negligible probability. †

As a last remark, we think that the proof of [28, Lemma 5] can be modified likewise: when α_0 is a multiple of (wlog) p , $[\alpha_1 - 2\alpha_0[x]_N]_N$ appears (from $\mathcal{A}$'s view) uniformly distributed in $[\alpha_1 + p\mathbb{Z}_q]_N$, and is still in $[0, B]$ with probability $\frac{B/p}{N/p} = \frac{B}{N} \leq \text{neg}(\lambda)$. This allows to remove game G_4 in the proof of [28, Theorem 3] and thus avoids reliance on the CNI assumption. In essence, we positively answers B. Libert's intuition: "we do not know if the Composite Non Invertibility assumption is strictly necessary or if it is an artifact of the proof" [28, p. 21].

B.4 Proof of Prop. 17

Proof. The proof starts by one step of game hopping.

First game hop. Let G_0 be the vCCA game against DPEG and G_1 be the same game as G_0 where we modify the challenger such that, when handling a decryption request on ciphertext c , the new challenger invokes the LOH extractor

to verify that

$$\text{extract}(c, \text{aux}) \neq \emptyset,$$

rather than checking $\text{DPEG.ImVer}(\text{sk}, c) = \text{True}$. Clearly, the two games cannot be distinguished, unless the adversary is able to create a ciphertext such that $\text{DPEG.ImVer}(\text{sk}, c) = \text{True}$ and which is not an affine combination of the well-formed ciphertexts that it generated, in violation with the assumption that DPEG has the LOH property.

Final reduction. To finalize the proof, we then show that, from an adversary $\mathcal{A}$ against G_1 , we can build an adversary $\mathcal{B}$ against the CPA security of DPEG which uses $\mathcal{A}$ as a subroutine and achieves at least $\mathcal{A}$'s advantage. For the reduction to work, we assume that $\mathcal{A}$ and $\mathcal{B}$ agree on a consistent numbering of the ciphertexts (remark then that ct^Δ is the first generated well-formed ciphertext and is assumed to have number 0). The reduction then proceeds as follows,

- When receiving the *single* challenge request over messages $m_0 \neq m_1 \in \mathbb{Z}_N$ from $\mathcal{A}$, it transfers it as is to the CPA challenger to get ciphertext $\text{ct}^* = \text{DPEG.Enc}(\text{pk}, m_\gamma)$ (for unknown bit γ), which it sends back to $\mathcal{A}$.
- When $\mathcal{A}$ issues a decryption request over ciphertext $\text{ct} \in \mathbb{Z}_{N^2}^3$, $\mathcal{B}$ runs the LOH extractor to get

$$(\pi, \beta) = \text{extract}(\text{ct}, \text{aux}).$$

Recall that since the only well-formed ciphertexts that $\mathcal{A}$ can generate from scratch (i.e. without using ct^Δ) are encryptions of 0, $\beta = 0$. Then, when $\pi = \emptyset$, $\mathcal{B}$ returns $\perp$ to $\mathcal{A}$. When it is not the case, then $\mathcal{B}$ returns $\perp$ whenever $\pi_1 \neq 0$, or π_0 when this is not the case ($\pi_1 = 0$).

Further remark that the LOH extractor works only on ciphertexts generated by the adversary (aux is essentially the trace of execution of $\mathcal{A}$) and *not* over those generated by the challenger. Thus, following Def. 7, we have $m = 2$ and the set $(c_0, \dots, c_{m-1})$ (in that definition) is reduced to $(\text{ct}^\Delta, \text{ct}^*)$ where ct^* is the challenge ciphertext. Hence, for a ciphertext ct independent of ct^* (i.e. such that $\pi_1 = 0$), $\text{DPEG.Dec}(\text{sk}, \text{ct}) = \pi_0$. This follows from the perfect correctness of DPEG and the fact that, by definition of DPEG's encryption function, Lemma 13 as well as Lemma 14 (PEG₂ satisfies Property 6 in Def. 9), any well-formed ciphertext $\text{ct} = (c_0, c_1, c_2)$ which encrypts a linear combination $\sum_i \alpha_i m_i$ is such that there exists r such that

$$c_0 = (\text{ct}^\Delta.c_0)^{\sum_i \alpha_i m_i} g^{2Nr}, \quad c_1 = (\text{ct}^\Delta.c_1)^{\sum_i \alpha_i m_i} h^r, \quad \text{and} \quad c_2 = (\text{ct}^\Delta.c_2)^{\sum_i \alpha_i m_i} h^{r'},$$

yielding $\pi_0 = \sum_i \alpha_i m_i$.

Lastly, since ct^* -dependent ciphertexts are such that $\pi_1 \neq 0$, the above reduction duly replies $\perp$ for all decryption requests on ciphertexts which are byproducts of the challenge ciphertext, consistently with the vCCA game decryption oracle specification. $\square$

B.5 Additional technical lemmas for Sect. 5.1

Recall the notations in Sect. 5.1.

Lemma 34. *Let $\mathcal{E}_H$ and $\mathcal{E}'_H$ denote the schemes in the proof of Prop. 19. Then $\mathcal{E}'_H$ is vCCA secure under the assumption that $\mathcal{E}_H$ is.*

Proof. Let G_0 denote the vCCA game against $\mathcal{E}'_H$. We first consider game G_1 which is the same as G_0 except that we modify the challenger such that the challenge ciphertext is generated (in G_1) as $\mathcal{E}_H.\text{Enc}(\mathcal{E}_H.\text{pk}, m_\gamma)$ rather than (in G_0) as $\mathcal{E}'_H.\text{Enc}(\mathcal{E}'_H.\text{pk}, m_\gamma)$. From the adversary's view, the two games are computationally indistinguishable since ciphertexts under both $\mathcal{E}_H$ and $\mathcal{E}'_H$ are indistinguishable from uniform over the same ciphertext domain $\mathcal{C}$.

To complete the proof, we show that given an adversary $\mathcal{A}$ against G_1 , we can build an adversary $\mathcal{B}$ against the vCCA security of $\mathcal{E}_H$ that uses $\mathcal{A}$ as subroutine and achieves the same advantage. The reduction $\mathcal{B}$ is simple: after getting $\mathcal{E}_H.\text{pk}$ from its challenger, it uses it to generate $\mathcal{E}'_H.\text{pk}$ following Eq. (17) and forwards $\mathcal{E}'_H.\text{pk}$ (but *not* $\mathcal{E}_H.\text{pk}$ to $\mathcal{A}$). Then, $\mathcal{B}$ forwards all of $\mathcal{A}$'s requests, which include one challenge request and polynomially many decryption requests, as is to its challenger. $\square$

Lemma 35. *Let $\mathcal{E}_H = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{Eval})$ denote a vCCA secure homomorphic scheme with function domain $\mathcal{F}$ restricted to linear functions and ciphertext domain $\mathcal{C} \supset \mathcal{C}_E \supseteq \mathcal{C}_F$, with $|\mathcal{C}_E| \leq 2^{-\lambda}|\mathcal{C}|$, that is correct, compact and satisfies property (12). Under the additional assumptions that ciphertexts under $\mathcal{E}_H$ are indistinguishable from uniform over $\mathcal{C}$ and that $(\mathcal{C}, \oplus)$ is an Abelian group, where $\oplus$ denotes $\mathcal{E}_H$'s homomorphic addition operator, we have that for any $m_0, \dots, m_{K-1} \in \mathcal{E}_H.\mathcal{P}$ and any PPT adversary $\mathcal{A}$,*

$$P \left(\begin{array}{l} c_e \in \mathcal{E}_H.\mathcal{C}_E \wedge \\ \{c'_0, \dots, c'_{L-1}\} \notin \{c_0, \dots, c_{K-1}\} \end{array} \left| \begin{array}{l} (pk, sk) := \mathcal{E}_H.\text{KeyGen}(1^\lambda) \\ (c_0, \dots, c_{K-1}) = (\mathcal{E}_H.\text{Enc}(pk, m_0), \dots, \mathcal{E}_H.\text{Enc}(pk, m_{K-1})) \\ c_e := \mathcal{A}(c_0, \dots, c_{K-1}) \\ (\alpha, c'_0, \dots, c'_{L-1}) := \text{extract}(c_e, \text{aux}) \end{array} \right. \right) \leq \text{neg}(\lambda), \quad (26)$$

Proof. Consistently with Eq. (26), we now define G_0 as the game, parameterized by polynomially many messages $m_0, \dots, m_{K-1}$, in which a challenger first runs $\mathcal{E}_H.\text{KeyGen}(1^\lambda)$ to get (pk, sk) , with neither of them being communicated to a PPT adversary $\mathcal{A}$. The challenger then gives,

$$(c_0, \dots, c_{K-1}) = (\mathcal{E}_H.\text{Enc}(pk, m_0), \dots, \mathcal{E}_H.\text{Enc}(pk, m_{K-1})) \quad (27)$$

to $\mathcal{A}$. The adversary wins G_0 if it generates a ciphertext c that belongs to $\mathcal{E}_H.\mathcal{C}_E$ which cannot be expressed as a linear combination of ciphertexts $c_0, \dots, c_{K-1}$. Remark that, from the statement, $\mathcal{E}_H$ is able only to homomorphically evaluate linear combinations of elements of $\mathcal{C}_F$ and is vCCA secure, hence, from Prop. 18, the vCCA extractor is that of a complete and knowledge sound SNARK over the language in Eq. (13). Hence, when, given $\mathcal{A}$'s trace of execution, the

extractor fails to extract a linear combination of the $c_0, \dots, c_{K-1}$ for ciphertext c , none exists (with overwhelming probability).

Game G_1 . We now consider game G_1 , also parameterized by polynomially many messages $m_0, \dots, m_{K-1}$, in which the challenger first runs $\mathcal{E}_H.\text{KeyGen}(1^\lambda)$ to get (pk, sk) , with neither of them being communicated to $\mathcal{A}$, but then communicates K uniformly picked elements from $\mathcal{C}$ instead of (27).

Indistinguishability between G_0 and G_1 . We now show that, given an adversary $\mathcal{A}_0$ than can distinguish between G_0 and G_1 , we can build an adversary $\mathcal{B}_0$ against the ciphertext pseudorandomness of $\mathcal{E}_H$ that uses $\mathcal{A}_0$ as a subroutine and achieves the same advantage as $\mathcal{A}_0$. The challenger against the pseudorandomness of $\mathcal{E}_H$ initially runs $\mathcal{E}_H.\text{KeyGen}(1^\lambda)$ to get (pk, sk) , communicates pk to $\mathcal{B}_0$ and picks its challenge bit $\gamma \xleftarrow{\$} \{0, 1\}$. The challenger also provides an encryption oracle that replies with genuine ciphertexts when $\gamma = 0$ and with uniformly picked elements in $\mathcal{C}$, $u_0, \dots, u_{L-1}$, otherwise. The reduction $\mathcal{B}_0$ then only performs K encryption requests towards its challenger and forwards the resulting challenger outputs directly to $\mathcal{A}_0$. It is clear that $\mathcal{B}_0$ exactly simulates G_0 when $\gamma = 0$ in its challenger, and G_1 otherwise.

In this last game, $\mathcal{A}$'s view is independent from $\mathcal{E}_H.\text{pk}$. Hence, the distribution of $\mathcal{A}$'s output is independent of the structure of $\mathcal{C}_E$, which depends on pk . Since $|\mathcal{C}_E| \leq 2^{-\lambda}|\mathcal{C}|$, even given polynomially many uniform samples from $\mathcal{C}$, a PPT adversary can output an element of $\mathcal{C}_E$ only with negligible probability¹⁴, i.e.,

$$P(c \in \mathcal{E}_H.\mathcal{C}_E | (\text{pk}, \text{sk}) := \mathcal{E}_H.\text{KeyGen}(1^\lambda), c := \mathcal{A}(u_0, \dots, u_{L-1})) \leq \text{neg}(\lambda).$$

□

C The mDPEG depth-1 multiplicative scheme

We can then apply this blueprint from DPEG as the underlying scheme, yielding scheme mDPEG which is defined as follows:

- mDPEG.KeyGen: run DPEG.KeyGen(1^λ) to get (pk, sk) .
- mDPEG.Enc: given $m \in \mathbb{Z}_N$ and pk , pick uniformly $b \in \mathbb{Z}_N$ and return¹⁵

$$(b, \text{DPEG.Enc}([m - b]_N)) = (c_0, c_1) \in \mathbb{Z}_N \times \mathbb{Z}_{N^2}^3. \quad (28)$$

¹⁴ Stricto sensu, this is true under the additional requirement that the only, if any, trivially valid ciphertext (i.e. that do not decrypt to $\perp$ under a non-negligible proportion of key pairs) is the neutral element of the Abelian group. For example, for DPEG, there exists only one such ciphertext, $(1, 1, 1)$, that decrypts to 0 under all key pairs. To the best of our knowledge, all known vCCA secure constructions satisfy this assumption (in particular, the generic constructions in [30, 10] have no such trivial ciphertexts).

¹⁵ Remark that, still following [22, Sect. 4.2], we slightly deviate from the original Catalano-Fiore blueprint which rather encrypt by doing $([m - b]_N, \text{DPEG.Enc}(b)) = (c_0, c_1)$. These two variants are equivalent, but Eq. (28) is more convenient for handling challenge requests in reductions.

- **mDPEG.Add**: given two (first form) ciphertexts $(c_0, c_1) \in \mathbb{Z}_N \times \mathbb{Z}_{N^2}^3$ and $(c'_0, c'_1) \in \mathbb{Z}_N \times \mathbb{Z}_{N^2}^3$ return (first form) ciphertext

$$(c''_0, c''_1) = ([c_0 + c'_0]_N, \text{DPEG.Add}(c_1, c'_1)).$$

- **mDPEG.Mul**: given two (first form) ciphertexts $(c_0, c_1) \in \mathbb{Z}_N \times \mathbb{Z}_{N^2}^3$ and $(c'_0, c'_1) \in \mathbb{Z}_N \times \mathbb{Z}_{N^2}^3$ return (second form) ciphertext

$$(c''_0, c''_1, c''_2) = (\text{DPEG.Add}(\alpha, \alpha', \alpha''), c_1, c'_1) \in \mathbb{Z}_{N^2}^9,$$

where $\alpha = \text{DPEG.Enc}([c_0 c'_0]_N)$, $\alpha' = \text{DPEG.MulByConst}_{c'_0}(c_1)$ and $\alpha'' = \text{DPEG.MulByConst}_{c_0}(c'_1)$.

- **mDPEG.Dec₁** (first form): given $(c_0, c_1) \in \mathbb{Z}_N \times \mathbb{Z}_{N^2}^3$, if $\text{DPEG.Dec}(\text{sk}, c_1) = \perp$ then return $\perp$. Otherwise return $[c_0 + \text{DPEG.Dec}(\text{sk}, c_1)]_N$.
- **mDPEG.Dec₂** (second form): given $(c_0, c_1, c_2) \in \mathbb{Z}_{N^2}^9$, if $\exists i \in \llbracket 0, 2 \rrbracket : \text{DPEG.Dec}(\text{sk}, c_i) = \perp$ then return $\perp$. Otherwise return,

$$[\text{DPEG.Dec}(\text{sk}, c_0) + \text{DPEG.Dec}(\text{sk}, c_1)\text{DPEG.Dec}(\text{sk}, c_2)]_N.$$

It is then easy to check that, for all $m, m' \in \mathbb{Z}_N$,

$$\text{mDPEG.Dec}_2(\text{sk}, \text{mDPEG.Mul}(\text{mDPEG.Enc}(\text{pk}, m), \text{mDPEG.Enc}(\text{pk}, m'))) = [mm']_N.$$

The CCA1 security of mDPEG then follows from that of DPEG from the next proposition which is a slight adaptation of [22, Prop. 4].

Proposition 36. *mDPEG is CCA1 secure under the DCR assumption.*

Proof. We proceed by showing that if there exists a CCA1 adversary $\mathcal{A}$ against mDPEG, then there exists a CCA1 adversary $\mathcal{B}$ against DPEG that uses $\mathcal{A}$ as a subroutine and achieves at least $\mathcal{A}$'s advantage. This is done via a simple reduction, $\mathcal{B}$, proceeds as follows:

- When $\mathcal{A}$ issues a challenge request with messages $m_0, m_1 \in \mathbb{Z}_N$, then $\mathcal{B}$ uniformly picks $b \in \mathbb{Z}_n$ and issue a challenge request with $[m_0 - b]_N, [m_1 - b]_N$ towards the CCA1 challenger against DPEG to get $c_1 = \text{DPEG.Enc}(m_\gamma - b)$ (for unknown bit γ). It then returns (b, c_1) to $\mathcal{A}$.
- When $\mathcal{A}$ issues a decryption request with (first form) ciphertext $(c_0, c_1) \in \mathbb{Z}_N \times \mathbb{Z}_{N^2}^3$, $\mathcal{B}$ first issues a decryption request over c_1 towards its challenger to get either $\perp$, in which case it also returns $\perp$, or $\mu \neq \perp$, in which case it returns $[c_0 + \mu]_N$ to $\mathcal{A}$.
- When $\mathcal{A}$ issues a decryption request with (second form) ciphertext $(c_0, c_1, c_2) \in \mathbb{Z}_{N^2}^9$, $\mathcal{B}$ issues decryption requests over c_0, c_1 and c_2 towards its challenger. If one or more $\perp$ reply is received, $\mathcal{B}$ also returns $\perp$. Otherwise, let μ_0, μ_1 and μ_2 denote the three replies from the challenger, $\mathcal{B}$ then returns $[\mu_0 + \mu_1 \mu_2]_N$ to $\mathcal{A}$.

The claim then follows from Prop. 16. $\square$

Proposition 37. *mDPEG is vCCA secure under the DCR and LOH assumption.*

Proof. The reduction in the proof of Prop. 36 towards a vCCA challenger (rather than a CCA1 challenger) against DPEG establishes that, if there exists a vCCA adversary $\mathcal{A}$ against mDPEG, then there exists a vCCA adversary $\mathcal{B}$ against DPEG that uses $\mathcal{A}$ as a subroutine and achieves at least $\mathcal{A}$'s advantage. The claim then follows from Prop. 17. $\square$

D Additional preliminaries

D.1 Computational assumptions

Definition 38 (DDH). *Given a cyclic group of prime order p , the Decision Diffie-Hellman assumption states that the distributions $\{(g, g^a, g^b, g^{ab}) : a, b \xleftarrow{\$} \mathbb{Z}_p\}$ and $\{(g, g^a, g^b, g^c) : a, b, c \xleftarrow{\$} \mathbb{Z}_p\}$ are computationally indistinguishable.*

Definition 39 (DCR, [28,35]). *Let $N = pq$, the Decision Composite Residuosity assumption states that the distributions $\{[z^N]_{N^2} : z \xleftarrow{\$} \mathbb{Z}_N^*\}$ and $\{z \xleftarrow{\$} \mathbb{Z}_{N^2}^*\}$ are computationally indistinguishable.*

Compared to the above two, the Composite Non-Invertibility (CNI) assumption is less standard although it is falsifiable and has been around for several years [26, Assumption 4.4]. The name CNI has been coined by Libert [28, Def. 1]. Informally, this assumption states that given a RSA modulus $N = pq$, an adversary cannot create an encryption of a multiple of either p or q under the Paillier scheme.

Definition 40 (CNI, [28,26]). *Let $\text{Pai} = (\text{KeyGen}, \text{Enc}, \text{Dec})$ denote the Paillier scheme (Sect. E), the Composite Non-Invertibility assumption states that for any PPT adversary $\mathcal{A}$ we have,*

$$P\left(\mu = \text{Pai.Dec}(c_e) \wedge \left| \begin{array}{l} ((N, y), \omega) := \text{Pai.KeyGen}(1^\lambda) \\ c := \mathcal{A}(N) \end{array} \right. \right) \leq \text{neg}(\lambda).$$

D.2 CPA^D security

Another security notion which is foundational for homomorphic encryption is that of CPA^D security [27]. This notion is a slight generalization of CPA security¹⁶, in which the CPA adversary is further granted access to a decryption oracle accepting only well-formed ciphertexts, that is either well-formed fresh ciphertexts obtained by genuinely running the encryption function or ciphertexts

¹⁶ Contrary to CPA, the CPA^D game by default allows for multiple challenge requests. While single- and multiple challenge variants of CPA security are equivalent [2,1], this is not the case for CPA^D [10].

derived from fresh well-formed ciphertexts via legit homomorphic evaluations. To do so, the CPA^D game provides encryption and evaluation oracles (even in the public-key setting) which purpose is to fill a game state with well-formed ciphertexts for handling subsequent evaluation or decryption requests on state indices. For *correct* encryption schemes, it is known that CPA^D security collapses onto CPA security [27]. When this assumption does not hold, which is the case for LWE- or RLWE-based schemes, CPA^D security has to be dealt with explicitly [27,17,18].

D.3 S. Schäge’s SHPKE scheme class

We recall the definition of the class of SHPKE recently introduced by Schäge [38] and over which no black-box reduction can establish CCA1 security under such a falsifiable assumption. This impossibility result has been later extended to cover the weaker notion of PCA1 security in [39].

Definition 41 (Semi-Homomorphic public-key encryption (SHPKE) reproduced from [38,39]). *A public-key encryption scheme $\text{PKE} = (\text{KeyGen}, \text{Enc}, \text{Dec})$ is semi-homomorphic if the following properties are fulfilled.*

1. *For any security parameter λ and all secret key/public key pairs $(pk, sk) \leftarrow \text{PKE.KeyGen}(1^\lambda)$, it is possible to define finite, cyclic groups $(G, +)$ and $(H, \cdot)$ such that G is the plaintext space, and $H = \text{PKE.Enc}(pk, G; \text{COIN})$ that is embedded in the ciphertext space $\mathcal{C} \supseteq H$. Also, G and H have an efficient membership test.*
2. *The set G grows exponentially in the security parameter λ .*
3. *There exists an efficient algorithm that draws integer t uniformly at random from $t \leftarrow [|G|]$ with only statistically small error probability δ .*
4. *Let $m, m' \in G$ and take any scalar $t \in [|G|]$. Then it holds that*

$$\begin{aligned} \text{PKE.Dec}(sk, \text{PKE.Enc}(pk, m') \cdot \prod_{i=1}^t \text{PKE.Enc}(pk, m)) &= m' + \sum_{i=1}^t m \\ &= m' + tm. \end{aligned}$$

5. *If $m \in G$ and $r \in D$ are taken uniformly random, then $\text{PKE.Enc}(pk, m; r)$ is distributed like a uniformly sampled ciphertext $c \in H$.*

It is important to point out that a key property of the SHPKE class is that H is required to have an efficient public membership test or, equivalently, that valid ciphertexts under a SHPKE can be publicly efficiently recognized.

D.4 vCCA extractor soundness (from [30])

Recall the definition of vCCA security (Def. 5, p. 10). Then [30, Def. 5] also provides the following definition.

Definition 42 (Soundness [30]). Given $\mathcal{E}_H = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{Eval})$ and $\mathcal{F}' \subseteq \mathcal{F}$, a $\mathcal{F}'$ -extraction algorithm *extract* is sound if for any PPT adversary $\mathcal{A}$, any $(pk, sk) \leftarrow \mathcal{E}_H.\text{KeyGen}(1^\lambda)$ and any $c \leftarrow \mathcal{A}(pk)$, $(f, c_0, \dots, c_{L-1}) \leftarrow \text{extract}(c, \text{aux})$ is such that,

- c_i is a fresh ciphertext, $\forall i \in \llbracket 0, L-1 \rrbracket$.
- The distribution of $\mathcal{E}_H.\text{Dec}(c)$ is indistinguishable from that of

$$f(\mathcal{E}_H.\text{Dec}(c_0), \dots, \mathcal{E}_H.\text{Dec}(c_{L-1})).$$

With the conventions that,

$$f(\mathcal{E}_H.\text{Dec}(c_0), \dots, \mathcal{E}_H.\text{Dec}(c_{L-1})) = \perp,$$

if $\exists i \in \llbracket 0, L-1 \rrbracket : \mathcal{E}_H.\text{Dec}(c_i) = \perp$ and that $\text{extract}(c, \text{aux}) = (id, c)$ for any fresh ciphertext c .

We point the following subtleties with the above definitions:

- “fresh” ciphertext does not mean “fresh well-formed” ciphertext (i.e. a ciphertext in the image of the encryption function) but rather a ciphertext that was not generated by the adversary running *Eval*.
- For a fresh ciphertext c , [30] states that the extractor can (consistently) return either $\perp$ or (id, c) . We remark, that if the extractor returns $\perp$, then the challenge ciphertext is not filtered out by the second step vCCA decryption oracle (which behavior is left unspecified when the extractor returns $\perp$ [30, Fig. 3]). So we assume the second option only.

We remark that the definition of vCCA security (Def. 5) naturally does not require the $\mathcal{F}'$ -extraction algorithm to be sound (since the existence of a sound extractor is a property that emerges from the cryptographic scheme). We also remark that, although the soundness of the extraction algorithm is sufficient to achieve vCCA security, it is not obvious at all from the definition of the notion that it is necessary.

D.5 Preliminaries on SNARGs and SNARKs

We follow the definitions in [25]. A SNARG system Π for a boolean relation $\mathcal{R} : X \times W \rightarrow \{\text{True}, \text{False}\}$ of a $\mathcal{NP}$ language,

$$L = \{x \in X : \exists w \in W, \mathcal{R}(x, w) = \text{True}\},$$

is a triplet of PPT algorithms $\Pi = (\text{Setup}, \text{Prove}, \text{Verify})$ defined as,

- **Setup:** on input 1^λ , outputs $(\text{crs}, [\text{sk}])$, where crs is a public common reference string and sk is an optional secret verification key.
- **Prove:** given crs , a statement $x \in X$ and a witness $w \in W$ such that $\mathcal{R}(x, w) = \text{True}$, produces a proof π for x .

- **Verif**: given crs and optionally sk as well as a statement x and a proof π , decides whether π is a valid proof for x .

When **Verif** need some secret input sk , the SNARG system is *designated verifier*. When no secret information is required to verify proofs, it is *publicly verifiable*. Then Π is a SNARG for L if the following properties holds:

1. *Completeness*: for all $(x, w) \in X \times W : \mathcal{R}(x, w) = \text{True}$,

$$P \left(\Pi.\text{Verif}(\text{crs}, [\text{sk}], x, \pi) = \text{False} \mid \begin{array}{l} (\text{crs}, [\text{sk}]) := \Pi.\text{Setup}(1^\lambda) \\ \pi := \Pi.\text{Prove}(\text{crs}, x, w) \end{array} \right) \leq \text{neg}(\lambda).$$

2. *Soundness*: for all PPT adversary $\mathcal{A}$,

$$P \left(\begin{array}{l} \Pi.\text{Verif}(\text{crs}, [\text{sk}], x, \pi) = \text{True} \\ \wedge x \notin L \end{array} \mid \begin{array}{l} (\text{crs}, [\text{sk}]) := \Pi.\text{Setup}(1^\lambda) \\ (x, \pi) := \mathcal{A}(\text{crs}) \end{array} \right) \leq \text{neg}(\lambda).$$

3. *Succinctness*: the length of a proof is $|\pi| = \text{poly}(\lambda)(|x| + |w|)^{o(1)}$.

The SNARG Π is further a SNARK for L when properties 1 and 3 hold and property 2 is replaced by the stronger property that:

- *Knowledge Soundness*: for all PPT adversary $\mathcal{A}$, there exists a PPT extractor **extract** such that,

$$P \left(\begin{array}{l} \Pi.\text{Verif}(\text{crs}, [\text{sk}], x, \pi) = \text{True} \\ \wedge \mathcal{R}(x, w) = \text{False} \end{array} \mid \begin{array}{l} ([\text{crs}], \text{sk}) := \Pi.\text{Setup}(1^\lambda) \\ (x, \pi) := \mathcal{A}(\text{crs}) \\ w := \text{extract}(x, \pi, \text{aux}) \end{array} \right) \leq \text{neg}(\lambda).$$

Note that this definition does not cover zk -SNARKs as we do not need any zero-knowledge property in this paper. Also, the above definition does not cover the notion of simulation-extractability (see e.g. [33, Sect. 5] for details) that is also not needed in this work.

E The Paillier scheme

In this section, we remind the Paillier scheme. We focus of the folklore “new school” variant which departs slightly from the original scheme presented in [35].

- **Pai.KeyGen**: choose a RSA modulus $N = (2p' + 1)(2q' + 1)$, with p' and q' two distinct Sophie Germain primes of suitable size. Then, uniformly pick $y \in \mathbb{Z}_{N^2}$ ¹⁷. Then $\text{pk} = (N, y)$ forms the public key. Further let $\omega = \text{lcm}(p-1, q-1)$ which is the private decryption key, sk .

¹⁷ Stricto sensu, y (resp. r) should be picked in $\mathbb{Z}_{N^2}^*$ (resp. $\mathbb{Z}_N^*$), i.e. $\{x \in \mathbb{Z}_{N^2} : \gcd(x, N^2) = 1\}$ (resp. $\{x \in \mathbb{Z}_N : \gcd(x, N) = 1\}$). However, the probability of picking a y (resp. r) uniformly in $\mathbb{Z}_{N^2}$ (resp. $\mathbb{Z}_N$) such that $\gcd(y, N^2) \neq 1$ (resp. $\gcd(y, N) \neq 1$) is negligible in the security parameter λ .

- **Pai.Enc**: given $m \in \mathbb{Z}_N$ and (N, y) , picks r uniformly in $\mathbb{Z}_N$ ¹⁸ and return $[(1 + N)^m y^{rN}]_{N^2}$. Alternatively, one may equivalently return $[(1 + mN)y^{rN}]_{N^2}$.
- **Pai.Dec**: given $c \in \mathbb{Z}_{N^2}$ and ω , let,

$$L(x) = \frac{x - 1}{N}, \quad (29)$$

and return

$$\frac{L([c^\omega]_{N^2})}{\omega} \bmod N.$$

It should be emphasized that the division in Eq. (29) is an integer division when all other divisions should be understood as multiplications by an inverse relatively to the appropriate modulus (i.e. either N or N^2). It should further be emphasized that, although the division in Eq. (29) is exact for well-formed ciphertexts, *the scheme specification usually does not indicate what to do when $x - 1$ is not a multiple of N* . When this occurs (because an ill-formed ciphertext is inputted to the decryption function), one may either decide to use $\lfloor \frac{x-1}{N} \rfloor$ or to return $\perp$. When only CPA security is at stake, both options can equally be safely chosen.

¹⁸ See footnote 17.

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